

Iterative Exploration-Driven Sparse SDP Clustering via Thompson Sampling

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Abstract

High-dimensional sparse clustering is a combinatorial NP-hard problem that arises from the coupling between cluster assignment and variable selection. We demonstrate that semidefinite programming (SDP) relaxation of K-means is robust to variable over-selection by establishing minimax separation bounds. Leveraging this robustness, we propose a block-coordinate ascent framework that alternates between SDP-based clustering and conservative variable selection. To address the tendency of deterministic greedy methods to become trapped in local optima, we formulate the variable selection step as a bandit problem. Crucially, to reliably evaluate feature utility and generate stochastic rewards for Thompson sampling even under imperfect intermediate cluster assignments, we employ a robust maximum mean discrepancy (MMD) permutation test. This approach introduces adaptive memory by aggregating historical variable-selection outcomes into posterior distributions, and selects features via posterior sampling, enabling stochastic exploration that promotes the inclusion of underexplored features and facilitates escape from local maxima. We establish conditions for consistent variable selection and exact cluster recovery, and extend the method to settings with unknown covariance through a scalable estimation procedure. Synthetic experiments and a real-data application in document clustering demonstrate that the proposed memory-driven randomized approach consistently outperforms state-of-the-art sparse clustering methods.

Keywords: high-dimensional data; K-means; multi-armed bandit; permutation test; semidefinite relaxation; variable selection

1 Introduction

Cluster structure in high-dimensional data often depends on a small subset of variables, such as a few genes distinguishing disease subtypes (Golub et al. 1999). While leveraging this sparsity improves clustering in domains such as genomics (Lu et al. 2017), neuroimaging (Mishra et al. 2017, Namgung et al. 2024), and finance (Nystrup et al. 2021), it is challenging because variable selection and clustering are highly interdependent discrete problems: identifying signals is difficult without knowing clusters, and vice versa. We adopt a statistical perspective on this coupled combinatorial optimization problem. Because our primary goal is accurate clustering, we treat variable inclusion indicators as nuisance parameters. This paper formally shows that consistent and efficient estimation of cluster assignments does not require precise identification or optimal convergence of these nuisance components.

We use the planted structure model of Giraud & Verzelen (2019) to formalize the sparsity structure. Assume that the number of clusters K is known. Let $G^* := (G_1^*, \dots, G_K^*)$ denote non-random partition of observation indices $\{1, \dots, n\}$. Let $\mathbf{c}_1^*, \dots, \mathbf{c}_K^* \in \mathbb{R}^p$ and $\Sigma^* \in \mathbb{R}^{p \times p}$ denote the unknown cluster centers and the common covariance matrix, respectively. We observe a data matrix $\mathbf{X} \in \mathbb{R}^{p \times n}$, with columns $\mathbf{X}_1, \dots, \mathbf{X}_n$ independently generated as

$$\mathbf{X}_i = \mathbf{c}_k^* + \mathbf{W}_i \text{ if } i \in G_k^*, \quad \mathbb{E}[\mathbf{W}_i] = 0, \quad \text{Cov}(\mathbf{W}_i) = \Sigma^*. \quad (1)$$

The goal is to recover G^* from the observations $\mathbf{X}$ alone, treating the $\mathbf{c}_k^*$ and Σ^* as nuisance parameters. Let $\Omega^* := (\Sigma^*)^{-1}$. For two-class Gaussian mixtures, the Bayeoptimal classification boundary, which maximizes the signal-to-noise ratio, is proportional to $\Omega^*(\mathbf{c}_1^* - \mathbf{c}_2^*)$ (Fan et al. 2013). Motivated by this, the high-dimensional clustering literature (Cai et al. 2019, Azizyan et al. 2015) assumes an unknown set of signal variables, denoted by S^* , and quan-

tifies clustering signal strength via the squared ℓ_2 distance:

$$S^* := \bigcup_{1 \leq k_1 \neq k_2 \leq K} \text{supp}(\Omega^*(\mathbf{c}_{k_1}^* - \mathbf{c}_{k_2}^*)) \subset \{1, \dots, p\}, \quad \Delta_{S \cap S^*}^2 := \min_{1 \leq k_1 \neq k_2 \leq K} \|(\Omega^*(\mathbf{c}_{k_1}^* - \mathbf{c}_{k_2}^*))_{S \cap S^*}\|_2^2, \quad (2)$$

The interdependence of variable selection and clustering naturally motivates alternating updates. However, many existing implementations of this approach (Witten & Tibshirani 2010, Cai et al. 2019) lack safeguards against convergence to poor local solutions. To address this, our method combines *partial relaxation*, which enlarges the search space to make the optimization more tractable, with a *bandit-driven random search* to escape local optima.

For the partial relaxation, we build on recent theoretical advances in semidefinite programming (SDP) relaxations for K -means in statistics literature. Under a sub-Gaussian noise assumption, we extend existing minimax theory (Chen & Yang 2021) to incorporate variable selection and to quantify the sensitivity of SDP K -means to the (possibly randomly) chosen variable set. In particular, we establish a uniform minimax separation bound that guarantees exact cluster recovery whenever the signal strength

$$\Delta_{S \cap S^*}^2 \gtrsim \log n + \frac{|S| \log p}{n} + \sqrt{\frac{|S| \log p}{n}}. \quad (3)$$

This result shows that SDP-relaxed K -means is robust to moderate variable over-selection and mild under-selection: accurate clustering requires only that a moderately sized variable set captures sufficient signal, rather than exactly recovering the true signal variable set.

Motivated by the insight that perfect variable selection is unnecessary for optimal clustering, we develop a block coordinate optimization algorithm that alternates between SDP-relaxed K -means and stochastic variable selection. We formulate variable selection as a multi-armed bandit problem and solve it via Thompson Sampling, modeling each variable's

clustering utility as a latent Bernoulli parameter and estimating it through an iteratively updated Beta posterior. This framework introduces two key innovations for sparse clustering: (i) *active exploration*, where variable inclusion indicators are sampled from their posterior distributions, encouraging the exploration of underrepresented variables and helping the algorithm escape poor local optima; and (ii) *memory*, where accumulated evidence from past successes and failures in variable selection stabilizes learning and mitigates overfitting to transient under- or over-selection in any single iteration.

For bandit formulation, we first cast variable selection as a maximum mean discrepancy (MMD)-based two-sample testing problem, assessing whether the clusters differ in distribution along each individual variable. To obtain stochastic rewards at each iteration, we employ permutation tests. MMD tests based on permutation procedures are known to be optimal for two-sample testing under various forms of data corruption (Schrab & Kim 2025, Mun et al. 2025, Kim & Schrab 2026). Specifically, we adopt the adversarial corruption framework of Schrab & Kim (2025) to ensure robustness of the variable selection step in the presence of imperfect clustering from earlier iterations of the algorithm.

Finally, for unknown covariance structure, we avoid full covariance estimation by employing the “innovated transformation” $\Omega^* \mathbf{X}$ (Fan & Lv 2016), which estimates only the necessary precision components, ensuring computational efficiency.

1.1 Existing Literature

Unsupervised dimension reduction methods such as PCA may emphasize variation that is unrelated to the sparse clustering signal and, as a result, hinder rather than facilitate downstream clustering (Chang 1983). The sparse clustering literature therefore differs primarily in how the clustering signal is defined and how it is extracted. Existing approaches can be broadly classified into two categories: two-step procedures and iterative methods.

We focus our review on iterative methods, as they are more directly related to our work. For comprehensive discussions of two-step procedures, we refer readers to [Jin & Wang \(2016\)](#) and [Witten & Tibshirani \(2010\)](#). In particular, [Jin et al. \(2017\)](#) show that within the two-step framework, accurate clustering can still be achieved without perfect variable selection under certain sparsity and signal regimes.

Compared to two-step methods, iterative approaches jointly update variable selection and cluster assignments, typically alternating clustering with regularized estimation of model parameters such as variable weights ([Witten & Tibshirani 2010](#), [Huo & Tseng 2017](#)), cluster centers ([Sun et al. 2012](#)), and variable rankings ([Arias-Castro & Pu 2017](#), [Zhang et al. 2020](#)). Regularized EM methods follow the same principle, updating labels in the E-step and sparse model parameters in the step ([Dong et al. 2024](#), [Pan & Shen 2007](#), [Zhou et al. 2009](#)). Notably, [Cai et al. \(2019\)](#) derive a minimax rate for excess clustering risk on new observations via ℓ_1 -regularized EM estimation of transformed cluster center differences.

Casting variable selection as a reinforcement learning problem is intractable, as the state space grows exponentially with the number of features. To address this, bandit-based methods have been proposed, including Upper Confidence Trees for classification ([Gaudel & Sebag 2010](#)) and Thompson sampling for regression ([Liu & Ročková 2023](#)). We extend these ideas to unsupervised clustering. Our bandit-based approach to solving mixed-integer programs is also related to recent learning-to-optimize literature ([Scavuzzo et al. 2024](#)), where reinforcement learning is integrated into branch-and-bound to learn variable selection policies or primal heuristics from signals generated during the optimization process.

1.2 Notations

Vectors and matrices are denoted in bold, with entries in corresponding non-bold. For a matrix $\mathbf{A}$, single and double subscripts denote columns and (row, column) entries; sub-

scripts may be indices or index sets. For vectors, subscripts denote elements or sub-vectors, and parentheses avoid ambiguity, for example, $(\tilde{\mathbf{c}}_1^*)_S$. We write $\mathbf{A} \geq 0$, $\mathbf{A} > 0$, and $\mathbf{A} \succeq 0$ for elementwise nonnegativity, positivity, and positive semidefiniteness, respectively. For a set S , let $|S|$ denotes its cardinality. For an integer $a > 0$, let $\mathbf{1}_a \in \mathbb{R}^a$ denote the *all-one vector*, and $[a]$ denote $\{1, \dots, a\}$. For a set $G_k \subset [n]$, let $\mathbf{1}_{G_k} \in \{0, 1\}^n$ denote the indicator vector that takes the value 1 at indices in G_k and 0 elsewhere. When referring to sections, figures and equations in the Supplementary Material, we add an “S” before the number. For example, Section 11.

2 SDP K -Means with variable selection

We first review the SDP relaxation of K -means and its high-dimensional limitations via a minimax analysis. We then incorporate sparsity through a best-subset SDP formulation, discuss its challenges, and establish robustness to variable over-selection.

2.1 SDP K -means

The K -means objective maximization over all possible partitions G is formulated as the following integer program (Zhuang et al. 2023):

$$\max_{\mathbf{H} \in \{0,1\}^{n \times K}} \langle \mathbf{X}^\top \Omega^* \mathbf{X}, \mathbf{H} \mathbf{D} \mathbf{H}^\top \rangle, \text{ s.t. } \mathbf{H} \mathbf{1}_K = \mathbf{1}_n, \quad (4)$$

where $\mathbf{D} := \text{diag}(|G_1|, \dots, |G_K|)$, $H_{i,k} := \mathbb{1}(i \in G_k)$ for $i \in [n]$ and $k \in [K]$ and $\langle \cdot, \cdot \rangle$ denotes the Frobenius inner product. With the change of variable $\mathbf{Z} = \mathbf{H} \mathbf{D} \mathbf{H}^\top$ satisfying $\mathbf{Z}^\top = \mathbf{Z}$, $\mathbf{Z} \succeq 0$, $\text{tr}(\mathbf{Z}) = K$, $\mathbf{Z} \mathbf{1}_n = \mathbf{1}_n$, and $\mathbf{Z} \geq 0$, we obtain the SDP relaxation of (4):

$$\max_{\mathbf{Z} \in \mathbb{R}^{n \times n}} \langle \mathbf{X}^\top \Omega^* \mathbf{X}, \mathbf{Z} \rangle \quad \text{s.t.} \quad \mathbf{Z}^\top = \mathbf{Z}, \mathbf{Z} \succeq 0, \text{tr}(\mathbf{Z}) = K, \mathbf{Z} \mathbf{1}_n = \mathbf{1}_n, \mathbf{Z} \geq 0. \quad (5)$$

The SDP K -means algorithm recovers cluster labels by applying spectral clustering to the solution of (5). When Σ^* is known to be $\mathbf{I}_p$, $\mathbf{W}_i$ are Gaussians, and the dimensionality p is fixed, SDP K -means achieves the minimax separation under the exact cluster recovery loss (Chen & Yang 2021), where the separation is defined as in (2) with $S = [p]$. Since the minimax separation $\log n + \sqrt{(\log n)^2 + (Kp \log n)/n}$ grows with p at a square-root rate, methods that ignore sparsity (including SDP K -means) may fail when p is comparable to or larger than n , unless separation scales accordingly. This motivates adaptations of SDP to incorporate sparsity in high dimensions and studying the corresponding separation rate.

2.2 Incorporating Sparsity via Best Subset SDP K -Means

We extend the SDP relaxation in (5) to incorporate variable selection. For exposition, we consider two clusters and a known precision matrix Ω^* , and define the transformed data $\widetilde{\mathbf{X}} := \Omega^* \mathbf{X}$. As discussed in Section 1, such transformation is motivated by the Bayeoptimal decision rule in classification literature. Then $\mathbf{X}^\top \Omega^* \mathbf{X} = \widetilde{\mathbf{X}}^\top \Sigma^* \widetilde{\mathbf{X}}$, and $\widetilde{\mathbf{X}} = \widetilde{\mathbf{C}}^* + \widetilde{\mathbf{W}}$, where $\widetilde{\mathbf{C}}^*$ is the transformed cluster center matrix and $\widetilde{\mathbf{W}}$ is the transformed noise matrix. So, the SDP objective separates into a signal term $(\widetilde{\mathbf{C}}^*)^\top \Sigma^* \widetilde{\mathbf{C}}^*$ and noise terms independent of $\widetilde{\mathbf{C}}^*$. Under sparsity assumption (2), the signal concentrates on S^* :

$$(\widetilde{\mathbf{C}}^*)^\top \Sigma^* \widetilde{\mathbf{C}}^* = (\widetilde{\mathbf{C}}_{S^*}^*)^\top \Sigma_{S^*, S^*}^* \widetilde{\mathbf{C}}_{S^*}^* + \text{terms involving features in } [p] \setminus S^*.$$

Motivated by this, for a candidate variable set S we approximate the signal component $(\widetilde{\mathbf{C}}_{S^*}^*)^\top \Sigma_{S^*, S^*}^* \widetilde{\mathbf{C}}_{S^*}^*$ using its empirical analogue $\widetilde{\mathbf{X}}_S^\top \Sigma_{S, S}^* \widetilde{\mathbf{X}}_S$ and solve

$$\max_{\mathbf{Z} \in \mathbb{R}^{n \times n}} \langle \widetilde{\mathbf{X}}_S^\top \Sigma_{S, S}^* \widetilde{\mathbf{X}}_S, \mathbf{Z} \rangle \quad \text{s.t.} \quad \mathbf{Z}^\top = \mathbf{Z}, \mathbf{Z} \succeq 0, \mathbf{Z} \geq 0, \text{tr}(\mathbf{Z}) = K, \mathbf{Z} \mathbf{1}_n = \mathbf{1}_n. \quad (6)$$

See Section 22.1 for empirical evidence that this objective indeed captures the clustering

signal. We refer to the clustering procedure, which maps $(\widetilde{\mathbf{X}}, S)$ to cluster labels, as *subset SDP* (Algorithm 1); it serves as the core subroutine of our block coordinate framework.

To jointly optimize over the partition $\mathbf{Z}$ and the feature set S , we introduce binary indicators $\mathbf{y} \in \{0, 1\}^p$, where $y_j = 1$ if variable j is selected. This yields the *Best Subset SDP*:

$$\begin{aligned} \max_{\mathbf{Z}, \mathbf{y}} \quad & \sum_{j_1=1}^p \sum_{j_2=1}^p y_{j_1} y_{j_2} \Sigma_{j_1, j_2}^* \langle \widetilde{\mathbf{X}}_{j_1} \widetilde{\mathbf{X}}_{j_2}^\top, \mathbf{Z} \rangle \\ \text{s.t.} \quad & \mathbf{Z}^\top = \mathbf{Z}, \mathbf{Z} \succeq 0, \mathbf{Z} \geq 0, \text{tr}(\mathbf{Z}) = K, \mathbf{Z} \mathbf{1}_n = \mathbf{1}_n, y_j \in \{0, 1\}, j = 1, \dots, p, \sum_{j=1}^p y_j \leq |S^*|. \end{aligned} \quad (7)$$

This is a mixed-integer program with bilinear coupling between $\mathbf{Z}$ and $\mathbf{y}$. Global optimization is intractable since optimal variable selection depends on the partition and vice versa, even under simplified covariance structures. To address this, we propose a block coordinate framework alternating between SDP-based clustering and variable selection. Section 2.3 provides theoretical justification, showing that exact recovery is achieved as long as the selected subsets retain sufficient signal-to-noise ratio, even if imperfectly specified.

2.3 Robustness of Subset SDP to Variable Over-Selection

When $\Sigma^* = \sigma^2 \mathbf{I}_p$ and $\mathbf{W}_i$ are Gaussian, SDP K -means restricted to an arbitrary S solves:

$$\max_{\mathbf{Z} \in \mathbb{R}^{n \times n}} \langle \mathbf{X}_{S, \cdot}^\top \mathbf{X}_{S, \cdot}, \mathbf{Z} \rangle \quad \text{s.t.} \quad \mathbf{Z}^\top = \mathbf{Z}, \mathbf{Z} \succeq 0, \text{tr}(\mathbf{Z}) = K, \mathbf{Z} \mathbf{1}_n = \mathbf{1}_n, \mathbf{Z} \geq 0. \quad (8)$$

Algorithm 1 Subset SDP K -means for $K = 2$: **SDPclust**

Require: Transformed and truncated data matrix $\widetilde{\mathbf{X}}_{S, \cdot}$, covariance sub-matrix $\Sigma_{S, S}$

- 1: $\hat{\mathbf{Z}} \leftarrow \arg \max_{\mathbf{Z} \in \mathbb{R}^{n \times n}} \langle \widetilde{\mathbf{X}}_{S, \cdot}^\top \Sigma_{S, S} \widetilde{\mathbf{X}}_{S, \cdot}, \mathbf{Z} \rangle$
 $\quad \text{s.t.} \quad \mathbf{Z}^\top = \mathbf{Z}, \mathbf{Z} \succeq 0, \text{tr}(\mathbf{Z}) = 2, \mathbf{Z} \mathbf{1}_n = \mathbf{1}_n, \mathbf{Z} \geq 0$
 - 2: Output $\hat{G} = (\hat{G}_1, \hat{G}_2) \leftarrow$ spectral clustering result on $\hat{\mathbf{Z}}$
-

Our analysis proceeds in two steps: (i) if the separation in $S \cap S^*$ (as in (2)) exceeds a threshold depending on n , p , and subset size $|S|$, exact recovery holds uniformly over all S with high probability; (ii) we characterize the corresponding minimax optimal regime.

For $\psi = (\mathbf{c}_1, \dots, \mathbf{c}_K, \mathbf{H})$, with $\mathbf{c}_k \in \mathbb{R}^p$ cluster centers and $\mathbf{H} \in \{0, 1\}^{n \times K}$ cluster assignment matrix, let $\mathbb{P}_\psi$ denote the joint distribution of n independent Gaussian vectors with means $\mathbf{c}_1, \dots, \mathbf{c}_K$ and covariance $\sigma^2 \mathbf{I}_p$. For all $1 \leq k_1 \neq k_2 \leq K$, assume $\mathbf{c}_{k_1} - \mathbf{c}_{k_2}$ has at most s nonzero entries and its squared ℓ_2 norm exceeds a threshold Δ^2 . The parameter space $\Psi(s, \Delta, K) := \mathcal{C}(s, \Delta, K) \times \mathcal{H}(K)$ is defined as the product of the following sets:

$$\begin{aligned} \mathcal{C}(s, \Delta, K) &:= \{(\mathbf{c}_1, \dots, \mathbf{c}_K) : \mathbf{c}_k \in \mathbb{R}^p, \|\mathbf{c}_{k_1} - \mathbf{c}_{k_2}\|_2^2 \geq \Delta^2, \text{ for all } 1 \leq k_1 \neq k_2 \leq K, \\ &\quad \cup_{k_1, k_2} \text{supp}(\mathbf{c}_{k_1} - \mathbf{c}_{k_2}) = S^* \subset [p], |S^*| \leq s\}, \\ \mathcal{H}(K) &:= \{\mathbf{H} = (H_{i,k}) \in \{0, 1\}^{n \times K} : \sum_{k=1}^n H_{i,k} = 1 \text{ for all } i = 1, \dots, n\}. \end{aligned}$$

For a given clustering algorithm $\hat{\mathbf{H}} : \mathbb{R}^{p \times n} \rightarrow \mathcal{H}(K)$, we assess its performance using the worst-case probability of exact cluster recovery over the distribution class $\Psi(s, \Delta, K)$:

$$\sup_{\psi \in \Psi(s, \Delta, K)} \min_{\pi \in \text{perm}(K)} \mathbb{P}_\psi(\hat{\mathbf{H}}(\mathbf{X}) \neq \pi(\mathbf{H})), \quad (9)$$

where $\text{perm}(K)$ is the set of label permutations of $[K]$ and $\pi(\mathbf{H})$ is the matrix after label permutation specified by $\pi \in \text{perm}(K)$. For brevity, we omit $\min_{\pi \in \text{perm}(K)}$ henceforth.

Given a problem instance $\psi^* = (\mathbf{c}_1^*, \dots, \mathbf{c}_K^*, \mathbf{H}^*) \in \Psi(s, \Delta, K)$, the true cluster assignment $\mathbf{H}^*$ has one-to-one correspondence to the block-diagonal matrix $\mathbf{Z}^* := \mathbf{H}^* \mathbf{D}^* (\mathbf{H}^*)^\top = \sum_{k=1}^K |G_k^*|^{-1} \mathbf{1}_{G_k^*} \mathbf{1}_{G_k^*}^\top$, up to label permutations. Let $\mathcal{S}$ denote the collection of variable

subsets with sufficiently strong signal:

$$\begin{aligned} \mathcal{S} &:= \{S : S \subset [p], |S| \leq \sqrt{p}, \Delta_{S \cap S^*}^2 \gtrsim \bar{\Delta}_S^2\}, \\ \bar{\Delta}_S^2 &:= \sigma^2 \left(\log n + \frac{|S| \log p}{m} + \sqrt{\frac{|S| \log p}{m}} \right), \quad m := 2 \min_{1 \leq k_1 \neq k_2 \leq K} \left\{ \left(\frac{1}{|G_{k_1}^*|} + \frac{1}{|G_{k_2}^*|} \right)^{-1} \right\}. \end{aligned} \quad (10)$$

Theorem 1 gives the separation required for SDP K -means to achieve exact recovery:

Theorem 1. *Assume $m \geq C_1 n / \log n$ with some universal constant $C_1 > 0$ and $|G_k^*| \geq 2$ for all $k \in [K]$. Let $\hat{\mathbf{Z}}(S)$ be the solution of (8) corresponding to the subset $S \in \mathcal{S}$. Then*

$$\mathbb{P}(\hat{\mathbf{Z}}(S) = \mathbf{Z}^*, \forall S \in \mathcal{S}) \geq 1 - C_3 K/n,$$

where $C_2 > 0$ is some constant. Furthermore, if $(s \log p)/n = o(1)$ and $n \geq 2$, no clustering rule can guarantee exact recovery uniformly over the class $\bar{\Theta} := \Theta(s, C_3 \sigma^2 \log n, K)$ with $C_3 > 0$ some small constant. In other words, there exists a constant $C_4 > 0$ such that

$$\inf_{\hat{\mathbf{H}}: \mathbb{R}^{p \times n} \rightarrow \mathcal{H}(K)} \sup_{\psi \in \bar{\Theta}} \mathbb{P}_\psi(\hat{\mathbf{H}}(\mathbf{X}) \neq \mathbf{H}) \geq C_5.$$

The proof of Theorem 1 is given in Section 11. The stated probability in Theorem 1 holds uniformly over all variable subsets in $\mathcal{S}$, justifying iterative procedures that solve a sequence of subset SDPs: once a “good” S is reached, clustering is accurate. The assumption $m \geq C_1 n / \log n$ prevents overly imbalanced clusters and is standard in clustering theory (Cai et al. 2019, Jin & Wang 2016, Löffler et al. 2021). Since m is a harmonic mean, this implies $K \lesssim \log n$, allowing slowly diverging known K , which is milder than K bounded above by a constant assumption in prior work (Verzelen & Arias-Castro 2017, Jin & Wang 2016). The second claim of Theorem 1 shows no clustering rule can guarantee exact recovery for all S with signal strength of order $\sigma^2 \log n$. When $(|S| \log p)/m \lesssim \log n$, the separation

bound matches this lower bound up to constants, establishing minimax optimality.

Theorem 1 also indicates that in the sparse regime where $|S^*| \ll p$, variable selection can be critical for clustering accuracy. For example, when $K = 2$, $|G_1^*| = |G_2^*|$ and $(\mathbf{c}_1^* - \mathbf{c}_2^*)_{S^*} = c_0 \mathbf{1}_{|S^*|}$, then for any set S satisfying $|S| \lesssim \sqrt{p}$ and $|S| \lesssim (n \log n) / \log p$, exact recovery holds if and only if

$$|S \cap S^*| c_0^2 \gtrsim \log n + \frac{|S| \log p}{n} + \sqrt{\frac{|S| \log p}{n}} \asymp \log n. \quad (11)$$

This condition is stringent if S is a very noisy estimate of S^* (i.e., missing too many signal variables), but a slight overestimate of S^* may not be very damaging. An important message of Theorem 1 is that recovering the exact sparse set is not essential for accurate clustering. Rather, identifying any sparse feature set with sufficiently large cumulative signal strength is key. Such a “good” set can be any S in $\mathcal{S}$, which motivates the iterative algorithm in the next subsection, where we aim to obtain some set S in $\mathcal{S}$ rather than a specific one.

3 Iterative Block Coordinate Optimization Framework

For computational tractability, we propose a block coordinate optimization framework which alternates between variable selection and clustering. For simplicity of exposition, we assume that Ω^* is known for now. The procedure initializes with, for example, spectral clustering on all variables to estimate cluster $\hat{G}^0$, then alternates between two blocks:

- **Selection block:** Given the current cluster $\hat{G}^{t-1}$, variables are scored by their mean differences between clusters and selected accordingly, yielding a new variable set $\hat{S}^t$.

- **Clustering block:** Given $\hat{S}^t$, we (i) compute $\tilde{\mathbf{X}} = \Omega^* \mathbf{X}$ and (ii) solve the subset SDP (Algorithm 1), obtaining $\hat{G}^t = \text{SDPclust}(\tilde{\mathbf{X}}_{\hat{S}^t, \cdot}, \Sigma_{\hat{S}^t, \hat{S}^t}^*)$, which is then used in the selection block of the next iteration.

These two blocks are shown as large dotted boxes in Figure 1; the algorithm should be interpreted as a wrapper and depends on the specific procedures used within them.

Within the selection block, we specify a feature scoring rule and a selection rule, considering (i) Greedy Selection and (ii) Thompson sampling. Figure 1 provides a schematic overview of the two algorithms, shown on the left and right of the selection block, respectively. Section 3.1 first presents Greedy Selection as a baseline. Section 3.2.1 highlights its main limitation: the premature exclusion of weak but genuine signals due to imperfectly estimated cluster in the current iteration, especially in ultra-high dimensions. Section 3.2.2 then addresses this issue with a Thompson sampling-based update that introduces controlled exploration, thereby improving feature selection power under unstable early screening.

We first present the algorithms under known covariance, as in Theorem 1; extensions to the unknown case appear in Section 3.4. Figure 1 summarizes both settings. For clarity, we focus on the case $K = 2$, though the methods naturally generalize to larger K .

3.1 Greedy Selection via Permutation Testing

This section presents the greedy selection procedure used in the selection block in Section 3. Given the previous cluster estimates $\hat{G}^{t-1}$, the procedure consists of two sub-steps described as follows. The full selection block appears in Algorithm 2.

- **Reward step.** Each variable j receives a random reward $Y_j^t \in \{0, 1\}$ based on its clustering power, evaluated via p-value from a permutation test adjusted by false discovery rate (FDR)-control.

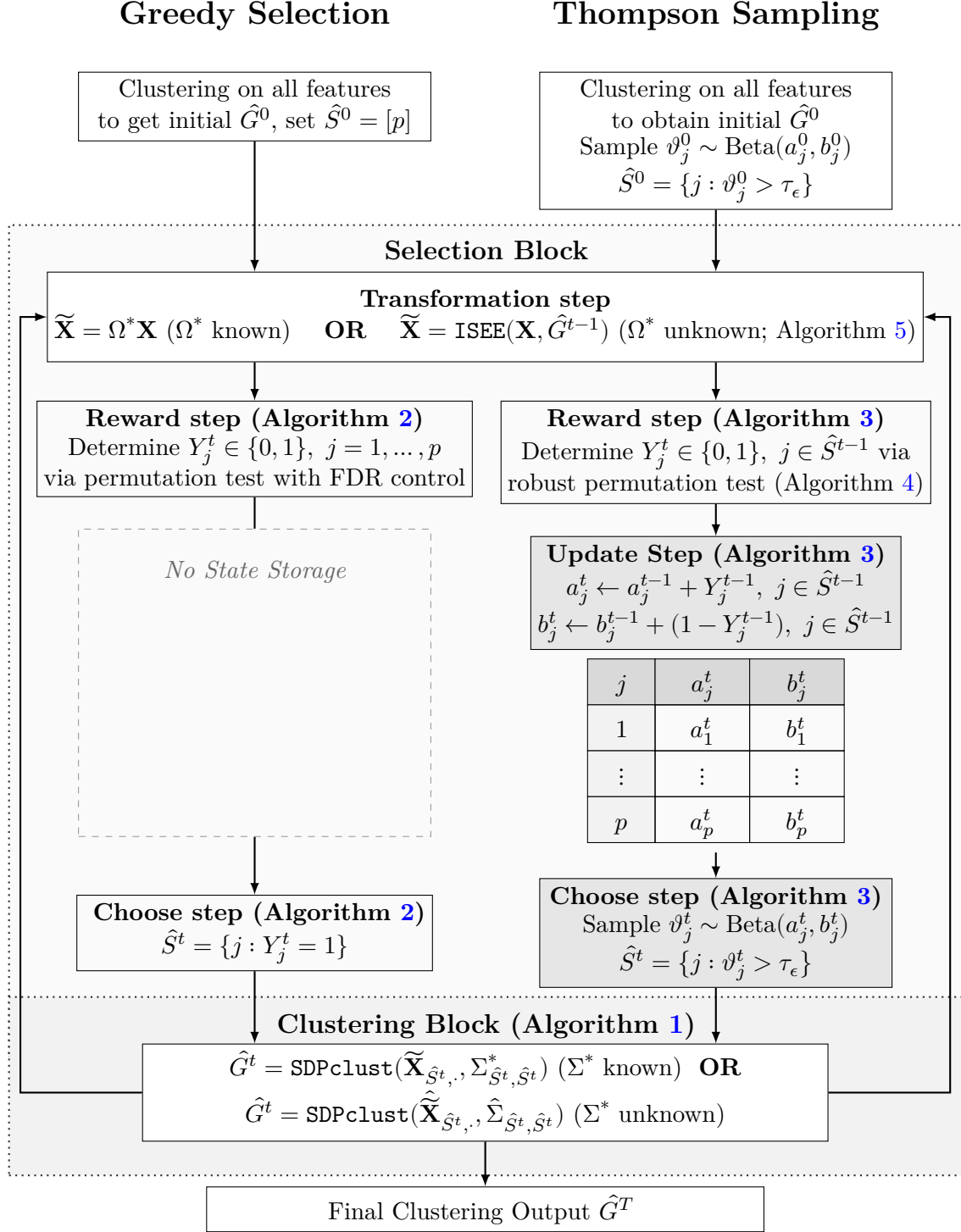


Figure 1: Schematic comparison of the block coordinate optimization algorithms. For Thompson sampling algorithm $\tau_\epsilon = \log(1/\epsilon) / \log((\epsilon + 1)/\epsilon)$.

- **Choose step.** All variables with $Y_j^t = 1$ are deterministically selected.

Variable selection relies on signal strength. The innovated transformation $\widetilde{\mathbf{X}} = \Omega^* \mathbf{X}$ maximizes the signal to noise ratio in Gaussian classification (Fan et al. 2013) and, by extension, clustering (Cai et al. 2019). Consequently, variable importance is captured by the Bayes rule coefficients $\beta^* := \Omega^*(\mathbf{c}_1^* - \mathbf{c}_2^*)$. Because true cluster labels are unknown, we iteratively estimate β^* using the previous step label estimates $\hat{G}^{t-1}$. To avoid reliance on distributional assumptions and nuisance parameters, we use a permutation test. For any cluster partition G , we estimate β_j^* by its empirical counterpart:

$$\Lambda(G, \widetilde{\mathbf{X}}_j) := \left| \frac{1}{|G_1|} \sum_{i_1 \in G_1} \widetilde{X}_{i_1, j} - \frac{1}{|G_2|} \sum_{i_2 \in G_2} \widetilde{X}_{i_2, j} \right|. \quad (12)$$

To evaluate the significance of the observed statistic, denoted as $\Lambda(\hat{G}^{t-1}, \widetilde{\mathbf{X}}_j)$, we generate a null distribution by permuting the cluster assignments while preserving the cluster sizes. Let π_b be a uniformly sampled random permutation of the n sample indices $\{1, \dots, n\}$. We construct the b -th permuted partition, $\hat{G}^{t-1, (b)} := (\hat{G}_1^{t-1, (b)}, \hat{G}_2^{t-1, (b)})$, as follows:

$$\hat{G}_1^{t-1, (b)} := \{\pi_b(1), \dots, \pi_b(|\hat{G}_1^{t-1}|)\}, \quad \hat{G}_2^{t-1, (b)} := \{\pi_b(|\hat{G}_1^{t-1}| + 1), \dots, \pi_b(n)\}.$$

Repeating this process generates B random permutations, yielding a set of permuted partitions $\{\hat{G}^{t-1, (b)}\}_{b=1}^B$. These are used to compute the null statistics $\Lambda(\hat{G}^{t-1, (b)}, \widetilde{\mathbf{X}}_j)$. By comparing the observed statistic $\Lambda(\hat{G}^{t-1}, \widetilde{\mathbf{X}}_j)$ against this empirical null distribution, we calculate an empirical p -value for each feature.

To address the multiplicity problem inherent in high-dimensional variable selection, we apply a FDR control to the resulting statistics. While any valid FDR control method can be employed, our specific choices are as follows. When the covariance is known, we use the Benjamini–Hochberg (BH) procedure (Benjamini & Hochberg 1995). When the

Algorithm 2 Greedy variable selection at the t -th iteration (Selection Block)

Require: Data $\tilde{\mathbf{X}}$, clusters $\hat{G}^{t-1}$, permutations B , FDR target q , FDR control method Q

- 1: $\lambda_j^{(0)} \leftarrow \Lambda(\hat{G}^{t-1}, \mathbf{X}_j)$ for $j = 1, \dots, p$ $\triangleright$ Observed statistic computed by (12)
- 2: **for** $b = 1, \dots, B$ **do** $\triangleright$ Generate null distribution
- 3: Sample random permutation π_b and construct $\hat{G}^{t-1,(b)} = (\hat{G}_1^{t-1,(b)}, \hat{G}_2^{t-1,(b)})$
- 4: $\lambda_j^{(b)} \leftarrow \Lambda(\hat{G}^{t-1,(b)}, \tilde{\mathbf{X}}_j)$ for $j = 1, \dots, p$ $\triangleright$ Permuted statistics computed by (12)
- 5: $(Y_1^t, \dots, Y_p^t) \leftarrow Q(\{\lambda_j^{(b)}\}_{j=1, b=0}^{p, B}; q)$ $\triangleright$ Reward step
- 6: **return** $\hat{S}^t \leftarrow \{j \in [p] \mid Y_j^t = 1\}$ $\triangleright$ Selection step

covariance is unknown, tests can be highly correlated across variable indices; thus, we adopt the Significance Analysis of Microarrays (SAM) framework (Tusher et al. 2001), which leverages permutations to directly estimate the empirical FDR.

3.2 Variable Selection via Thompson Sampling and Robust Test

Our greedy algorithm selects variables using mean differences $\Lambda(\hat{G}^{t-1}, \tilde{\mathbf{X}}_j)$ in (12), which cannot distinguish between signal and noise variables if the labels $\hat{G}^{t-1}$ are inaccurate. Therefore informative but weak features may be prematurely discarded, and there is no mechanism to promote re-selecting them. Moreover, the update $\hat{S}^t$ in Algorithm 2 depends only on current rewards. In particular, *nearly significant* features under noisy cluster assignments do not accumulate evidence across iterations. This lack of memory can yield a self-reinforcing loop of “poor labels $\Rightarrow$ low test power $\Rightarrow$ under selection of S^* (which is critically damaging as shown in Theorem 1) $\Rightarrow$ poor SDP estimated labels.” Increasing the FDR level mitigates this but inflates $|\hat{S}^t|$, potentially violating the regime (e.g. $|\hat{S}^t| \leq \sqrt{p}$) where uniform SDP recovery holds. These issues suggest that the selection should (i) *remember* past evidence on variable relevance and (ii) *occasionally re-select* uncertain variables so that improved labels can later validate them. Accordingly, Section 3.2.1 replaces memoryless greedy selection with Thompson Variable Selection, which maintains variable-wise posterior memory and enables randomized exploration.

3.2.1 Thompson Sampling Variable Selection (TVS) and Robust Test

Without FDR control, we can view the permutation-test outcome in Algorithm 2, conditional on $\mathbf{X}$ sampled from model (1), as follows:

1. They form Bernoulli random variables whose success probabilities depend on the variable index j and the previous cluster estimates $\hat{G}^{t-1}$.
2. They correspond to permutation-based two-sample tests on corrupted samples (Schrab & Kim 2025), using the empirical maximum mean discrepancy (MMD) with a linear kernel as the test statistic. Here, corruption happens due to imperfect clustering $\hat{G}^{t-1}$.

Thus, each permutation test can be viewed as a Bernoulli random variable, whose success probability can be analyzed through the lens of two-sample testing under corrupted samples. We exploit this Bernoulli structure to formulate the selection block in our block coordinate optimization as a multi-armed bandit problem, while leveraging the corrupted testing framework to enhance the permutation procedure so that the resulting Bernoulli rewards achieve adequate power and effectively guide variable selection.

For ease of presentation, we defer discussion of the corrupted testing component to Section 3.2.3 and focus first on the Bernoulli perspective. Casting the problem as a bandit model allows us to incorporate memory and controlled exploration, as discussed in Section 3.2.1. In particular, we model variable selection as a *multi-play* bandit problem (Komiyama et al. 2015, Liu & Ročková 2023): at each iteration, the agent selects a subset of arms (variables), observes Bernoulli rewards, and aggregates this feedback with past evidence to guide future actions. This formulation enables systematic use of historical information and occasional re-selection of uncertain variables. We formalize this approach and present the TVS algorithm in Section 3.2.2, and describe how the corrupted two-sample testing framework is used to

construct reliable Bernoulli rewards in Section 3.2.3.

3.2.2 Bernoulli rewards and TVS algorithm

We formally define the Bernoulli rewards using an arbitrary randomized test that depends on the data and the cluster partition.

Definition 2 (Bernoulli reward). *For $j = 1, \dots, p$, let Λ_j be an arbitrary randomized test such that, for any cluster partition $\hat{G}^{t-1}$ and random seed U^t , $\Lambda_j(\hat{G}^{t-1}, \tilde{\mathbf{X}}; U^t) \in \{0, 1\}$ is measurable with respect to the data $\tilde{\mathbf{X}}$ and U^t . Given $\hat{G}^{t-1}$, we view the test outcome as a Bernoulli reward $Y_j(\hat{G}^{t-1}) := \Lambda_j(\hat{G}^{t-1}, \tilde{\mathbf{X}}; U^t)$ with mean reward $\theta_j(\hat{G}^{t-1}) := \mathbb{P}(Y_j(\hat{G}^{t-1}) = 1 \mid \tilde{\mathbf{X}}, \hat{G}^{t-1})$, where the probability is taken over the randomness of U^t .*

We use independent randomizations U^t across iterations t and features j , where the dependence of U^t on j is suppressed whenever no confusion. The specific test Λ_j we use for the TVS algorithm is robust MMD test, explained in Section 3.2.3. To provide an intuitive explanation of the TVS algorithm, we first consider the oracle setting where the clustering block consistently outputs the true partition G^* . Then conditional on $(\tilde{\mathbf{X}}, G^*)$, the rewards $\{Y_j^t(G^*)\}_{t \geq 1}$ are i.i.d. Bernoulli $(\theta_j(G^*))$ for each fixed j . Denote the *oracle* mean reward as $\theta_j^{\text{orc}} := \theta_j(G^*) = \mathbb{P}(Y_j(G^*) = 1 \mid \tilde{\mathbf{X}}, G^*)$. Motivated by Liu & Ročková (2023), we define the *oracle* utility of a variable subset S for a given regularization parameter $\epsilon > 0$ as:

$$r_\epsilon^{\text{orc}}(S) := \sum_{j \in S} \left\{ \theta_j^{\text{orc}} \log \left(\frac{\epsilon + 1}{\epsilon} \right) + \log \epsilon \right\}. \quad (13)$$

This formulation establishes a straightforward criterion for determining when the inclusion of a candidate variable j is beneficial. Specifically, adding j improves the utility if and only if its oracle mean reward exceeds a ϵ -dependent cutoff:

$$r_\epsilon^{\text{orc}}(S \cup \{j\}) \geq r_\epsilon^{\text{orc}}(S) \quad \Longleftrightarrow \quad \theta_j^{\text{orc}} > \tau_\epsilon, \quad \tau_\epsilon := \frac{\log(1/\epsilon)}{\log(\frac{\epsilon+1}{\epsilon})}.$$

Accordingly, we define the oracle variable subset:

$$S_\epsilon^* := \arg \max_{S \subseteq [p]} r_\epsilon^{\text{orc}}(S) = \{j \in [p] : \theta_j^{\text{orc}} > \tau_\epsilon\}. \quad (14)$$

Here, the inclusion cutoff τ_ϵ controls the sparsity-exploration trade-off. A smaller cutoff promotes exploration, whereas a larger value enforces stricter sparsity. Unlike [Liu & Ročková \(2023\)](#), who fix $\tau_\epsilon = 0.5$, we consider a range of τ_ϵ values to systematically investigate these different regimes. At the oracle level, the TVS algorithm seeks to recover the target subset S_ϵ^* . As we demonstrate in Section 3.3, under suitable conditions on the θ_j^{orc} , this oracle subset aligns with the true signal set ($S_\epsilon^* = S^* \in \mathcal{S}$). This equivalence motivates an iterative algorithm which approximates the oracle TVS procedure.

In implementation, since G^* and θ_j^{orc} 's are unknown, we replace them with estimates from the previous iteration. At iteration t , rewards are evaluated using $\hat{G}^{t-1}$ and the randomized testing rule Λ_j with exogenous randomness U^t :

$$Y_j^t = Y_j(\hat{G}^{t-1}) = \Lambda_j(\hat{G}^{t-1}, \tilde{\mathbf{X}}; U^t), \quad \mathbb{E}[Y_j^t \mid \tilde{\mathbf{X}}, \hat{G}^{t-1}] = \theta_j(\hat{G}^{t-1}).$$

If $\hat{G}^{t-1} = G^*$ (up to label permutation) for some $t-1 \geq 0$, then $\theta_j(\hat{G}^{t-1}) = \theta_j(G^*) = \theta_j^{\text{orc}}$. This “oracle coupling” links the algorithm to the oracle case analysis. The iterative algorithm maintains feature-wise Beta posteriors $\text{Beta}(a_j^t, b_j^t)$, where a_j^t and b_j^t count significant ($Y_j^t = 1$) and non-significant ($Y_j^t = 0$) tests up to time t , respectively. At iteration t , given $(a^{t-1}, b^{t-1}, \hat{G}^{t-1}, \hat{S}^{t-1})$, the TVS update as follows:

- **Reward & Update step.** Compute the reward $Y_j^t = \Lambda_j(\hat{G}^{t-1}, \tilde{\mathbf{X}}; U^t)$ using the previous cluster $\hat{G}^{t-1}$ via permutation test (Algorithm 4 in Section 3.2.3) with type I

Algorithm 3 Thompson Variable Selection at the t -th Iteration (Selection Block)

Require: Transformed data $\tilde{\mathbf{X}}$, partition $\hat{G}^{t-1}$, variable set $\hat{S}^{t-1}$, priors $\{(a_j^{t-1}, b_j^{t-1})\}_{j=1}^p$, cutoff τ_ϵ , type I error rate $\tau_\epsilon - \alpha$, permutation test **rMMD** (Algorithm 4 with misclustering limit r)

- 1: $a_j^t \leftarrow a_j^{t-1}, \quad b_j^t \leftarrow b_j^{t-1}$ for all $j \notin \hat{S}^{t-1}$
- 2: **for** $j \in \hat{S}^{t-1}$ **do**
- 3: $Y_j^t \leftarrow \text{rMMD}(\hat{G}^{t-1}, \tilde{\mathbf{X}}_j, r, \tau_\epsilon - \alpha, B)$ $\triangleright$ **Reward Step** (Algorithm 4; only for $j \in \hat{S}^{t-1}$)
- 4: $a_j^t \leftarrow a_j^{t-1} + Y_j^t, \quad b_j^t \leftarrow b_j^{t-1} + (1 - Y_j^t)$ $\triangleright$ **Update Step** (only for $j \in \hat{S}^{t-1}$)
- 5: Sample $\vartheta_j^t \sim \text{Beta}(a_j^t, b_j^t)$ for all $j \in [p]$ $\triangleright$ **Choose Step**
- 6: **return** $\hat{S}^t \leftarrow \{j : \vartheta_j^t > \tau_\epsilon\}$, Updated Priors $\{(a_j^t, b_j^t)\}_{j=1}^p$

error rate $\tau_\epsilon - \alpha$ with a margin $\alpha \in (0, \tau_\epsilon)$. Then only update the variables in $\hat{S}^{t-1}$:

$$a_j^t = a_j^{t-1} + Y_j^t \mathbb{1}(j \in \hat{S}^{t-1}), \quad b_j^t = b_j^{t-1} + (1 - Y_j^t) \mathbb{1}(j \in \hat{S}^{t-1}), \quad j = 1, \dots, p.$$

- **Choose step.** Sample $\vartheta_j^t \sim \text{Beta}(a_j^t, b_j^t)$ and set $\hat{S}^t = \{j : \vartheta_j^t > \tau_\epsilon\}$.

The resulting set $\hat{S}^t$ is passed to the clustering block to obtain $\hat{G}^t$. The TVS rule appears in Algorithm 3, with the full procedure illustrated in the right panel of Figure 1. The use of α is to encourage exploration for reducing under-selection, as motivated by Theorem 1.

The cutoff τ_ϵ is crucial to the success of the TVS algorithm, as illustrated in Figure 2. The simulation shows that ϵ strongly affects clustering accuracy and that recovering more true positives can improve performance even at the cost of additional false positives. This supports Theorem 1 and contrasts with the regression setting in Liu & Ročková (2023), which fixes $\tau_\epsilon = 0.5$ and emphasizes false-positive control over true-positive recovery. Motivated by these findings, we select ϵ via grid search to encourage satisfaction of Assumption 2. Since the true clustering accuracy is unavailable, we choose ϵ by maximizing the silhouette index (Rousseeuw 1987), a standard measure of clustering quality computed on the union of variables selected for each candidate ϵ .

3.2.3 Bernoulli reward via robust permutation MMD test

At iteration t , treating the estimated partition $\hat{G}^{t-1}$ as fixed, the permutation test in Algorithm 2 can be framed as a two-sample hypothesis test under the data corruption model introduced by Schrab & Kim (2025). Let $n_1 = |\hat{G}_1^{t-1}|$ and $n_2 = |\hat{G}_2^{t-1}|$. Suppose nature draws n_1 and n_2 samples from the true data-generating distributions P_j and Q_j for feature j of $\tilde{\mathbf{X}}$. An adversary then corrupts the data by swapping r samples drawn from P_j with r samples drawn from Q_j , with the identities of these corrupted samples remaining unknown.

Using this corrupted data, we test the null hypothesis $H_0 : P_j = Q_j$, which implies that feature j possesses no discriminatory power for clustering. To quantify the divergence between the two samples, we use the Maximum Mean Discrepancy (MMD) with a linear kernel $k(x, x') = xx'$. The population squared MMD simplifies to the squared difference of the true transformed means (details in Section 12):

$$\text{MMD}_k^2(P_j, Q_j) := \mathbb{E}_{P_j}[k(\tilde{X}, \tilde{X}')] - 2\mathbb{E}_{P_j, Q_j}[k(\tilde{X}, \tilde{X}')] + \mathbb{E}_{Q_j}[k(\tilde{X}', \tilde{X}'')] = (\Omega^*(\mathbf{c}_1^* - \mathbf{c}_2^*))_j^2.$$

If the true, uncorrupted partition G^* is known, the V-statistic empirical MMD plug-in estimator $\widehat{\text{MMD}}_k^2(G^*, \mathbf{X}_j)$ exactly reduces to the square of our test statistic $\Lambda(G^*, \mathbf{X}_j)$ defined in (12) (see Section 12 for details). In practice, however, G^* is unknown, so we evaluate this statistic on the estimated partition as $\Lambda(\hat{G}^{t-1}, \mathbf{X}_j)$. With this statistic, we run the robust permutation kernel two-sample test of Schrab & Kim (2025) and use its binary output as the reward of Algorithm 3 (see Algorithm 4 for the full procedure). This procedure tightens the standard permutation test's rejection rule based on the corruption

limit r and the test statistic's sensitivity Γ_j defined as:

$$\Gamma_j(\hat{G}^{t-1}) := \sup_{\pi \in \Pi_n} \sup_{\tilde{\mathbf{X}}_j, \chi_j: d_{\text{ham}}(\tilde{\mathbf{X}}_j, \chi_j) \leq 1} |\Lambda(\hat{G}^{t-1}, \tilde{\mathbf{X}}_j^\pi) - \Lambda(\hat{G}^{t-1}, \chi_j^\pi)|.$$

Here, Π_n is the set of all n -permutations, the superscript π denotes permuted vectors, and $d_{\text{ham}}(\tilde{\mathbf{X}}_j, \chi_j) \leq 1$ implies that two data sets $\tilde{\mathbf{X}}_j$ and χ_j differ by at most one observation. To bound the linear kernel's sensitivity, we apply min-max scaling to restrict each feature to $[0, 1]$ during the reward step, which results in $\Gamma_j(\hat{G}^{t-1}) \leq 1/\min(|\hat{G}_1^{t-1}|, |\hat{G}_2^{t-1}|)$.

We motivate the use of Algorithm 4 via Theorem 34 which establishes uniform type I and II error guarantees when the cluster separation is sufficiently large. This result is closely related to our later Assumption 2. However, the theorem is stated with respect to the data-generating distribution, whereas our TVS algorithm conditions on data $\tilde{\mathbf{X}}$, so it should not be used as a rigorous justification of the oracle margin assumption (Cf. Assumption 2).

Remark 3. *We do not employ the rMMD test for the greedy algorithm in Section 3.1 because its conservativeness yields excessively small variable sets. As shown in Theorem 1, this under-selection degrades clustering performance in the subsequent iteration. In contrast, TVS decouples variable selection from the immediate reward. Although the rMMD test produces conservative rewards, the accumulated evidence in the Beta posteriors acts as a buffer. This prevents a strict reward from disproportionately restricting the next selection*

Algorithm 4 Robust MMD Permutation Test for Feature j : rMMD

Require: Current partition $\hat{G}^{t-1}$ with sizes n_1, n_2 , scaled feature j data $\tilde{\mathbf{X}}_j$, misclustering limit r , significance level $\tau_\epsilon - \alpha$, permutation count B .

- 1: $\pi_0, \pi_1, \dots, \pi_B \leftarrow$ Identity permutation and i.i.d. random permutations of $[n]$.
 - 2: **for** $b = 0, 1, \dots, B$ **do**
 - 3: Construct permuted partition $\hat{G}^{t-1, (b)}$ using π_b .
 - 4: Compute $\Lambda_j^{(b)} \leftarrow \Lambda(\hat{G}^{t-1, (b)}, \tilde{\mathbf{X}}_j) \triangleright \Lambda$ defined in (12); $\Lambda_j^{(0)}$ is the observed statistic
 - 5: Compute the $(1 - \tau_\epsilon + \alpha)$ -quantile $q_{\tau_\epsilon - \alpha}$ of the empirical distribution $\{\Lambda_j^{(b)}\}_{b=0}^B$.
 - 6: **return** 1 if $\Lambda_j^{(0)} > q_{\tau_\epsilon - \alpha} + 2r \frac{1}{\min(n_1, n_2)} \triangleright \frac{1}{\min(n_1, n_2)} = \text{global sensitivity } \Gamma_j(\hat{G}^{t-1})$
-

step, thereby enabling the moderate over-selection necessary for successful clustering.

3.3 Theory of Block Coordinate Optimization with TVS

We establish theoretical guarantees for block coordinate optimization using TVS under known covariance (Algorithm 3 for selection and Algorithm 2 for clustering). Our analysis conditions on the observed data $\mathbf{X}$ and unifies the modularized clustering and variable selection components. This modularity allows the framework to accommodate alternative clustering methods beyond SDP K-means. Recall from (10) and Theorem 1 that $\mathbf{Z}^* = \sum_{k=1}^K |G_k^*|^{-1} \mathbf{1}_{G_k^*} \mathbf{1}_{G_k^*}^\top$ denotes block-diagonal matrix that has one-to-one correspondence to the true cluster partition $G^* = (G_1^*, G_2^*)$, and $\mathcal{S}$ denotes the family of “good” subsets for which exact cluster recovery holds. We formalize the “oracle coupling” that links the SDP clustering performance to the multi-armed bandit rewards.

Assumption 1. Assume the event $\mathcal{E}_{\text{SDP}} := \{ \widehat{\mathbf{Z}}(S) = \mathbf{Z}^*, \forall S \in \mathcal{S} \}$ holds.

By Theorem 1, we have $\mathbb{P}(\mathcal{E}_{\text{SDP}}) \geq 1 - C_3 K/n$, where the randomness arises entirely from the data generating model in (1). The following lemma (proof in Section 14.1) shows that when this event holds, the agent receives true oracle rewards.

Lemma 4. Conditioned on $\mathcal{E}_{\text{SDP}}$, if the selection block chooses an active set $\widehat{S}^t \in \mathcal{S}$ at iteration t , the clustering block yields the true partition: $\widehat{\mathbf{Z}}(\widehat{S}^t) = \mathbf{Z}^*$ and $\widehat{G}^t = G^*$ (up to label permutation). Consequently, for every $j \in \widehat{S}^t$, the subsequent rewards match their oracle counterparts: $\theta_j(\widehat{G}^t) = \theta_j^{\text{orc}}$ and $Y_j(\widehat{G}^t) = Y_j^{\text{orc},t}$, where $Y_j^{\text{orc},t} \stackrel{i.i.d.}{\sim} \text{Bernoulli}(\theta_j^{\text{orc}})$.

This lemma implies that once the algorithm selects a “good” feature set, the bandit feedback relies purely on context-free oracle means. To ensure the algorithm can reliably distinguish signal from noise under these oracle conditions, we impose a margin assumption.

Assumption 2 (Oracle margin). Assume there exist $\epsilon > 0$ and $\alpha \in (0, 1 - \tau_\epsilon)$ with τ_ϵ

defined in (13) such that

$$\min_{j \in S^*} \theta_j^{\text{orc}} \geq \tau_\epsilon + \alpha, \quad \max_{j \notin S^*} \theta_j^{\text{orc}} \leq \tau_\epsilon - \alpha. \quad (15)$$

Section 14.2 provides a standard multivariate Gaussian example of a situation in which Assumption 14.2 holds. Compared to Liu & Ročková (2023), this identifiability condition is weaker in our clustering setting: it needs not hold uniformly over all subsets $S \subset [p]$ or arbitrary partitions G , nor does it require the algorithm to recover S^* at every iteration t . During early iterations, rewards are computed using the estimated partition $\hat{G}^{t-1}$, resulting in means of $\theta_j(\hat{G}^{t-1})$, which Assumption 2 does not constrain. Instead, the assumption requires that once the TVS algorithm identifies a set in $\mathcal{S}$ and achieves $\hat{G}^{t-1} = G^*$, the rewards coincide with the oracle means, enabling the Thompson sampling analysis to apply. The next lemma (proof in Section 17) shows that under these conditions, the TVS procedure eventually always outputs the true signal variable set.

Theorem 5. *Assume that in the TVS algorithm, selecting a variable j at time t yields an oracle Bernoulli reward $Y_j^{\text{orc},t} \sim \text{Bernoulli}(\theta_j^{\text{orc}})$ with algorithm parameters $\tau_\epsilon, \alpha, a_j^0, b_j^0$ that satisfy Assumption 2. Let $\{\hat{S}^t\}_{t \geq 1}$ denote the sequence of active feature sets produced by the selection block (Algorithm 3) over $t = 1, \dots, T$. Then conditional on data matrix $\mathbf{X}$ (thus p is fixed), we have*

$$\mathbb{P}\left(\liminf_{t \rightarrow \infty} \hat{S}^t = S^* \mid \mathbf{X}\right) = 1.$$

To extend the oracle analysis to our iterative algorithm, we require a “basin of attraction” condition ensuring the procedure eventually and persistently visits a set in $\mathcal{S}$.

Assumption 3. *Let $\{\hat{S}^t\}_{t \geq 1}$ be sequence of active feature sets produced by the selection block (Algorithm 3) over $t = 1, \dots, T$. Assume the following event holds with probability at*

least $1 - \delta_{\text{tail}}$ for some $\delta_{\text{tail}} \in [0, 1)$:

$$\mathcal{E}_{\text{tail}} := \left\{ \exists T_0 < \infty \text{ such that } \hat{S}^t \in \mathcal{S} \ \forall t \geq T_0 \right\}. \quad (16)$$

This assumption is partially supported by Theorem 35, which shows that TVS selects noise variables with low probability. We now synthesize these components to establish the variable selection consistency of our TVbased block coordinate optimization algorithm, conditioned on $(\mathbf{X}, \mathcal{E}_{\text{SDP}} \cap \mathcal{E}_{\text{tail}})$. It is important to first clarify the logical progression of the algorithm’s success: exact cluster recovery strictly precedes exact variable selection. Once the active set $\hat{S}^t$ enters the “good” set $\mathcal{S}$, the algorithm recovers the true cluster labels, which in turn provides the stable oracle rewards necessary to achieve ultimate variable selection consistency. By Assumption 3, there exists a finite timestamp T_0 such that the algorithm enters $\mathcal{S}$ and continues to output $\hat{S}^t \in \mathcal{S}$ for all $t \geq T_0$. Assuming that the test randomizations $\{U^t\}_{t \geq 1}$ used in the permutation tests are independent across t and independent of the data $\mathbf{X}$, Assumption 1 and Lemma 4 guarantee that the resulting rewards align perfectly with the oracle. Specifically, for all $t \geq T_0$ and all $j \in [p]$, we have:

$$Y_j^t = Y_j^{\text{orc}} \quad \text{and} \quad \mathbb{E}[Y_j^t \mid \mathbf{X}, \mathcal{E}_{\text{SDP}} \cap \mathcal{E}_{\text{tail}}] = \theta_j^{\text{orc}}.$$

Consequently, we can apply Theorem 5 to the time-shifted process $\{(\hat{S}^t, a_j^t, b_j^t)\}_{t \geq T_0}$, treating the random posteriors at time T_0 as the new initial priors (i.e., $\text{Beta}(a_j^{T_0}, b_j^{T_0})$ for each arm j). Because the asymptotic analysis of Theorem 5 does not depend on the initial priors, this time-shifted process achieves the variable selection consistency:

$$\mathbb{P}\left(\liminf_{t \rightarrow \infty} \hat{S}^t = S^* \mid \mathcal{E}_{\text{SDP}} \cap \mathcal{E}_{\text{tail}}\right) = 1.$$

3.4 Unknown Covariance Case

We address the unknown covariance case by adopting a strategy that avoids estimating the full precision matrix Ω^* . At iteration t , we directly estimate the innovated data matrix $\widetilde{\mathbf{X}}$ using the current cluster labels $\hat{G}^{t-1}$. This estimate is then used in the reward step (Algorithm 2 or 3) and the clustering block (Algorithm 1). The direct estimation of $\widetilde{\mathbf{X}}$ is done via the Innovated Scalable Efficient Estimation (ISEE) procedure (Fan & Lv 2016), a nodewise regression approach originally developed for high-dimensional sparse precision matrix estimation. We modify it to selectively estimate only the required quantities for SDPcluster procedure (Algorithm 1), avoiding full precision matrix recovery.

The ISEE procedure exploits two observations. First, the innovated vector $\widetilde{\mathbf{X}}_i = \Omega^* \mathbf{X}_i \in \mathbb{R}^p$ is Gaussian with $\text{Cov}(\widetilde{\mathbf{X}}_i) = \Omega^* \Sigma^* \Omega^* = \Omega^*$. Second, for any subset $A \subset [p]$ of size 2, and $i \in G_1^*$, we obtain $\widetilde{\mathbf{X}}_{A,i} = (\tilde{\mathbf{c}}_1^*)_A + \tilde{\mathcal{E}}_{A,i}$ by using the conditional distribution property of multivariate Gaussian and Shur complement:

$$\underbrace{\mathbf{X}_{A,i}}_{\text{response} \in \mathbb{R}^{|A|}} = \underbrace{(\mathbf{c}_1^*)_A + (\Omega_{A,A}^*)^{-1} \Omega_{A,A^c}^* (\mathbf{c}_1^*)_{A^c}}_{:= (\alpha_1)_A \in \mathbb{R}^{|A|} \text{ (intercept)}} - \underbrace{(\Omega_{A,A}^*)^{-1} \Omega_{A,A^c}^*}_{\text{slope} \in \mathbb{R}^{|A|} \times (p-|A|)} \underbrace{\mathbf{X}_{A^c,i}}_{\text{predictor} \in \mathbb{R}^{p-|A|}} + \underbrace{\mathbf{E}_{A,i}}_{\text{residual} \in \mathbb{R}^{|A|}}, \quad (17)$$

where $\mathbf{E}_{A,i} \sim \mathcal{N}(0, (\Omega_{A,A}^*)^{-1})$. This implies (Fan & Lv 2016):

$$(\tilde{\mathbf{c}}_1^*)_A = \Omega_{A,A}^* (\alpha_1)_A \quad \text{and} \quad \tilde{\mathcal{E}}_{A,i} = \Omega_{A,A}^* \mathbf{E}_{A,i}. \quad (18)$$

Section 19 provides detailed derivations for (17) and (18). Since $p - |A|$ can be comparable or larger than n , we fit (17) via high-dimensional regression (e.g., lasso) to estimate $\Omega_{A,A}^*$, $\mathbf{E}_{A,i}$ and $(\alpha_1)_A$, which further yields plug-in estimate of $\widetilde{\mathbf{X}}_{A,i}$. Repeating over A in a partition $\mathcal{A}$ (blocks of size 2 or 3) and stacking gives an estimate of the full matrix $\widetilde{\mathbf{X}}_{G_1^*} \in \mathbb{R}^{|G_1^*| \times p}$. A similar expression holds for $i \in G_2^*$, with $\mathbf{c}_1^*$ and α_1 replaced by $\mathbf{c}_2^*$ and α_2 , respectively. Following the same steps as described above, we can obtain the estimate of

Algorithm 5 ISEE Transformation Subroutine (ISEE)

Require: Data $\mathbf{X} \in \mathbb{R}^{p \times n}$, cluster assignments $\hat{G} = (\hat{G}_1^{t-1}, \hat{G}_2^{t-1})$

```

1: Partition  $[p]$  into disjoint subsets  $A_1, \dots, A_L$  of size 2 or 3
2: for  $\ell = 1, \dots, L$  do
3:   for  $k \in \{1, 2\}$  do ▷ Estimate parameters for each cluster
4:     Regress  $\mathbf{X}_{A_\ell}$  on  $\mathbf{X}_{A_\ell^c}$  using samples  $i \in \hat{G}_k^{t-1}$  (e.g., via Lasso)
5:     Obtain: Intercept  $\hat{\alpha}_{(k)}$  and residuals  $\{\hat{\mathbf{E}}_{A_\ell, i} : i \in \hat{G}_k^{t-1}\}$ 
6:      $\hat{\mathbf{E}}_{A_\ell} \leftarrow [\hat{\mathbf{E}}_{A_\ell, i}]_{i=1}^n$  ▷ Pool residuals across clusters
7:      $\hat{\Omega}_{A_\ell A_\ell} \leftarrow (\frac{1}{n} \hat{\mathbf{E}}_{A_\ell} \hat{\mathbf{E}}_{A_\ell}^\top)^{-1}$  ▷ Estimate block precision
8:     for  $i = 1, \dots, n$  do ▷ Reconstruct innovated data
9:       Let  $k$  be the cluster index of sample  $i$ 
10:       $\hat{\tilde{\mathbf{X}}}_{A_\ell, i} \leftarrow \hat{\Omega}_{A_\ell A_\ell} (\hat{\alpha}_{(k)} + \hat{\mathbf{E}}_{A_\ell, i})$ 
11: return  $\hat{\tilde{\mathbf{X}}} \leftarrow [\hat{\tilde{\mathbf{X}}}_{A_1}, \dots, \hat{\tilde{\mathbf{X}}}_{A_L}]^\top$ 

```

the submatrix $\tilde{\mathbf{X}}_{G_2^*} \in \mathbb{R}^{|G_2^*| \times p}$. In practice, true labels are unknown; at iteration t , we use $\hat{G}^{t-1}$ as a proxy, perform the above regressions within each estimated cluster, and output $\hat{\tilde{\mathbf{X}}}^t = \text{ISEE}(\mathbf{X}, \hat{G}^{t-1})$. The full procedure is detailed in Algorithm 5. Section 22.2 numerically demonstrates the benefits of ISEE bypassing the full covariance structure estimation.

Remark 6. *Integrating covariance estimation creates a fundamental tension: it requires large sample sizes, yet clustering difficulty scales with $\log n$ (Theorem 1). Consequently, this integration degrades TVS performance (Figure 4). From a reinforcement learning perspective, the partition $\hat{G}$ acts as a “state” determining reward probabilities. Covariance estimation makes reward probabilities overly sensitive to $\hat{G}$. To address this issue, we propose two practical strategies. When the precision matrix can be reliably estimated, typically in settings with moderate to large n and small to moderate p , where well-established theory ensures consistency, we use a greedy algorithm with ISEE. In contrast, in the small- n , large- p regime, reliable covariance estimation is infeasible. In this case, we apply our known-covariance method with the identity matrix as a plug-in estimate of the covariance.*

4 Numerical Studies

We evaluate our algorithms on both synthetic and real datasets. The SDP is solved using a custom first-order ADMM solver. For all experiments, we report clustering accuracy averaged over 200 independent replications. We benchmark our methods against several widely used iterative approaches, including sparse K -means (SKM; [Witten & Tibshirani 2010](#)), CHIME ([Cai et al. 2019](#)), sparse alternate sum clustering (SAS; [Arias-Castro & Pu 2017](#)) along with a two-step method (IFPCA; [Jin & Wang 2016](#)), and a complete convex relaxation method sparse convex clustering (SCVX; [Wang et al. 2018a](#)). Details on hyperparameter settings, the solver, and competing methods are provided in [Section 21](#).

4.1 Simulation studies

Synthetic datasets are generated from model [\(1\)](#) with two symmetric clusters:

$$\mathbf{X}_i = \mathbf{W}_i + \mathbf{c}^*, \quad i = 1, \dots, n/2, \quad \mathbf{X}_i = \mathbf{W}_i - \mathbf{c}^*, \quad i = n/2 + 1, \dots, n. \quad (19)$$

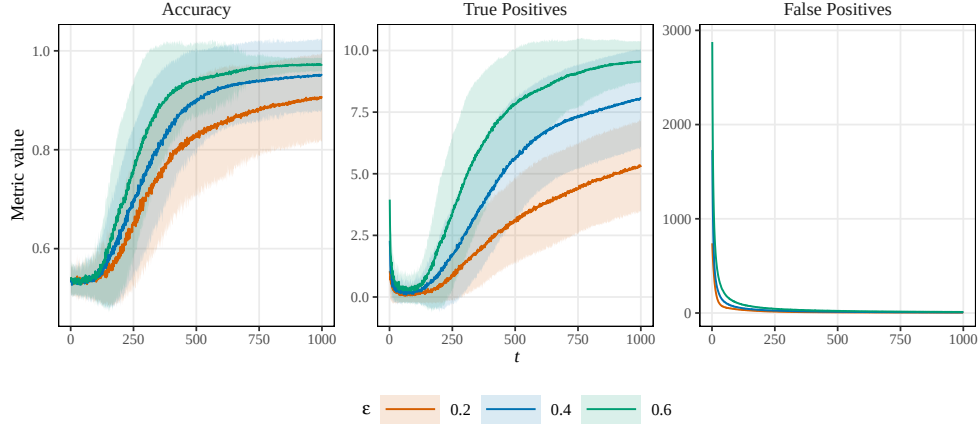
For the known-covariance setting, we generate $\mathbf{W}_i$ with entrywise i.i.d. Gaussian and Laplace distributions, each standardized to have unit variance. In the unknown-covariance setting, $\mathbf{W}_i$ is obtained by premultiplying these unit-variance Gaussian and Laplace vectors by $(\Sigma^*)^{1/2}$. We fix the support of $\Omega^* \mathbf{c}^*$ as $S^* = [10]$ and vary the ambient dimension p . We consider two forms of Σ^* , specified via its inverse: a chain graph and AR(1) (details provided in [Section 21](#)). All entries of $(\Omega^* \mathbf{c}^*)_{S^*}$ have equal magnitude, chosen so that the signal strength Δ_{S^*} defined in [\(2\)](#) remains constant as p increases. Details on parameter settings are provided in [Table 3](#).

We study the sensitivity of the TVS algorithm to the regularization parameter ϵ under model [\(19\)](#), using the setting in line 2 of [Table 3](#), with Δ_{S^*} (defined in [\(2\)](#)) fixed at 4 and

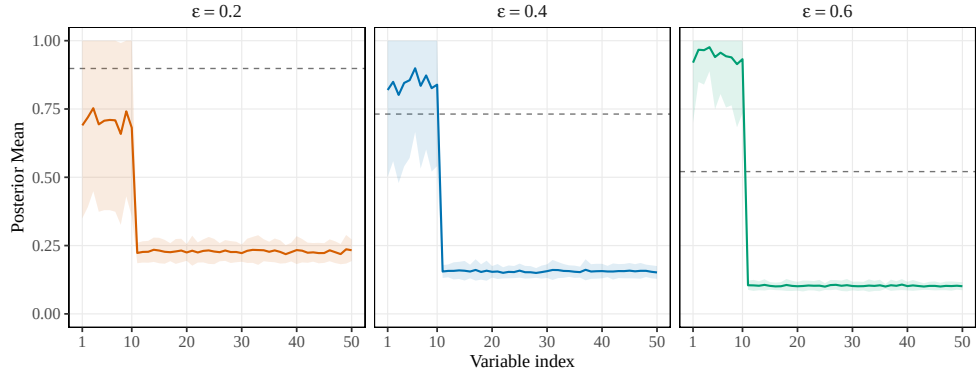
ambient dimension $p = 8000$. Although Liu & Ročková (2023) recommends $\epsilon \approx 0.618$ (corresponding to $\tau_\epsilon = 0.5$), we consider $\epsilon \in 0.2, 0.4, 0.6$ to examine the exploration–exploitation trade-off. Figure 2a shows the average clustering accuracy of $\hat{G}^t$ and the numbers of true and false positives for $\hat{S}^t$ over $t = 1, \dots, 1000$, averaged across 100 replications. The more permissive choice $\epsilon = 0.6$ achieves the best overall performance in this high-dimensional setting: although it yields more false positives, it also identifies substantially more true positives, resulting in higher clustering accuracy. Figure 2b plots the posterior means $a_j^{1000}/(a_j^{1000} + b_j^{1000})$ for the first 100 variables together with the threshold τ_ϵ . For $\epsilon = 0.4$ and 0.6 , the threshold clearly separates signal and noise features, suggesting that an appropriate choice of ϵ yields strong performance and supporting Assumption 2.

Next, we run experiments under the synthetic data with known covariance setting (line 1-3 of Table 3). Figure 3-(a) compares non-sparse methods (spectral clustering, hierarchical clustering, and full-feature SDP K -means) with our greedy algorithm initialized by each method (i.e., using its cluster assignment as $\hat{G}^0$). The result shows that non-sparse methods fail to adapt to sparsity, whereas Algorithm 2 improve upon them and remains robust in high dimensions. Next, Figures 3-(b) and 3-(c) demonstrate the competitiveness of our approach compared to existing sparse clustering methods under both Gaussian and Laplace noise settings. Notably, IFPCA and SAS exhibits switching performance between Gaussian noise and Laplace noise. Importantly, the TVS algorithm consistently outperforms competing methods across both Gaussian and Laplace noise settings, as well as across all dimensions. This strong performance underscores the benefits of incorporating memory and active exploration, and highlights the robustness of the TVS approach to local optima and deviations from underlying noise assumptions.

Next, we evaluate the unknown covariance setting (lines 3–6 of Table 3) using variants of our iterative algorithms with and without ISEE. In most cases (6 out of 8 settings), TVS



(a) Clustering accuracy and variable selection over $t = 1, \dots, 1000$.



(b) Posterior inclusion probabilities at $T = 1000$ for the first 50 variables.

Figure 2: Performance trajectories and posterior probabilities across varying exploration rates $\epsilon \in \{0.2, 0.4, 0.6\}$ with fixed ambient dimension $p = 8000$. **(a)** Average trajectories of clustering accuracy ($\hat{G}^t$) and true/false positives ($\hat{S}^t$). **(b)** Posterior inclusion probabilities $a_j^T / (a_j^T + b_j^T)$. Horizontal dashed lines indicate the inclusion thresholds $\tau_\epsilon = \log(1/\epsilon) / \log((\epsilon + 1)/\epsilon)$ specific to each rate. In both panels, solid lines denote the mean and shaded bands denote ± 1 standard deviation across 100 independent replications.

without ISEE outperforms other methods, indicating that bypassing covariance estimation is beneficial when the sample size is small, the dimension is large, or the separation is weak. However, in a small- p , large- n , low-separation regime presented in Figures 4-(c) and 4-(d), TVS without ISEE performs poorly, while the greedy algorithm with ISEE performs well. This suggests that when the sample size is sufficiently large relative to the dimension, making covariance estimation reliable, the greedy approach with ISEE is effective. In contrast, TVS with ISEE underperforms in most settings, as discussed in Remark 6.

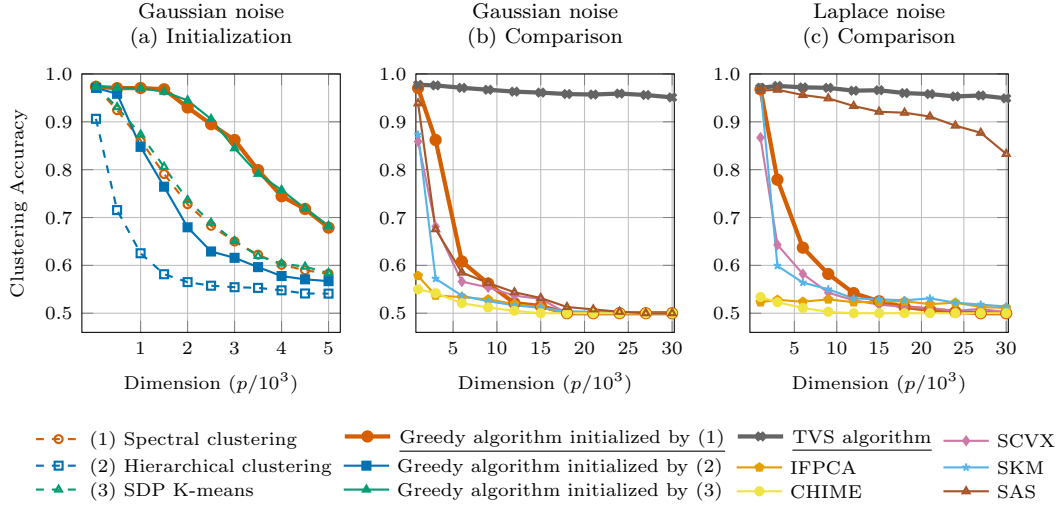


Figure 3: Clustering accuracy under known covariance settings. In (a), we compare the non-sparse baselines (dotted curves) with the greedy algorithms initialized by each of them (solid curves). In (b) and (c), we compare the greedy and TVS algorithms (thick lines in the plots and underlined in the legends) against sparsity-aware baselines. Full parameter settings are provided in Table 3.

We include additional simulations in the supplementary material, to numerically demonstrate that: the subset SDP (6) captures the clustering signal (Section 22.1), ISEE bypassing the full covariance structure estimation improves the clustering performance (Section 22.2), robustness of our algorithms to mild violations of covariance homogeneity (Section 22.3) and exact sparsity (Section 22.4), alternative SDP solver for scalability to large sample sizes (Section 22.5), and comparison to subspace clustering approaches (Section 22.6).

4.2 Real data analysis

We evaluate the proposed methods on a subset of the 20 Newsgroups corpus from the UCI Machine Learning Repository, consisting of $n = 400$ documents sampled from the semantically distinct discussion groups `sci.space` and `rec.autos`. This pair of groups provides a challenging setting for sparse variable selection: although the two domains share substantial mechanical terminology, their underlying topics remain fundamentally different. Standard preprocessing was applied, including lowercasing, punctuation and

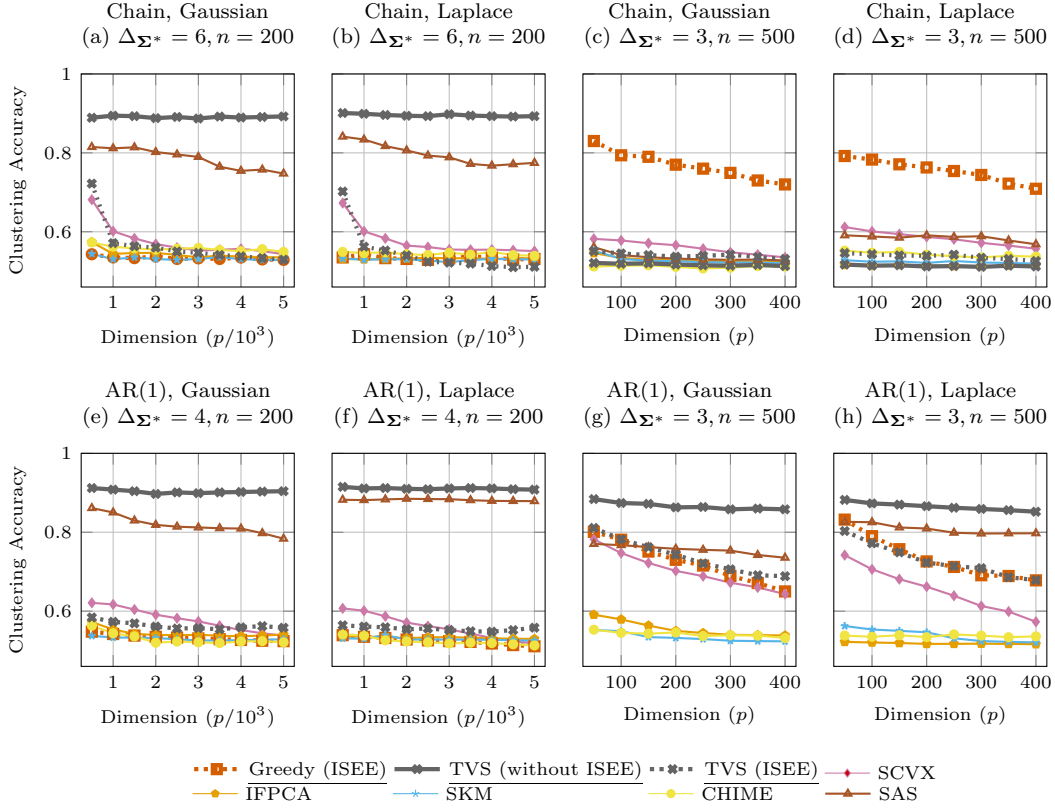


Figure 4: Clustering accuracy under unknown covariance settings. Our proposed algorithms are highlighted with thick lines in the plots and underlined in the legends. In (a), we compare the greedy algorithms with non-sparse baselines; the dotted curves represent the non-sparse methods, corresponding to the initializations prior to the first variable selection block. In (b) and (c), we compare the greedy and TVS algorithms against sparsity-aware baselines. Full parameter settings are provided in Table 3.

number removal, stop-word filtering, and whitespace normalization (see Section 21.3 for details). Documents were represented using term frequency-inverse document frequency (TF-IDF) features, and terms appearing in fewer than 1% of documents were removed. The resulting TF-IDF matrix has dimension 1331×400 with approximately 97.3% zero entries and a median of 24 nonzero terms per document. Since TF-IDF vectors are nonnegative and document lengths vary considerably, each document vector was ℓ_2 -normalized prior to clustering. The true newsgroup labels were used as the ground-truth partition. For our algorithms, we do not estimate the covariance structure because $n \ll p$ ($n = 400$, $p = 1331$). As observed Figure 4, bypassing covariance estimation is a well-justified approach in this high-dimensional setting. Table 1 reports clustering accuracy across competing

Table 1: Clustering accuracy on the 20 Newsgroups dataset.

Method	SKM	SAS	IF-PCA	CHIME	SCVX	Greedy	TVS
Accuracy	0.505	0.508	0.523	0.504	0.520	0.753	0.735

methods. Both proposed approaches substantially outperform existing sparse clustering baselines. A key advantage of the TVS algorithm is that it provides a posterior-mean measure of variable importance. Under TVS, the ten variables with the highest posterior means are *auto*, *automatic*, *best*, *bought*, *brake*, *car*, *cars*, *chevy*, *clutch*, and *dealer*—most of which are specific to automotive terminology. While the highest-ranked words based on TF-IDF values often contain semantically uninformative terms such as “like” and “new,” the words selected by TVS successfully exclude them. Consequently, the TVS selections are substantially more discriminative and interpretable, maintaining a clear focus on the automotive category.

5 Discussion

We addressed the coupled challenge of high-dimensional variable selection and clustering. By treating variable inclusion as a nuisance parameter, we established uniform minimax separation bounds for exact recovery in SDP K -means. Crucially, our theory demonstrates that while variable selection is necessary in sparse regimes, SDP K -means is highly robust to moderate over-selection provided sufficient cumulative signal is retained.

Exploiting this robustness, we developed a block-coordinate framework alternating between SDP clustering and variable selection. To prevent stagnation in local optima (a common failure of greedy methods) we formulated variable selection as a multi-armed bandit problem. Solved via Thompson Sampling, this introduces *adaptive memory* and *active exploration* to prevent the premature discarding of weak signals. We further stabilized the bandit rewards using a robust permutation MMD test to handle data corruption from imperfect

intermediate clustering. Under a basin-of-attraction assumption, we proved the variable selection consistency of this framework. For unknown covariance settings, we adapted the ISEE procedure (Fan & Lv 2016) to bypass full precision matrix estimation.

Several future directions remain. First, establishing global convergence from arbitrary initializations, beyond our basin-of-attraction assumption, remains an open challenge. Second, theoretically justifying the robust MMD test under the iterative framework, where conditioning on observed data complicates standard two-sample testing bounds, is an important next step. Finally, extending minimax lower bounds to general unknown covariance matrices would complete the information-theoretic characterization of sparse clustering.

6 Disclosure statement

The authors have the following conflicts of interest to declare (or replace with a statement that no conflicts of interest exist).

7 Data Availability Statement

The 20 newsgroup dataset is available at <https://archive.ics.uci.edu/dataset/113/twenty+newsgroups>.

SUPPLEMENTARY MATERIAL

The Supplemental Materials contain proofs of Theorems, omitted algebraic derivations, omitted details for simulations and additional simulation results.

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8 Overview of Supplementary Materials

This supplementary material contains the technical proofs deferred from the main text, alongside additional supporting results. Throughout this document, references to sections, equations, or figures within the supplement are prefixed with “S” (e.g., Section 9), whereas references to the main manuscript remain un-prefixed (e.g., Theorem 1). The content is organized as follows:

- Section 9 provides a complete list of alphabetical notations used throughout the main text and supplementary materials for ease of reference.
- Section 10 contains the technical lemmas and definitions used in the proof of Theorem 1.
- Section 11 provides the proof of the separation bound for exact recovery by semidefinite relaxation of K -means, as presented in Theorem 1. Most of the section is devoted to the proof of the minimax upper bound. Section 11.6 presents the proof of the minimax lower bound.
- Section 12 provides the detailed algebraic derivations, omitted from Section 3.2.3, that connect the population and empirical maximum mean discrepancy to our population clustering parameters and proposed test statistic. It also presents a theoretical guarantee of the robust MMD statistic.
- Section 13 shows that the TVS algorithm selects noise variables only with low probability, under a mild assumption.
- Section 14 presents the proof of preliminary results that support the theoretical guarantees of the proposed TVS algorithm.
- Section 15 presents the regret bound of the TVS algorithm introduced in Section

3.2.1.

- Section 16 provides the proof for the regret bound presented in Section 15.
- Section 17 presents the proof of variable selection consistency (Theorem 5).
- Section 18 analyzes the expected number of times a signal arm is not pulled; this result is used in both Sections 15 and 17.
- Section 19 provides omitted details of the ISEE subroutine (Algorithm 5) introduced in Section 3.4.
- Section 20 provides the omitted details of our proposed greedy and TVS algorithms, including the solver for the SDP, the high-dimensional regression method used for ISEE, the stopping criterion, and the choice of regularization parameters.
- Section 21 provides additional information on the numerical studies presented in Section 4.
- Section 22 presents simulation studies that were omitted from the main text.

9 List of Notations in the Main Text

This section provides a comprehensive reference of the mathematical notations utilized throughout the main manuscript. For ease of navigation, the notations are divided into standard alphabetic symbols and Greek letters.

9.1 Alphabets

- a_j^t : First parameter (success count proxy) of the Beta posterior for the j -th variable at iteration t .

- b_j^t : Second parameter (failure count proxy) of the Beta posterior for the j -th variable at iteration t .
- B : Total number of random permutations generated for the hypothesis test.
- $\mathbf{c}_k^*$: True center of the k -th cluster (p -dimensional non-random vector).
- $\mathbf{C}^*$: True $p \times n$ cluster center matrix, stacking the true cluster center corresponding to each observation as column vectors.
- $\tilde{\mathbf{C}}^*$: Transformed true cluster center matrix, defined as $\Omega^* \mathbf{C}^*$.
- $\mathcal{C}(s, \Delta, K)$: Class of sparse cluster centers parameterized by sparsity s , minimum separation Δ , and K clusters, used in the minimax framework.
- c_0 : Entrywise signal magnitude used to illustrate the separation condition in the two-cluster example.
- $\mathbf{D}$: A $K \times K$ diagonal matrix containing the cluster sizes $|G_k|$ for a given partition G .
- $\mathcal{E}_{\text{SDP}}$: The high-probability event where subset SDP K -means achieves exact recovery for all $S \in \mathcal{S}$.
- $\mathcal{E}_{\text{tail}}$: The high-probability event where the active feature set eventually enters and remains within the valid collection $\mathcal{S}$.
- G : A generic partition $(G_1, \dots, G_K)$ of the observation index set $\{1, \dots, n\}$.
- G^* : The true partition $(G_1^*, \dots, G_K^*)$ of the observation index set.
- $\hat{G}^t$: Estimated cluster partition at iteration t ($\hat{G}^0$ denotes the initialization).
- $\mathbf{H}$: An $n \times K$ binary matrix representing a cluster partition, equivalent to G up to label permutation.

- $\hat{\mathbf{H}}$: A generic clustering function mapping the observation matrix to a partition in $\mathcal{H}(K)$.
- $\mathcal{H}(K)$: The space of all valid $n \times K$ cluster assignment matrices.
- K : Number of clusters.
- m : Harmonic mean-based cluster imbalance indicator, defined as $2 \min_{k_1 \neq k_2} (|G_{k_1}^*|^{-1} + |G_{k_2}^*|^{-1})^{-1}$.
- n : Total number of observations.
- P_j, Q_j : True data-generating distributions for clusters 1 and 2 for feature j within the robust two-sample testing framework.
- p : Ambient data dimension (total number of features).
- $q_{\tau_\epsilon - \alpha}$: The $(1 - \tau_\epsilon + \alpha)$ -quantile of the empirical null distribution used in the robust MMD test.
- r : Misclustering limit parameter used to dictate the tolerance threshold in the robust MMD test.
- $r_\epsilon^{\text{orc}}(S)$: The oracle utility function for a candidate variable subset S .
- s : Maximum allowable number of true signal variables ($|S^*| \leq s$).
- S : A generic candidate subset of the variable index set.
- S^* : The true set of signal variables.
- S_ϵ^* : The maximizer of oracle regret given regularization parameter ϵ in TVS algorithm.
- $\mathcal{S}$: The collection of “good” variable subsets that satisfy both the size limit and the signal strength threshold required for exact recovery.
- $\hat{S}^t$: Variable selection result at the t -th iteration.

- T : Total number of algorithm iterations.
- U^t : Random seed used for generating data permutations at iteration t .
- $\mathbf{W}_i$: Random noise vector corresponding to the i -th observation.
- $\mathbf{W}$: The $p \times n$ random noise matrix.
- $\mathbf{X}_i$: The i -th p -dimensional observation vector.
- $\mathbf{X}$: The $p \times n$ observed data matrix.
- $\widetilde{\mathbf{X}}$: Transformed data matrix, defined as $\Omega^* \mathbf{X}$.
- y_j : Binary indicator denoting whether variable j is included in the best subset SDP formulation.
- $\mathbf{y}$: Vector of binary inclusion indicators $(y_1, \dots, y_p)^\top$.
- Y_j^t : Binary reward for the j -th variable at iteration t , determined via the permutation test.
- $Y_j(\hat{G}^{t-1})$: Equivalent to Y_j^t , explicitly denoting that the reward depends on the cluster estimation from the previous step.
- $\mathbf{Z}$: An $n \times n$ positive semidefinite matrix representation of a cluster partition, used in the SDP relaxation.
- $\mathbf{Z}(S)$: Solution of the subset SDP objective corresponding to variable set S .
- $\mathbf{Z}^*$: The $\mathbf{Z}$ -matrix form of the true cluster partition G^* .

9.2 Greek Alphabets

- α : A margin parameter ($0 < \alpha < 1 - \tau_\epsilon$) used to define the significance level $\tau_\epsilon - \alpha$ in the robust test and to establish the oracle margin separation.

- β^* : The Bayes rule coefficient vector $\Omega^*(\mathbf{c}_1^* - \mathbf{c}_2^*)$, representing the optimal classification boundary in Gaussian models.
- $\Gamma_j(\hat{G}^{t-1})$: Global sensitivity of the robust MMD test statistic for feature j under the estimated partition, used to bound the impact of adversarial sample swaps.
- Δ_S^2 : The squared clustering signal contained strictly within variable set S .
- $\bar{\Delta}_S^2$: The minimum signal strength threshold required for a variable set S to guarantee exact recovery.
- Δ : The minimum true clustering signal Δ_{S^*} for instances bounded within the minimax framework.
- ϵ : The primary threshold parameter regulating the exploration-exploitation trade-off in the Thompson Variable Selection (TVS) cutoff.
- $\theta_j(\hat{G}^{t-1})$: The success probability parameter (expected reward) of the Bernoulli test outcome for variable j , conditioned on the previous cluster estimation.
- θ_j^{orc} : The oracle mean reward, defined as $\theta_j(G^*)$.
- ϑ_j^t : Beta random variable drawn from $\text{Beta}(a_j^t, b_j^t)$ for the j -th variable, which is thresholded during the “choose step” of the TVS algorithm.
- $\Lambda(G, \tilde{\mathbf{X}}_j)$: The explicit test statistic function (empirical mean difference) evaluating variable j under partition G in our greedy algorithm.
- $\lambda_j^{(b)}$: The evaluated test statistic for variable j computed using the b -th permuted partition ($\Lambda_j^{(0)}$ represents the unpermuted, observed statistic).
- $\Lambda_j(G, \tilde{\mathbf{X}}_j; U)$: an arbitrary randomized test statistic evaluating variable j under partition G and randomness U , in our TVS algorithm. We propose to use **rMMD** (Algorithm 4) for this statistic.

- π_b : The b -th uniform random permutation of the sample indices.
- Σ^* : The $p \times p$ common true covariance matrix of the observations.
- σ^2 : The entrywise variance of the observations when operating under an isotropic covariance assumption ($\Sigma^* = \sigma^2 \mathbf{I}_p$).
- τ_ϵ : The beta random number inclusion cutoff applied in the TVS algorithm, defined as $\log(1/\epsilon)/\log((\epsilon + 1)/\epsilon)$.
- ψ : The parameter tuple $(\mathbf{c}_1, \dots, \mathbf{c}_K, \mathbf{H})$ detailing a specific sparse clustering problem instance.
- $\Psi(s, \Delta, K)$: The complete distribution class of Gaussian mixture models parameterized by sparsity s , signal strength Δ , and K clusters.
- Ω^* : The $p \times p$ true precision matrix, defined as $(\Sigma^*)^{-1}$.

10 Technical Lemmas and Definitions

This section presents technical lemmas and definitions that are used throughout the main proofs, beginning with the tail bounds for the chi-square distribution.

Lemma 7 (Lemma 1 of [Laurent & Massart 2000](#)). *If Z has a chi-square distribution with p degrees of freedom, then for all $t > 0$, we have*

$$\mathbb{P}(Z \geq p + 2\sqrt{pt} + 2t) \leq e^{-t}, \quad \text{and} \quad \mathbb{P}(Z \leq p - 2\sqrt{pt}) \leq e^{-t}.$$

Before presenting the next two lemmas, we formally define the sub-exponential norm.

Definition 8. *Let $\psi_1(x) := e^x - 1$. The sub-exponential norm of a random variable X ,*

denoted by $\|X\|_{\psi_1}$, is defined as

$$\|X\|_{\psi_1} := \inf \left\{ \lambda > 0 : \mathbb{E} \left[\psi_1 \left(\frac{|X|}{\lambda} \right) \right] \leq 1 \right\}.$$

Using this definition, Lemma 9 provides an upper bound on the subexponential norm of a chi-square random variable.

Lemma 9 (Lemma VIII.2 of [Chen & Yang 2021](#)). *Let $\mathbf{X}_1, \mathbf{X}_2 \stackrel{i.i.d.}{\sim} \mathcal{N}(0, I_q)$. Then there exists a constant $C_5 > 0$ such that*

$$\|\|\mathbf{X}_1\|_2^2 - q\|_{\psi_1} + \|\langle \mathbf{X}_1, \mathbf{X}_2 \rangle\|_{\psi_1} \leq C_5 \sqrt{q}.$$

Next, Lemma 10 provides a maximal inequality for sub-exponential norm.

Lemma 10 (Lemma 2.2.2 of [Van Der Vaart & Wellner 2023](#)). *For random variables $X_1, \dots, X_m$, there exists a constant C_7 such that*

$$\left\| \max_{1 \leq i \leq m} X_i \right\|_{\psi_1} \leq C_7 \log(m+1) \max_{1 \leq i \leq m} \|X_i\|_{\psi_1}.$$

Next, Lemma 11 provides an upper bound on the ℓ_2 operator norm of an $m \times n$ random matrix with i.i.d. Gaussian entries.

Lemma 11 (Corollary 5.35 of [Vershynin 2012](#)). *Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ have i.i.d. $\mathcal{N}(0, \sigma^2)$ entries. Then, for any $t > 0$, we have*

$$\mathbb{P}(\|\mathbf{A}\|_{\text{op}} \geq \sigma(\sqrt{m} + \sqrt{n} + \sqrt{2t})) \leq e^{-t}.$$

Next, Lemma 12 provides a concentration inequality for empirical processes indexed by an

unbounded function class.

Lemma 12 (Theorem 4 of [Adamczak 2008](#)). *Let $X_1, \dots, X_n$ be independent random variables taking values in a measurable space $(S, \mathcal{B})$, and let $\mathcal{F}$ be a countable class of measurable functions $f : S \rightarrow \mathbb{R}$. Assume that for all $f \in \mathcal{F}$ and all $i = 1, \dots, n$, it holds that $\mathbb{E}[f(X_i)] = 0$ and*

$$\left\| \sup_{f \in \mathcal{F}} |f(X_i)| \right\|_{\psi_1} < \infty,$$

Define

$$Z := \sup_{f \in \mathcal{F}} \left| \sum_{i=1}^n f(X_i) \right|, \quad \gamma^2 := \sup_{f \in \mathcal{F}} \sum_{i=1}^n \text{Var}[f(X_i)], \quad \text{and} \quad M_{\psi_1} := \left\| \max_{i \in [n]} \sup_{f \in \mathcal{F}} |f(X_i)| \right\|_{\psi_1}$$

Then there exists a constant $C_7 > 0$ such that for all $t > 0$, we have

$$\mathbb{P}(Z \geq 2\mathbb{E}[Z] + t) \leq \exp\left(-\frac{t^2}{3\gamma^2}\right) + 3\exp\left(-\frac{t}{C_7 M_{\psi_1}}\right).$$

Next, Lemma 13 presents a concentration inequality for the suprema of the Gaussian process.

Lemma 13 (Theorem 5.8 of [Boucheron et al. 2016](#)). *Let $(X_t)_{t \in \mathcal{T}}$ be an almost surely continuous, centered Gaussian process indexed by a totally bounded set $\mathcal{T}$. Define*

$$\gamma^2 := \sup_{t \in \mathcal{T}} \text{Var}(X_t^2), \quad \text{and} \quad Z := \sup_{t \in \mathcal{T}} X_t.$$

Then, we have $\text{Var}(Z) \leq \gamma^2$. Moreover, for any $t > 0$,

$$\mathbb{P}(Z - \mathbb{E}[Z] \geq t) \leq \exp\left(-\frac{t^2}{2\gamma^2}\right).$$

The next lemma, Dudley's integral inequality, bounds the expected supremum of a Gaussian

process.

Lemma 14 (Theorem 8.1.3 of [Vershynin 2018](#)). *Let (T, d) be a metric space, and let $N(T, d, \varepsilon)$ denote the ε -covering number of T with respect to d . Let $(X_t)_{t \in T}$ be a mean-zero random process on a metric space with sub-Gaussian increments. Then*

$$\mathbb{E} \sup_{t \in T} X_t \leq C_8 K_{\psi_2} \int_0^\infty \sqrt{\log N(T, d, \varepsilon)} d\varepsilon,$$

where C_8 is an absolute constant, and K_{ψ_2} is the sub-Gaussian parameter.

Finally, Lemma 15 presents the uniform Hanson-Wright inequality for Gaussian quadratic forms.

Lemma 15 (Lemma VIII.4 of [Chen & Yang 2021](#)). *Let $\mathbf{X} \sim \mathcal{N}(0, I_q)$ and $\mathcal{A}$ be a bounded class of $q \times q$ matrices. Define the random variable*

$$Z = \sup_{\mathbf{A} \in \mathcal{A}} \left(\mathbf{X}^\top \mathbf{A} \mathbf{X} - \mathbb{E}[\mathbf{X}^\top \mathbf{A} \mathbf{X}] \right),$$

which corresponds to the supremum of a centered random process. Then, there exists a constant $C_9 > 0$ such that for any $t > 0$,

$$\mathbb{P}(|Z - \mathbb{E}[Z]| \geq t) \leq 2 \exp \left[-C_9 \min \left(\frac{t^2}{\|\mathbf{X}\|_{\mathcal{A}}^2}, \frac{t}{\sup_{\mathbf{A} \in \mathcal{A}} \|\mathbf{A}\|_{\text{op}}} \right) \right],$$

where $\|\mathbf{X}\|_{\mathcal{A}} := \mathbb{E} [\sup_{\mathbf{A} \in \mathcal{A}} \|(\mathbf{A} + \mathbf{A}^T)\mathbf{X}\|_2]$.

11 Proof of Theorem 1

The first three sections (Sections 11.2-11.4) prove the first claim of the theorem,

$$\mathbb{P}(\hat{\mathbf{Z}}(S) = \mathbf{Z}^*, \forall S \in \mathcal{S}) \geq 1 - C_3 K/n.$$

Then Section 11.6 proves the second claim of the theorem:

$$\inf_{\hat{\mathbf{H}}: \mathbb{R}^{p \times n} \rightarrow \mathcal{H}(K)} \sup_{\psi \in \overline{\Theta}} \mathbb{P}_{\psi}(\hat{\mathbf{H}}(\mathbf{X}) \neq \mathbf{H}) \geq C_5.$$

The proof of the first claim follows the strategy of Theorem II.1 of Chen & Yang (2021), with modifications to account for the sparsity assumption. The proof proceeds in three key steps.

1. **Reformulation into high-probability bound problem (Section 11.2):** For each S , we construct three statistics that bound a dual certificate λ^S , which verifies exact cluster recovery by the SDP problem (8) corresponding to S . A nontrivial gap between these bounds guarantees the existence of such a certificate (Corollary 22).
2. **First bounds and construction of the dual variable (Section 11.3).** Under the separation and set size conditions imposed by $\mathcal{S}$, we derive uniform high-probability bounds on the two statistics defined in the first step and determine the value of λ^S accordingly (Lemma 25).
3. **Second bounds and conclusion (Section 11.4).** We establish a uniform high-probability bound on the third statistic and verify that the previously chosen value of λ^S exceeds this bound, thereby confirming the existence of a dual certificate and completing the proof.

For notational simplicity, we adopt the following matrix-based notations of our model $\mathbf{X}_i = \mathbf{c}_k^* + \varepsilon_i$ for each $i \in G_k^*$, where $\varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2 \mathbf{I}_p)$:

Definition 16 (Matrix-based notations). *We view the data as a matrix $\mathbf{X} \in \mathbb{R}^{p \times n}$ and write $\mathbf{X} = \mathbf{M}^* + \mathcal{E}$, where $\mathbf{M}^*$ contains the cluster means column-wise, and $\mathcal{E}$ collects the noise vectors. For any support set $S \subset [p]$, we denote the i th observation and its noise*

restricted to S by

$$\mathbf{X}_{S,i} \quad \text{and} \quad \mathcal{E}_{S,i},$$

respectively. Given true clusters $G_1^*, \dots, G_K^*$, we define the empirical mean and average noise of cluster k on support S as

$$\bar{\mathbf{X}}_{S,k}^* := \frac{1}{|G_k^*|} \sum_{i \in G_k^*} \mathbf{X}_{S,i}, \quad \bar{\mathcal{E}}_{S,k}^* := \frac{1}{|G_k^*|} \sum_{i \in G_k^*} \mathcal{E}_{S,i}.$$

11.1 List of Notations

Before detailing the intermediate steps for the proof of Theorem 1, we define the specific notations utilized throughout this section. To facilitate the reading of the proof, these notations are organized into three thematic categories: variables associated with the semidefinite programming (SDP) relaxation and its dual formulation, variables defining the data matrix and true cluster model, and the bounding and reformulation statistics constructed for the proof.

11.1.1 SDP and Dual Variables

- $\mathbf{Q}^S$: dual variable corresponding to the PSD constraint of the SDP associated with the variable set S .
- λ^S : dual variable corresponding to the trace constraint in the SDP associated with the variable set S .
- α^S : dual variable corresponding to the symmetry constraint in the SDP associated with the variable set S .
- $\hat{\alpha}^S(\lambda^S)$: for a given value of λ^S , the unique value of α^S determined by condition (21).
- $\mathbf{B}^S$: dual variable corresponding to the entrywise nonnegativity constraint in the

SDP associated with the variable set S .

- $\dot{\mathbf{B}}^S(\lambda^S)$: for a given value of λ^S , the value of $\mathbf{B}^S$ implied by condition (22).
- $\mathbf{W}^S := \lambda^S \mathbf{I}_n - \mathbf{B}^S + \frac{1}{2} \mathbf{1}_n (\alpha^S)^\top + \frac{1}{2} \alpha^S \mathbf{1}_n^\top - \mathbf{X}_{S,\cdot}^\top \mathbf{X}_{S,\cdot}$: a function of the dual variables, introduced to simplify notation.
- $\dot{\mathbf{W}}^S(\lambda^S)$: the constructed function of the parameterized dual variables based on λ^S .
- $\mathcal{L}(\mathbf{Z}, \mathbf{Q}^S, \lambda^S, \alpha^S, \mathbf{B}^S)$: the Lagrangian function associated with the primal SDP problem.
- $\mathbf{r}^{(k,l,S)}$: the unique vector determining the row-wise sum of the (k,l) off-diagonal block of the dual certificate $\dot{\mathbf{B}}^S$.
- $t_{lk}^S(\lambda^S)$: the sum of all elements in the (l,k) th off-diagonal block of the constructed dual matrix $\dot{\mathbf{B}}^S(\lambda^S)$.

11.1.2 Data and True Cluster Model

- $\mathbf{Z}^*$: a symmetric block-diagonal real matrix corresponding to the true cluster partition G^* .
- $\mathbf{M}^*$: the matrix containing the true cluster means column-wise.
- $\mathcal{E}$: the matrix collecting the random noise vectors.
- $\mathbf{X}_{S,i}$: the i th observation restricted to the indices corresponding to the variable set S .
- $\mathcal{E}_{S,i}$: the noise vector of the i th observation restricted to the variable set S .
- $\overline{\mathbf{X}}_{S,k}^*$: the average of the observations in true cluster k , restricted to the indices corresponding to the variable set S .

- $\bar{\mathcal{E}}_{S,k}^*$: the average of the noise of true cluster k , restricted to the indices corresponding to the variable set S .
- $\mathcal{S}$: the collection of support sets satisfying the required separation and set size conditions.
- $\Delta_{S \cap S^*}^2$: the minimum squared Euclidean distance between true cluster centers restricted to $S \cap S^*$.
- m : the minimum harmonic-mean-like quantity of the true cluster sizes, defined as $\min_{k \neq l} 2|G_l^*||G_k^*|/(|G_l^*| + |G_k^*|)$.

11.1.3 Bounding and Reformulation Statistics

- $D_{kli}^S(\mathbf{X})$: the difference between the squared Euclidean distances from observation $\mathbf{X}_{S,i}$ to the true cluster centers l and k .
- $U^S(\mathbf{X})$: a data-dependent upper bound for the dual variable λ^S .
- Γ_K : the linear subspace of $\mathbb{R}^n$ defined as the orthogonal complement to the span of the cluster indicator vectors $(\mathbf{1}_{G_k^*})$.
- $L_1^S(\mathbf{X})$: the supremum of the noise quadratic form process over the subspace Γ_K .
- $I_{kl}^S(\mathbf{v}, \mathbf{X})$: the fundamental component representing signal–noise interactions.
- $N_{kl}^S(\mathbf{v}, \mathbf{X})$: the fundamental component representing noise–noise interactions.
- $T_{1,kl}^S(\mathbf{v}), T_{2,kl}^S(\mathbf{v}), T_{3,kl}^S(\mathbf{v})$: composite random quantities representing combinations of signal–noise and noise–noise interactions.
- $L_2^S(\mathbf{X}, \lambda^S)$: a supremum quantity defining the lower bound condition for λ^S .
- $\dot{\lambda}^S$: a specific chosen value for the dual variable λ^S , defined as $\sigma^2|S| + m\Delta_{S \cap S^*}^2/4$.

11.2 Reformulation into high-probability bound problem

The primary goal of this section is to derive Corollary 22, which converts the optimality condition into high-probability bounds. Let $S \subset [p]$ be a support set, and consider the primal SDP problem (20) corresponding to S , recalled below:

$$\max_{\mathbf{Z} \in \mathbb{R}^{n \times n}} \langle \mathbf{X}_{S,\cdot}^\top, \mathbf{X}_{S,\cdot}, \mathbf{Z} \rangle \quad \text{s.t.} \quad \mathbf{Z}^\top = \mathbf{Z}, \mathbf{Z} \succeq 0, \text{tr}(\mathbf{Z}) = K, \mathbf{Z} \mathbf{1}_n = \mathbf{1}_n, \mathbf{Z} \geq 0. \quad (20)$$

Let $\mathcal{L}(\mathbf{Z}, \mathbf{Q}^S, \lambda^S, \alpha^S, \mathbf{B}^S)$ denote the associated Lagrangian function. Here, the Lagrange multipliers

$$\mathbf{Q}^S \in \mathbb{R}^{n \times n}, \quad \lambda^S \in \mathbb{R}, \quad \alpha^S \in \mathbb{R}^n, \quad \text{and} \quad \mathbf{B}^S \in \mathbb{R}^{n \times n}$$

correspond to the primal constraints

$$\mathbf{Z} \succeq 0, \quad \text{tr}(\mathbf{Z}) = K, \quad \mathbf{Z}^\top = \mathbf{Z}, \quad \text{and} \quad \mathbf{Z} \geq 0,$$

respectively. Note that $\mathbf{Q}^S \succeq 0$ and $\mathbf{B}^S \geq 0$. Based on this formulation, this section proceeds in three main steps to ultimately establish Corollary 22:

1. **Necessary conditions for dual certificates (Section 11.2.1):** We formalize sufficient conditions for the existence of dual certificates (Lemma 17) and leverage them to derive necessary conditions (Lemma 18) that guide the construction of dual variables.
2. **Reduction to a single dual variable construction (Section 11.2.2):** Leveraging the necessary conditions, we parameterize the dual variables in terms of λ^S , reducing the construction to a one-dimensional problem (Lemma 19).
3. **Reformulation into high-probability bound problem (Section 11.2.3):** We derive statistics that bound the dual certificate λ^S from above (Lemma 20) and below

(Lemma 21). This shows that a nontrivial gap between them guarantees the existence of dual variables satisfying the conditions of Lemma 17 (Corollary 22).

11.2.1 Necessary conditions for dual certificates

Let $S \subset [p]$ be a support set, and consider the corresponding primal SDP problem (20). Lemma 17 provides sufficient conditions under which the dual variables λ^S , α^S , and $\mathbf{B}^S$ form a valid dual certificate (with $\mathbf{Q}^S$ implicitly captured by the conditions). In particular, the lemma guarantees that the duality gap vanishes at the true cluster membership matrix

$$\mathbf{Z}^* = \sum_{k=1}^K \frac{1}{|G_k^*|} \mathbf{1}_{G_k^*} \mathbf{1}_{G_k^*}^\top.$$

Lemma 17 (Optimality conditions). *Let $S \subset [p]$ be a support set, and consider the corresponding primal SDP problem (20). The matrix $\mathbf{Z}^*$ is the unique solution to this problem if there exist dual variables $\lambda^S \in \mathbb{R}$, $\alpha^S \in \mathbb{R}^n$, and $\mathbf{B}^S \in \mathbb{R}^{n \times n}$ such that the following conditions are satisfied:*

- (C1) $\mathbf{B}^S \geq 0$;
- (C2) $\mathbf{W}^S := \lambda^S \mathbf{I}_n - \mathbf{B}^S + \frac{1}{2} \mathbf{1}_n (\alpha^S)^\top + \frac{1}{2} \alpha^S \mathbf{1}_n^\top - \mathbf{X}_{S,\cdot}^\top \mathbf{X}_{S,\cdot} \succeq 0$;
- (C3) $\langle \mathbf{W}^S, \mathbf{Z}^* \rangle = 0$;
- (C4) $\langle \mathbf{B}^S, \mathbf{Z}^* \rangle = 0$;
- (C5) $\mathbf{B}_{\mathbf{G}_k^*, \mathbf{G}_l^*}^S > 0$ for all $1 \leq k \neq l \leq K$.

In Lemma 17, conditions (C1) and (C2) ensure dual feasibility, (C3) and (C4) guarantee a zero duality gap, and (C5) ensures the uniqueness of the solution. The proof is provided in Section 11.5.1. Building on these conditions, we now establish necessary conditions for

dual certificates. Let $(\mathbf{B}_{G_k^* G_l^*}^S \mathbf{1}_{|G_l^*|})_i$ denote the i th entry of the row-wise sum of $\mathbf{B}_{G_k^* G_l^*}^S \in \mathbb{R}^{|G_k^*| \times |G_l^*|}$, which is the (k, l) th off-diagonal block of $\mathbf{B}^S$. Let $\bar{\mathbf{X}}_{S,k}^*$ denote the average of the observations belonging to the true cluster k , restricted to the entries corresponding to the variable set S . Leveraging these notations, we state the necessary conditions in the following lemma.

Lemma 18 (Necessary conditions for optimality). *Let $S \subset [p]$ be a support set, and consider the corresponding primal SDP problem (20). Suppose dual variables $\lambda^S \in \mathbb{R}$, $\alpha^S \in \mathbb{R}^n$, and $\mathbf{B}^S \in \mathbb{R}^{n \times n}$ satisfy conditions (C2), (C3), and (C4) of Lemma 17. Then, for all $1 \leq k \neq l \leq K$ and $i \in G_k^*$, the following statements hold:*

$$\alpha_i^S = 2\mathbf{X}_{S,i}^\top \bar{\mathbf{X}}_{S,k}^* - \frac{\lambda^S}{|G_k^*|} - \|\bar{\mathbf{X}}_{S,k}^*\|_2^2, \text{ and} \quad (21)$$

$$(\mathbf{B}_{G_k^* G_l^*}^S \mathbf{1}_{|G_l^*|})_i = -\frac{|G_k^*| + |G_l^*|}{2|G_k^*|} \lambda^S + \frac{|G_l^*|}{2} (\|\bar{\mathbf{X}}_{S,l}^* - \mathbf{X}_{S,i}\|_2^2 - \|\bar{\mathbf{X}}_{S,k}^* - \mathbf{X}_{S,i}\|_2^2), \quad (22)$$

The proof of Lemma 18 is provided in Section 11.5.2.

11.2.2 Reduction to a single dual variable construction

For any given $\lambda^S \in \mathbb{R}$, condition (21) uniquely determines the corresponding value of α^S , which we denote by $\dot{\alpha}^S(\lambda^S)$. We then turn to the dual variable $\mathbf{B}^S$ and construct $\dot{\mathbf{B}}^S(\lambda^S)$, designed to satisfy condition (22). The matrix $\dot{\mathbf{B}}^S(\lambda^S)$ is structured as a block matrix, with symmetric rank-one matrices as its off-diagonal blocks and zero matrices as its diagonal blocks. These constructions follow the proof of Theorem II.1 in Chen & Yang (2021), with modifications to account for varying support sets.

We now describe the construction of $\dot{\mathbf{B}}^S(\lambda^S)$. Fix any value of $\lambda^S \in \mathbb{R}$, and for notational simplicity, we suppress the dependence on λ^S in what follows. For each $k = 1, \dots, K$, define the diagonal block $\dot{\mathbf{B}}_{G_k^*, G_k^*}^S \in \mathbb{R}^{|G_k^*| \times |G_k^*|}$ to be the zero matrix. Next, consider the

off-diagonal blocks. Given λ^S , the data, and the true cluster structure, condition (22) uniquely determines the row-wise sum of the (k, l) off-diagonal block of the dual certificate. Denote this unique vector by $\mathbf{r}^{(k,l,S)} \in \mathbb{R}^{|G_k^*|}$. Then, for each $1 \leq k \neq l \leq K$, we define the off-diagonal block $\dot{\mathbf{B}}_{G_k^*, G_l^*}^S \in \mathbb{R}^{|G_k^*| \times |G_l^*|}$ as the symmetric rank-one matrix:

$$\dot{\mathbf{B}}_{G_k^*, G_l^*}^S := \frac{1}{\mathbf{1}_{|G_l^*|}^\top \mathbf{r}^{(l,k,S)}} \mathbf{r}^{(k,l,S)} (\mathbf{r}^{(l,k,S)})^\top, \quad (23)$$

This construction ensures that the row-sum condition in (22) is satisfied:

$$\begin{aligned} \dot{\mathbf{B}}_{G_k^*, G_l^*}^S \mathbf{1}_{|G_l^*|} &= \frac{1}{\mathbf{1}_{|G_l^*|}^\top \mathbf{r}^{(l,k,S)}} \{ \mathbf{r}^{(k,l,S)} (\mathbf{r}^{(l,k,S)})^\top \} \mathbf{1}_{|G_l^*|} \\ &= \frac{1}{\mathbf{1}_{|G_l^*|}^\top \mathbf{r}^{(l,k,S)}} \mathbf{r}^{(k,l,S)} \{ (\mathbf{r}^{(l,k,S)})^\top \mathbf{1}_{|G_l^*|} \} \\ &= \mathbf{r}^{(k,l,S)}. \end{aligned}$$

We summarize the construction of $\dot{\alpha}^S(\lambda^S)$ and $\dot{\mathbf{B}}^S(\lambda^S)$ in the following lemma.

Lemma 19 (Parametrized dual certificate candidates). *Fix the data matrix $\mathbf{X}$ and the true cluster structure $G_1^*, \dots, G_K^*$. Let $S \subset [p]$ be a support set, and consider the corresponding primal SDP problem (20). Fix a value of the dual variable $\lambda^S \in \mathbb{R}$. Define $\dot{\alpha}^S(\lambda^S) \in \mathbb{R}^n$ by setting, for each $k = 1, \dots, K$,*

$$\dot{\alpha}_{G_k^*}^S(\lambda^S) := 2\mathbf{X}_{S, G_k^*}^\top \bar{\mathbf{X}}_{S, k}^* - \frac{\lambda^S}{|G_k^*|} \mathbf{1}_{|G_k^*|} - \|\bar{\mathbf{X}}_{S, k}^*\|_2^2 \mathbf{1}_{|G_k^*|},$$

Next, define $\dot{\mathbf{B}}^S(\lambda^S) \in \mathbb{R}^{n \times n}$ as the block matrix:

$$\dot{\mathbf{B}}^S(\lambda^S) := \begin{bmatrix} \mathbf{0} & \dot{\mathbf{B}}_{G_1^*, G_2^*}^S(\lambda^S) & \cdots & \dot{\mathbf{B}}_{G_1^*, G_K^*}^S(\lambda^S) \\ \dot{\mathbf{B}}_{G_2^*, G_1^*}^S(\lambda^S) & \mathbf{0} & \cdots & \dot{\mathbf{B}}_{G_2^*, G_K^*}^S(\lambda^S) \\ \vdots & \vdots & \ddots & \vdots \\ \dot{\mathbf{B}}_{G_K^*, G_1^*}^S(\lambda^S) & \dot{\mathbf{B}}_{G_K^*, G_2^*}^S(\lambda^S) & \cdots & \mathbf{0} \end{bmatrix},$$

where for each $1 \leq k \neq l \leq K$, the off-diagonal block $\dot{\mathbf{B}}_{G_k^*, G_l^*}^S(\lambda^S) \in \mathbb{R}^{|G_k^*| \times |G_l^*|}$ is defined as

$$\dot{\mathbf{B}}_{G_k^*, G_l^*}^S(\lambda^S) := \frac{1}{\mathbf{1}_{|G_l^*|}^\top \mathbf{r}^{(l,k,S)}} \mathbf{r}^{(k,l,S)} (\mathbf{r}^{(l,k,S)})^\top, \quad (24)$$

with $\mathbf{r}^{(k,l,S)} \in \mathbb{R}^{|G_k^*|}$ defined componentwise by

$$r_i^{(k,l,S)} := -\frac{|G_k^*| + |G_l^*|}{2|G_k^*|} \lambda^S + \frac{|G_l^*|}{2} (\|\bar{\mathbf{X}}_{S,l}^* - \mathbf{X}_{S,i}\|_2^2 - \|\bar{\mathbf{X}}_{S,k}^* - \mathbf{X}_{S,i}\|_2^2). \quad (25)$$

Finally, we construct

$$\dot{\mathbf{W}}^S(\lambda^S) := \lambda^S \mathbf{I}_n - \dot{\mathbf{B}}^S(\lambda^S) + \frac{1}{2} \mathbf{1}_n (\dot{\alpha}^S(\lambda^S))^\top + \frac{1}{2} \dot{\alpha}^S(\lambda^S) \mathbf{1}_n^\top - \mathbf{X}_{S,\cdot}^\top \mathbf{X}_{S,\cdot}$$

Constructed $\dot{\alpha}^S(\lambda^S)$ and $\dot{\mathbf{B}}^S(\lambda^S)$ satisfy the conditions in Lemma 18.

11.2.3 Reformulation into high-probability bound problem

Next, we identify conditions of λ^S , that ensures $\dot{\mathbf{B}}^S(\lambda^S)$ and $\dot{\alpha}^S(\lambda^S)$ satisfy the conditions in Lemma 17. We begin with conditions (C1), (C3), (C4) and (C5). To this end, we introduce the following data-dependent quantities:

$$D_{kli}^S(\mathbf{X}) := \|\bar{\mathbf{X}}_{S,l}^* - \mathbf{X}_{S,i}\|_2^2 - \|\bar{\mathbf{X}}_{S,k}^* - \mathbf{X}_{S,i}\|_2^2,$$

and

$$U^S(\mathbf{X}) := \min_{1 \leq k \neq l \leq K} \left\{ \left(\frac{1}{|G_k^*|} + \frac{1}{|G_l^*|} \right)^{-1} \min_{i \in G_k^*} D_{kli}^S(\mathbf{X}) \right\}. \quad (26)$$

Based on these definitions, we now state an upper bound on λ^S :

Lemma 20 (Upper bound on λ^S). *Let $S \subset [p]$ be a support set, and consider the corresponding primal SDP problem (20). For any value of λ^S , corresponding $\dot{\alpha}^S(\lambda^S)$, $\dot{\mathbf{B}}^S(\lambda^S)$, and $\dot{\mathbf{W}}^S(\lambda^S)$ satisfy conditions (C3) and (C4) of Lemma 17. Moreover, if*

$$\lambda^S < U^S(\mathbf{X}),$$

then $\dot{\alpha}^S(\lambda^S)$ and $\dot{\mathbf{B}}^S(\lambda^S)$ also satisfy conditions (C1) and (C5) of Lemma 17.

Proof of Lemma 20 is provided in Section 11.5.3.

Next we move onto condition (C2) of Lemma 17. To this end, we define some notation.

We first define a linear subspace of $\mathbb{R}^n$ as:

$$\Gamma_K := \text{span} \left(\mathbf{1}_{G_k^*} : k \in [K] \right)^\perp,$$

where $\perp$ means orthogonal complement. For each $\mathbf{v} \in \Gamma_K$ and support set $S \subset [p]$, we define a collection of data-dependent random variables. We begin with the following supremum of noise quadratic form process:

$$L_1^S(\mathbf{X}) := \sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} \|\mathcal{E}_{S, \cdot} \mathbf{v}\|_2^2, \quad (27)$$

Next, we define two fundamental components representing signal–noise and noise–noise

interactions:

$$I_{kl}^S(\mathbf{v}, \mathbf{X}) := |G_l^*| \sum_{i \in G_k^*} v_i \langle (\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}, \mathcal{E}_{S \cap S^*, i} \rangle, \text{ and} \quad (28)$$

$$N_{kl}^S(\mathbf{v}, \mathbf{X}) := |G_l^*| \sum_{i \in G_k^*} v_i \langle \bar{\mathcal{E}}_{S, k}^* - \bar{\mathcal{E}}_{S, l}^*, \mathcal{E}_{S, i} \rangle. \quad (29)$$

Note that both definitions are asymmetric with respect to the first two indices. Using these definitions, we introduce three composite random quantities:

$$T_{1,kl}^S(\mathbf{v}) := I_{kl}^S(\mathbf{v}, \mathbf{X}) \cdot I_{lk}^S(\mathbf{v}, \mathbf{X}), \quad (30)$$

$$T_{2,kl}^S(\mathbf{v}) := N_{kl}^S(\mathbf{v}, \mathbf{X}) \cdot N_{lk}^S(\mathbf{v}, \mathbf{X}), \text{ and} \quad (31)$$

$$T_{3,kl}^S(\mathbf{v}) := N_{kl}^S(\mathbf{v}, \mathbf{X}) \cdot I_{lk}^S(\mathbf{v}, \mathbf{X}) + I_{kl}^S(\mathbf{v}, \mathbf{X}) \cdot N_{lk}^S(\mathbf{v}, \mathbf{X}). \quad (32)$$

Next, the following quantity represents the sum of all elements of $\dot{\mathbf{B}}_{G_l^*, G_k^*}^S$, (l, k) th off-diagonal block of $\dot{\mathbf{B}}^S(\lambda^S)$:

$$t_{lk}^S(\lambda^S) := \sum_{j \in G_l^*} r_j^{(l, k, S)} = \sum_{j \in G_l^*} \left\{ \frac{|G_k^*|}{2} D_{lkj}^S(\mathbf{X}) - \frac{|G_k^*| + |G_l^*|}{2|G_l^*|} \lambda^S \right\}. \quad (33)$$

Leveraging all these notations, we finally define:

$$L_2^S(\mathbf{X}, \lambda^S) := \sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} \left\{ \sum_{1 \leq k \neq l \leq K} \frac{1}{t_{lk}^S(\lambda^S)} \left(T_{1,kl}^S(\mathbf{v}) + T_{2,kl}^S(\mathbf{v}) + T_{3,kl}^S(\mathbf{v}) \right) \right\}, \quad (34)$$

Using this quantity, Lemma 21 states the lower bound condition on λ_S that guarantees condition (C2) of Lemma 17 is satisfied.

Lemma 21 (Lower bound on λ_S). *Let $S \subset [p]$ be a support set, and consider the corre-*

sponding primal SDP problem (20). If

$$\lambda^S > L_1^S(\mathbf{X}) + L_2^S(\mathbf{X}, \lambda^S),$$

then the constructed dual variables $\dot{\alpha}^S(\lambda^S)$ and $\dot{\mathbf{B}}^S(\lambda^S)$ satisfy condition (C2) of Lemma 17.

Proof of Lemma 21 is provided in Section 11.5.4. Combining Lemmas 20 and 21, we finally arrive at the following corollary:

Corollary 22 (Problem reformulation). *Let $S \subset [p]$ be a support set, and consider the corresponding primal SDP problem (20). If there exists λ^S such that*

$$L_1^S(\mathbf{X}) + L_2^S(\mathbf{X}, \lambda^S) < \lambda^S < U^S(\mathbf{X}),$$

then the dual variables constructed with this λ^S via Lemma 19 yield a valid certificate that satisfies all conditions of Lemma 17, verifying that $\mathbf{Z}^$ is the unique optimal point of the primal problem.*

11.3 First Bounds and Construction of the Dual Variable

This section is devoted to the derivation of Lemma 25, which establishes that, under the separation condition, there exists a λ^S that partially satisfies the condition stated in Corollary 22. We proceed in two main steps:

1. **High-probability bounds (Section 11.3.1)** : We derive a lower bound for $U^S(\mathbf{X})$ (Lemma 23) and an upper bound for $L_1^S(\mathbf{X})$ (Lemma 24), under the separation and set size condition of $\mathcal{S}$.
2. **Specifying the value of the dual variable (Section 11.3.2)**: We set the value

of λ^S to lie between the bounds established in the previous step (Lemma 25).

11.3.1 High-probability Bounds

Recall from (26) that $U^S(\mathbf{X})$ is defined as

$$U^S(\mathbf{X}) = \min_{1 \leq k \neq l \leq K} \left\{ \left(\frac{1}{|G_k^*|} + \frac{1}{|G_l^*|} \right)^{-1} \min_{i \in G_k^*} D_{kli}^S(\mathbf{X}) \right\},$$

where

$$D_{kli}^S(\mathbf{X}) = \|\bar{\mathbf{X}}_{S,l}^* - \mathbf{X}_{S,i}\|_2^2 - \|\bar{\mathbf{X}}_{S,k}^* - \mathbf{X}_{S,i}\|_2^2.$$

Recall from (10) that we define $\mathcal{S}$ as

$$\mathcal{S} := \left\{ S : S \subset [p], S \cap S^* \neq \emptyset, |S| \leq \sqrt{p}, \Delta_{S \cap S^*}^2 \gtrsim \sigma^2 \left(\log n + \frac{|S| \log p}{m} + \sqrt{\frac{|S| \log p}{m}} \right) \right\},$$

where

$$\Delta_{S \cap S^*}^2 = \min_{1 \leq k \neq l \leq K} \|(\mathbf{c}_l^* - \mathbf{c}_k^*)_{S \cap S^*}\|_2^2,$$

and

$$m = \min_{1 \leq k \neq l \leq K} \left(\frac{2|G_l^*||G_k^*|}{|G_l^*| + |G_k^*|} \right) \geq C_1 \frac{n}{\log n}.$$

The following lemma establishes a high-probability lower bound for $D_{kli}^S(\mathbf{X})$.

Lemma 23. *Uniformly across all index (k, l, i) ranging over all pairs $1 \leq k \neq l \leq K$ and $i \in G_k^*$, and all $S \in \mathcal{S}$, with probability at least $1 - C_6/n$, it holds that*

$$D_{kli}^S(\mathbf{X}) \gtrsim \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + \left(\frac{1}{|G_k^*|} + \frac{1}{|G_l^*|} \right) |S| - R_{kl}^S.$$

where $C_6 > 0$ is a constant and

$$R_{kl}^S := \sqrt{\frac{\xi^S}{|G_l^*|}} \|(\mathbf{c}_l^* - \mathbf{c}_k^*)_{S \cap S^*}\|_2 + \left(\frac{1}{|G_k^*|} + \frac{1}{|G_l^*|} \right) \sqrt{|S| \xi^S} + \frac{1}{|G_k^*|} \zeta^S,$$

and $\zeta^S := \log(n^2 |S|^2 \binom{p}{|S|})$.

The proof of Lemma 23 is presented in Section 11.5.5.

Next, Lemma 24 establishes a high-probability upper bound for $L_1^S(\mathbf{X})$, where we recall from (27):

$$L_1^S(\mathbf{X}) = \sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} \|\mathcal{E}_{S, \cdot} \mathbf{v}\|_2^2,$$

and $\mathcal{E}_{S, \cdot}$ is a submatrix of $\mathcal{E} \in \mathbb{R}^{p \times n}$, whose entries are i.i.d. Gaussian random variables.

Lemma 24. *Uniformly across all $S \subset [p]$ such that $|S| \leq \sqrt{p}$, with probability at least $1 - C_7/n$, where $C_7 > 0$ is a constant, it holds that*

$$L_1^S(\mathbf{X}) \lesssim \sigma^2(n + \log n + |S| \log p).$$

The proof of Lemma 24 is provided in Section 11.5.6.

11.3.2 Specifying the Value of the Dual Variable

Since $L_2^S(\mathbf{X}, \lambda^S)$ in (34) depends on λ^S , we first specify a value which lies between $L_1^S(\mathbf{X})$ and $U^S(\mathbf{X})$ with high probability, uniformly over $S \in \mathcal{S}$.

Lemma 25. *Let $\dot{\lambda}^S := \sigma^2 |S| + m \Delta_{S \cap S^*}^2 / 4$. Then uniformly over $S \in \mathcal{S}$, with probability at least $1 - C_9/n$, where $C_9 > 0$ is a constant, it holds that*

$$L_1^S(\mathbf{X}) \lesssim \dot{\lambda}^S \lesssim U^S(\mathbf{X}).$$

Proof of Lemma 25 is provided in Section 11.5.7.

11.4 Second high-probability bounds and conclusion

This section completes the proof by showing that, with high probability, $L_2^S(\mathbf{X}, \dot{\lambda}^S) \lesssim \dot{\lambda}^S$. As defined in equation (34), $L_2^S(\mathbf{X}, \dot{\lambda}^S)$ is the supremum of an empirical process involving statistics t_{lk}^S (33), $I_{kl}^S(\mathbf{v}, \mathbf{X})$, $I_{lk}^S(\mathbf{v}, \mathbf{X})$ (28), $N_{kl}^S(\mathbf{v}, \mathbf{X})$, and $N_{lk}^S(\mathbf{v}, \mathbf{X})$ (29). Therefore, the bound on $L_2^S(\mathbf{X}, \dot{\lambda}^S)$ follows directly from bounds on the t_{lk}^S , $I_{kl}^S(\mathbf{v}, \mathbf{X})$ and $N_{kl}^S(\mathbf{v}, \mathbf{X})$. Accordingly, the remainder of this section proceeds as follows:

1. **Bounding the off-diagonal block sum of the dual variable matrix (Section 11.4.1):** We derive a high-probability lower bound for t_{lk}^S (Lemma 26).
2. **Bounding the signal–noise interaction term (Section 11.4.2):** We derive a high-probability upper bound for $I_{kl}^S(\mathbf{v}, \mathbf{X})$ (Lemma 27) and show that the corresponding component of $L_2^S(\mathbf{X}, \dot{\lambda}^S)$ is dominated by $\dot{\lambda}^S$ (Lemma 28).
3. **Bounding the noise–noise interaction term (Section 11.4.3):** We derive a high-probability upper bound for $N_{kl}^S(\mathbf{v}, \mathbf{X})$ (Lemmas 29 and 30), and show that the corresponding component of $L_2^S(\mathbf{X}, \dot{\lambda}^S)$ is dominated by $\dot{\lambda}^S$ (Lemma 31).
4. **Conclusion (Section 11.4.4).**

11.4.1 Bounding the off-diagonal block sum of the dual variable matrix

With the value of $\dot{\lambda}^S$ set in Lemma 25, we provide a high-probability lower bound of $t_{lk}^S(\dot{\lambda}^S)$ defined in (33).

Lemma 26. *For any $1 \leq l \neq k \leq K$, if $|G_l^*| \geq 2$ and $|G_k^*| \geq 2$, uniformly over $S \in \mathcal{S}$, with probability at least $1 - C_6/n$, where C_0 is the constant used in Lemma 23, it holds that*

$$t_{lk}^S(\dot{\lambda}^S) \gtrsim |G_l^*| |G_k^*| (\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + |S|).$$

The proof of Lemma 26 is provided in Section 11.5.8.

11.4.2 Bounding the signal-noise interaction term

We first provide a high-probability upper bound of

$$I_{kl}^S(\mathbf{v}, \mathbf{X}) = |G_l^*| \sum_{i \in G_k^*} v_i \langle (\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}, \mathcal{E}_{S \cap S^*, i} \rangle,$$

defined in (28).

Lemma 27. *Uniformly across $S \in \mathcal{S}$ and (k, l) such that $1 \leq k \neq l \leq K$, with probability at least $1 - C_9/n$, where $C_9 > 0$ is a constant, it holds that*

$$\sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} I_{kl}^S(\mathbf{v}, \mathbf{X}) \lesssim |G_l^*| \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2 \left(|G_k^*| + \sqrt{|G_k^*| \log n} + \sqrt{|G_k^*| |S| \log p} + \log n + |S| \log p \right)^{1/2}.$$

The proof of Lemma 27 is provided in Section 11.5.9. Building on this result, we establish the first partial upper bound for $L_2^S(\mathbf{X}, \dot{\lambda}^S)$, corresponding to the term $T_{1,kl}^S(\mathbf{v}) = I_{kl}^S(\mathbf{v}, \mathbf{X}) \cdot I_{lk}^S(\mathbf{v}, \mathbf{X})$, defined in (30).

Lemma 28. *Uniformly across $S \in \mathcal{S}$, with probability at least $1 - C_{11}/n$, where $C_{11} > 0$ is a constant, it holds that*

$$\sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} \sum_{1 \leq k \neq l \leq K} \frac{T_{1,kl}^S(\mathbf{v})}{t_{lk}^S(\dot{\lambda}^S)} \lesssim \dot{\lambda}^S.$$

Proof of Lemma 28 is provided in Section 11.5.10.

11.4.3 Bounding the noise-noise interaction term

Since $N_{kl}^S(\mathbf{v}, \mathbf{X})$, defined in (29), is an inner product of dependent Gaussian variables, we decompose it into a sum of inner products between independent Gaussian variables.

$$\begin{aligned}
N_{kl}^S(\mathbf{v}, \mathbf{X}) &= |G_l^*| \sum_{i \in G_k^*} v_i \langle \bar{\mathcal{E}}_{S,k}^* - \bar{\mathcal{E}}_{S,l}^*, \mathcal{E}_{S,i} \rangle \\
&= |G_l^*| \sum_{i \in G_k^*} v_i \langle \bar{\mathcal{E}}_{S,k}^*, \mathcal{E}_{S,i} \rangle - |G_l^*| \sum_{i \in G_k^*} v_i \langle \bar{\mathcal{E}}_{S,l}^*, \mathcal{E}_{S,i} \rangle \\
&= \underbrace{\frac{|G_l^*|}{|G_k^*|} \sum_{\{(i,j) \in G_k^*\}} v_i \langle \mathcal{E}_{S,i}, \mathcal{E}_{S,j} \rangle}_{:= G_{1,kl}^S(\mathbf{v})} - \underbrace{|G_l^*| \sum_{i \in G_k^*} v_i \langle \bar{\mathcal{E}}_{S,l}^*, \mathcal{E}_{S,i} \rangle}_{:= G_{2,kl}^S(\mathbf{v})},
\end{aligned}$$

We now present high-probability upper bounds for $G_{1,kl}^S(\mathbf{v})$ and $G_{2,kl}^S(\mathbf{v})$ in order. Since all terms consist purely of noise random variables, no separation condition is needed in this case.

Lemma 29. *Uniformly across $S \in [p]$ such that $|S| \leq \sqrt{p}$ and (k, l) such that $1 \leq k \neq l \leq K$, with probability at least $1 - C_{11}/n$, where $C_{11} > 0$ is a constant, it holds that*

$$\sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} G_{1,kl}^S(\mathbf{v}) \lesssim |G_l^*| \sqrt{|S|(\log n + |S| \log p)}.$$

The proof of Lemma 29 is provided in Section 11.5.11.

Lemma 30. *Uniformly across $S \in [p]$ such that $|S| \leq \sqrt{p}$ and (k, l) such that $1 \leq k \neq l \leq K$, with probability at least $1 - C_{12}/n$, where $C_{12} > 0$ is a constant, it holds that*

$$\sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} |G_{2,kl}^S(\mathbf{v})| \lesssim \sqrt{|G_l^*|} \sqrt{|S| + 2\sqrt{|S|\xi^S} + 2\xi^S} \sqrt{|G_k^*|} + \sqrt{\xi^S},$$

where $\xi^S = \log(n|S|^2 \binom{p}{|S|})$.

The proof of Lemma 30 is provided in Section 11.5.12.

Lemma 31. *Uniformly across $S \in \mathcal{S}$ and (k, l) such that $1 \leq k \neq l \leq K$, with probability at least $1 - C_{13}/n$, where $C_{13} > 0$ is a constant, it holds that*

$$\sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} \sum_{1 \leq k \neq l \leq K} \frac{T_{2,kl}^S(\mathbf{v})}{t_{lk}^S(\dot{\lambda}^S)} \lesssim \dot{\lambda}^S.$$

Proof of Lemma 31 is provided in Section 11.5.13.

11.4.4 Conclusion

The inequality regarding $T_{3,kl}^S(\mathbf{v})$ is a direct consequence of Lemmas 28, and 31, by the Cauchy-Schwarz inequality. Therefore, by combining Lemmas 25, 28, and 31, we conclude that, uniformly over $S \in \mathcal{S}$, with probability at least $1 - C_{14}/n$, where $C_{14} > 0$ is a constant, the quantity $\dot{\lambda}^S$ constructed in Lemma 25 satisfies

$$L_1^S(\mathbf{X}) + L_2^S(\mathbf{X}, \dot{\lambda}^S) \lesssim \dot{\lambda}^S \lesssim U^S(\mathbf{X}).$$

Therefore, by Corollary 22, the dual variables constructed in Lemma 19 serve as dual certificates verifying that $\mathbf{Z}^*$ is the unique optimal solution to the primal SDP problem corresponding to S , stated in (20). This completes the proof of Theorem 1.

11.5 Proof of Supporting Lemmas for the Minimax Upper Bound Proof

This section presents omitted proof of lemmas presented in the proof of Theorem 1.

11.5.1 Proof of Lemma 17

The proof proceeds in three steps:

1. Conditions (C1) and (C2) from dual feasibility,
2. Conditions (C3) and (C4) from zero dual gap,
3. Condition (C5) from the block-diagonal structure of $\mathbf{Z}^* = \sum_{k=1}^K |G_k^*|^{-1} \mathbf{1}_{G_k^*} \mathbf{1}_{G_k^*}^\top$.

Proof. We proceed by following the steps outlined above in sequence.

11.5.1.1 1. Conditions (C1) and (C2) from dual feasibility. After simplification, the Lagrangian function becomes an affine function of $\mathbf{Z}$:

$$\begin{aligned}
\mathcal{L}(\mathbf{Z}, \mathbf{Q}^S, \lambda^S, \alpha^S, \mathbf{B}^S) &:= \langle \mathbf{X}_{S,\cdot}^\top, \mathbf{X}_{S,\cdot}, \mathbf{Z} \rangle + \langle \mathbf{Q}^S, \mathbf{Z} \rangle + \lambda^S(K - \text{tr}(\mathbf{Z})) + (\alpha^S)^\top \left(\mathbf{1}_n - \frac{1}{2}(\mathbf{Z} + \mathbf{Z}^\top) \mathbf{1}_n \right) + \langle \mathbf{B}^S, \mathbf{Z} \rangle \\
&= \lambda^S K + (\alpha^S)^\top \mathbf{1}_n + \langle \mathbf{Q}^S - \mathbf{W}^S, \mathbf{Z} \rangle,
\end{aligned}$$

where we use $\text{tr}(\mathbf{Z}) = \langle \mathbf{I}_n, \mathbf{Z} \rangle$ and recall from Lemma 17 that

$$\mathbf{W}^S = \lambda^S \mathbf{I}_n - \mathbf{B}^S + \frac{1}{2} \mathbf{1}_n (\alpha^S)^\top + \frac{1}{2} \alpha^S \mathbf{1}_n^\top - \mathbf{X}_{S,\cdot}^\top \mathbf{X}_{S,\cdot}.$$

Therefore, the Lagrange dual function, shown below, can be derived analytically, as a linear function is bounded only when the slope part is zero.

$$\begin{aligned}
g(\mathbf{Q}^S, \lambda^S, \alpha^S, \mathbf{B}^S) &:= \sup_{\mathbf{Z} \in \mathbb{R}^{n \times n}} \mathcal{L}(\mathbf{Z}, \mathbf{Q}^S, \lambda^S, \alpha^S, \mathbf{B}^S) \\
&= \lambda^S K + (\alpha^S)^\top \mathbf{1}_n + \sup_{\mathbf{Z} \in \mathbb{R}^{n \times n}} \langle \mathbf{Q}^S - \mathbf{W}^S, \mathbf{Z} \rangle,
\end{aligned}$$

As a result, we have:

$$g(\mathbf{Q}^S, \lambda^S, \alpha^S, \mathbf{B}^S) = \begin{cases} \lambda^S K + (\alpha^S)^\top \mathbf{1}_n, & \mathbf{Q}^S = \mathbf{W}^S \text{ (slope is } \mathbf{0}) \\ \infty, & \text{otherwise.} \end{cases}$$

Making the following domain of the dual function as an explicit linear equality constraint, and combining it with $\mathbf{Q}^S \succeq 0$ and $\mathbf{B}^S \succeq 0$, with some abuse of terminology, we refer to the following problem as the Lagrange dual problem associated with the primal SDP problem

$$\min_{\lambda^S \in \mathbb{R}, \alpha^S \in \mathbb{R}^n, \mathbf{B}^S \in \mathbb{R}^{n \times n}} \lambda^S K + (\alpha^S)^\top \mathbf{1}_n, \quad \text{s.t.} \quad \mathbf{B}^S \succeq 0, \mathbf{W}^S \succeq 0.$$

The constraints in this optimization problem correspond to conditions (C1) and (C2).

11.5.1.2 2. Conditions (C3) and (C4) from zero dual gap. The duality gap is simply the difference between the dual and primal objective function:

$$\begin{aligned} & \lambda^S K + (\alpha^S)^\top \mathbf{1}_n - \langle \mathbf{X}_{S,\cdot}^\top, \mathbf{X}_{S,\cdot}, \mathbf{Z} \rangle \\ & \stackrel{(i)}{=} \lambda^S \text{tr}(\mathbf{Z}) + \frac{1}{2} (\alpha^S)^\top (\mathbf{Z} + \mathbf{Z}^\top) \mathbf{1}_n - \langle \mathbf{X}_{S,\cdot}^\top, \mathbf{X}_{S,\cdot}, \mathbf{Z} \rangle \\ & \stackrel{(ii)}{=} \langle \lambda^S \mathbf{I}_n + \frac{1}{2} \mathbf{1}_n (\alpha^S)^\top + \frac{1}{2} \alpha^S \mathbf{1}_n^\top - \mathbf{X}_{S,\cdot}^\top, \mathbf{X}_{S,\cdot} - \mathbf{B}^S, \mathbf{Z} \rangle + \langle \mathbf{B}^S, \mathbf{Z} \rangle \\ & = \langle \mathbf{W}^S, \mathbf{Z} \rangle + \langle \mathbf{B}^S, \mathbf{Z} \rangle \end{aligned}$$

where step (i) uses primal constraints $\text{tr}(\mathbf{Z}) = K$, $\mathbf{Z}^\top = \mathbf{Z}$ and $\mathbf{Z} \mathbf{1}_n = \mathbf{1}_n$, step (ii) uses $\text{tr}(\mathbf{Z}) = \langle \mathbf{I}_n, \mathbf{Z} \rangle$, and added and subtracted $\langle \mathbf{B}^S, \mathbf{Z} \rangle$. This duality gap is zero if $\langle \mathbf{W}^S, \mathbf{Z} \rangle = 0$ (C3) and $\langle \mathbf{B}^S, \mathbf{Z} \rangle = 0$ (C4).

11.5.1.3 3. Condition (C5) from the block-diagonal structure. Finally, we derive condition (C5), which ensures that $\mathbf{Z}^*$ is the unique solution to the primal SDP. To

this end, we first show that it is the only primal feasible point with the following block-diagonal structure: where $\mathbf{Z}_{G_k^*, G_l^*} = 0$ for all $1 \leq k \neq l \leq K$:

$$\mathbf{Z} = \begin{bmatrix} \mathbf{Z}^{(1)} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathbf{Z}^{(2)} & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{Z}^{(K)} \end{bmatrix}.$$

We use proof by contradiction. Suppose there exists a primal feasible $\mathbf{Z} \neq \mathbf{Z}^*$ with the block diagonal structure described above. By the primal constraint $\mathbf{Z}\mathbf{1}_n = \mathbf{1}_n$, each block $\mathbf{Z}^{(k)}$ satisfies $\mathbf{Z}^{(k)}\mathbf{1}_{|G_k^*|} = \mathbf{1}_{|G_k^*|}$, which implies that $(1, n_k^{-1/2}\mathbf{1}_{|G_k^*|})$ is an eigenvalue-eigenvector pair of $\mathbf{Z}^{(k)}$. Since the trace of a matrix equals the sum of its eigenvalues, this implies $\text{tr}(\mathbf{Z}^{(k)}) \geq 1$ for every k . Combining this with the overall trace constraint $\text{tr}(\mathbf{Z}) = \sum_{k=1}^K \text{tr}(\mathbf{Z}^{(k)}) = K$, we deduce that $\text{tr}(\mathbf{Z}^{(k)}) = 1$ for every k . Additionally, each $\mathbf{Z}^{(k)}$ is positive semidefinite, due to the primal constraint $\mathbf{Z} \succeq 0$, along with the Schur complement condition for the positive semidefiniteness of block matrices. Then $\mathbf{Z}^{(k)}$ must take the form $n_k^{-1}\mathbf{1}_{|G_k^*|}\mathbf{1}_{|G_k^*|}^\top$, since 1 is the only nonzero eigenvalue of $\mathbf{Z}^{(k)}$, corresponding to the eigenvector $n_k^{-1/2}\mathbf{1}_{|G_k^*|}$. Consequently, $\mathbf{Z} = \mathbf{Z}^*$, a contradiction.

Given this block diagonal structure and the condition $\text{tr}(\mathbf{B}^S \mathbf{Z}^*) = 0$, we conclude that $\mathbf{Z}^*$ is the unique solution to the primal SDP problem if (C1)-(C4) is satisfied and $\mathbf{B}_{G_k^*, G_l^*}^S > 0$ for all $1 \leq k \neq l \leq K$. This corresponds to condition (C5) and completes the proof of Lemma 17. $\square$

11.5.2 Proof of Lemma 18

The proof proceeds in three steps:

1. Preliminary computations,

2. Necessary condition for α^S ,
3. Necessary condition for $\mathbf{B}^S$.

Proof. Throughout the proof, we assume that conditions (C1)–(C5) from Lemma 17 are satisfied. Recall that true cluster membership matrix is defined as $\mathbf{Z}^* = \sum_{k=1}^K |G_k^*|^{-1} \mathbf{1}_{G_k^*} \mathbf{1}_{G_k^*}^\top$, where $\mathbf{1}_{G_k^*}$, as defined in Section 1, is the indicator vector for membership in G_k^* .

11.5.2.1 Preliminary computations. We first summarize the results for reference. The derivations will follow.

$$((\mathbf{X}_{S,\cdot}^\top \mathbf{X}_{S,\cdot}) \mathbf{1}_{G_l^*})_{G_k^*} = |G_l^*| \mathbf{X}_{S,G_k^*}^\top \bar{\mathbf{X}}_{S,l}^*, \text{ for } (k,l) \in [K]^2 \text{ (no condition required)}, \quad (35)$$

$$\mathbf{1}_{G_k^*}^\top \mathbf{W}^S \mathbf{1}_{G_k^*} = 0 \text{ for } k \in [K] \text{ (by (C2) and (C3))}, \quad (36)$$

$$\mathbf{W}^S \mathbf{1}_{G_k^*} = \mathbf{0} \text{ for } k \in [K] \text{ (by (C2) and (C3))}, \quad (37)$$

$$\mathbf{B}_{G_k^*, G_k^*}^S = \mathbf{0} \text{ for } k \in [K] \text{ (by (C4))} \quad (38)$$

$$(\mathbf{B}^S \mathbf{1}_{G_k^*})_{G_k^*} = 0 \text{ for } k \in [K] \text{ (by (C4))}, \text{ and} \quad (39)$$

$$(\mathbf{B}^S \mathbf{1}_{G_l^*})_{G_k^*} = \mathbf{B}_{G_k^*, G_l^*}^S \mathbf{1}_{|G_l^*|} \text{ for } 1 \leq k \neq l \leq K \text{ (no condition required)}. \quad (40)$$

We start with (35). For $1 \leq k \leq K$ and $1 \leq l \leq K$, using only the definition of $\mathbf{1}_{G_k^*}$ and $\mathbf{1}_{G_l^*}$ given in Section 1, we have

$$\begin{aligned} ((\mathbf{X}_{S,\cdot}^\top \mathbf{X}_{S,\cdot}) \mathbf{1}_{G_l^*})_{G_k^*} &= (\mathbf{X}_{S,\cdot}^\top \mathbf{X}_{S,\cdot})_{G_k^*, G_k^*} (\mathbf{1}_{G_l^*})_{G_k^*} + (\mathbf{X}_{S,\cdot}^\top \mathbf{X}_{S,\cdot})_{G_k^*, (G_k^*)^c} (\mathbf{1}_{G_l^*})_{(G_k^*)^c} \\ &\stackrel{(i)}{=} (\mathbf{X}_{S,\cdot}^\top \mathbf{X}_{S,\cdot})_{G_k^*, (G_k^*)^c} (\mathbf{1}_{G_l^*})_{(G_k^*)^c} \\ &= (\mathbf{X}_{S,\cdot}^\top \mathbf{X}_{S,\cdot})_{G_k^*, G_l^*} \mathbf{1}_{|G_l^*|} \\ &= (\mathbf{X}_{S, G_k^*})^\top \mathbf{X}_{S, G_l^*} \mathbf{1}_{|G_l^*|} \\ &= |G_l^*| \mathbf{X}_{S, G_k^*}^\top \bar{\mathbf{X}}_{S,l}^*. \end{aligned}$$

where step (i) uses $(\mathbf{1}_{G_l^*})_{G_k^*} = 0$.

Using the definition of $\mathbf{Z}^*$ and condition (C3), which states that $\langle \mathbf{W}^S, \mathbf{Z}^* \rangle = 0$, we obtain:

$$\langle \mathbf{W}^S, \mathbf{Z}^* \rangle = \sum_{k=1}^K \frac{1}{|G_k^*|} \langle \mathbf{W}^S, \mathbf{1}_{G_k^*} \mathbf{1}_{G_k^*}^\top \rangle = \sum_{k=1}^K \frac{1}{|G_k^*|} \mathbf{1}_{G_k^*}^\top \mathbf{W}^S \mathbf{1}_{G_k^*} = 0.$$

By the condition (C2), which states that $\mathbf{W}^S \succeq 0$, we have $\mathbf{1}_{G_k^*}^\top \mathbf{W}^S \mathbf{1}_{G_k^*} \geq 0$ for all $k = 1, \dots, K$. This along with the equality above implies (36):

$$\mathbf{1}_{G_k^*}^\top \mathbf{W}^S \mathbf{1}_{G_k^*} = 0 \text{ for all } k = 1, \dots, K.$$

Given that $\mathbf{W}^S \succeq 0$, the function $\mathbf{v} \mapsto \mathbf{v}^\top \mathbf{W}^S \mathbf{v}$ is convex and bounded below by 0. Therefore, $\mathbf{1}_{G_k^*}^\top \mathbf{W}^S \mathbf{1}_{G_k^*} = 0$ implies that $\mathbf{1}_{G_k^*}$ minimizes $\mathbf{v}^\top \mathbf{W}^S \mathbf{v}$. By the first order condition, we obtain (37)

$$\mathbf{W}^S \mathbf{1}_{G_k^*} = \mathbf{0} \text{ for all } k = 1, \dots, K.$$

By condition (C4), which states that $\langle \mathbf{B}^S, \mathbf{Z}^* \rangle = 0$, and the block-diagonal structure of $\mathbf{Z}^*$, we obtain (38):

$$\mathbf{B}_{G_k^*, G_k^*}^S = \mathbf{0},$$

which implies (39):

$$(\mathbf{B}^S \mathbf{1}_{G_k^*})_{G_k^*} = \mathbf{B}_{G_k^*, G_k^*}^S (\mathbf{1}_{G_k^*})_{G_k^*} + \mathbf{B}_{G_k^*, G_k^*}^S (\mathbf{1}_{G_k^*})_{G_k^*}^C = 0.$$

On the other hand, since the off-diagonal block $\mathbf{B}_{G_k^*, G_l^*}^S$ may be nonzero for $k \neq l$, we can only conclude as (40):

$$(\mathbf{B}^S \mathbf{1}_{G_l^*})_{G_k^*} = \mathbf{B}_{G_k^*, G_l^*}^S \mathbf{1}_{|G_l^*|}.$$

11.5.2.2 Necessary condition for α^S . Our goal is to derive condition (21). Only using the definition of $\mathbf{W}^S$ given in Lemma 17, we compute:

$$\begin{aligned}\mathbf{W}^S \mathbf{1}_{G_k^*} &= \lambda^S \mathbf{1}_{G_k^*} + \frac{1}{2} \mathbf{1}_n (\alpha^S)^\top \mathbf{1}_{G_k^*} + \frac{1}{2} \alpha^S \mathbf{1}_n^\top \mathbf{1}_{G_k^*} - (\mathbf{X}_{S,\cdot}^\top \mathbf{X}_{S,\cdot}) \mathbf{1}_{G_k^*} - \mathbf{B}^S \mathbf{1}_{G_k^*} \\ &= \lambda^S \mathbf{1}_{G_k^*} + \frac{1}{2} \left(\sum_{i \in G_k^*} \alpha_i^S \right) \mathbf{1}_n + \frac{|G_k^*|}{2} \alpha^S - (\mathbf{X}_{S,\cdot}^\top \mathbf{X}_{S,\cdot}) \mathbf{1}_{G_k^*} - \mathbf{B}^S \mathbf{1}_{G_k^*}.\end{aligned}\quad (41)$$

Then the restriction of $\mathbf{W}^S \mathbf{1}_{G_k^*}$ to the indices in G_k^* is given by:

$$(\mathbf{W}^S \mathbf{1}_{G_k^*})_{G_k^*} \stackrel{(i)}{=} \lambda^S \mathbf{1}_{|G_k^*|} + \frac{1}{2} \left(\sum_{i \in G_k^*} \alpha_i^S \right) \mathbf{1}_{|G_k^*|} + \frac{|G_k^*|}{2} \alpha_{G_k^*}^S - |G_k^*| \mathbf{X}_{S,G_k^*}^\top \bar{\mathbf{X}}_{S,k}^* \stackrel{(ii)}{=} 0, \quad (42)$$

where step (i) uses (35) and (39), which require (C4), and step (ii) uses (37), which requires (C2) and (C3). Rearranging the terms, this computation implies:

$$\alpha_{G_k^*}^S = 2 \mathbf{X}_{S,G_k^*}^\top \bar{\mathbf{X}}_{S,k}^* - \left\{ \frac{2\lambda^S}{|G_k^*|} + \frac{1}{|G_k^*|} \left(\sum_{i \in G_k^*} \alpha_i^S \right) \right\} \mathbf{1}_{|G_k^*|}. \quad (43)$$

It remains to express $\sum_{i \in G_k^*} \alpha_i^S$ on the right-hand side solely in terms of the data and λ^S .

To this end, we compute:

$$\begin{aligned}\mathbf{1}_{G_k^*}^\top \mathbf{W}^S \mathbf{1}_{G_k^*} &\stackrel{(i)}{=} \mathbf{1}_{G_k^*}^\top \left\{ \lambda^S \mathbf{1}_{G_k^*} + \frac{1}{2} \left(\sum_{i \in G_k^*} \alpha_i^S \right) \mathbf{1}_n + \frac{|G_k^*|}{2} \alpha^S - (\mathbf{X}_{S,\cdot}^\top \mathbf{X}_{S,\cdot}) \mathbf{1}_{G_k^*} - \mathbf{B}^S \mathbf{1}_{G_k^*} \right\} \\ &\stackrel{(ii)}{=} \lambda^S |G_k^*| + |G_k^*| \sum_{i \in G_k^*} \alpha_i^S - |G_k^*|^2 \|\bar{\mathbf{X}}_{S,k}^*\|_2^2 \\ &\stackrel{(iii)}{=} 0,\end{aligned}\quad (44)$$

where step (i) uses (41) step (ii) uses (38), which require (C4), and step (iii) uses (36), which require (C2) and (C3). Rearranging the terms, this computation implies:

$$\sum_{i \in G_k^*} \alpha_i^S = |G_k^*| \|\bar{\mathbf{X}}_{S,k}^*\|_2^2 - \lambda^S. \quad (45)$$

This with (43) implies:

$$\alpha_{G_k^*}^S = 2\mathbf{X}_{S,G_k^*}^\top \bar{\mathbf{X}}_{S,k}^* - \left\{ \frac{\lambda^S}{|G_k^*|} + \|\bar{\mathbf{X}}_{S,k}^*\|_2^2 \right\} \mathbf{1}_{|G_k^*|}.$$

This corresponds precisely to condition 21.

11.5.2.3 Necessary condition for $\mathbf{B}^S$. Our goal is to derive condition (22). Fix $1 \leq k \neq l \leq K$. Similar to (41), only using the definition of $\mathbf{W}^S$ given in Lemma 17, we compute:

$$\mathbf{W}^S \mathbf{1}_{G_l^*} = \lambda^S \mathbf{1}_{G_l^*} + \frac{1}{2} \left(\sum_{i \in G_l^*} \alpha_i^S \right) \mathbf{1}_n + \frac{|G_l^*|}{2} \alpha^S - (\mathbf{X}_{S,\cdot}^\top \mathbf{X}_{S,\cdot}) \mathbf{1}_{G_l^*} - \mathbf{B}^S \mathbf{1}_{G_l^*}.$$

Then, the restriction of $\mathbf{W}^S \mathbf{1}_{G_l^*}$ to the indices in G_k^* is given by:

$$(\mathbf{W}^S \mathbf{1}_{G_l^*})_{G_k^*} \stackrel{(i)}{=} \frac{1}{2} \left(\sum_{i \in G_l^*} \alpha_i^S \right) \mathbf{1}_{|G_k^*|} + \frac{|G_l^*|}{2} \alpha_{G_k^*}^S - |G_l^*| \mathbf{X}_{S,G_k^*}^\top \bar{\mathbf{X}}_{S,l}^* - \mathbf{B}_{G_k^*,G_l^*}^S \mathbf{1}_{|G_l^*|} \stackrel{(ii)}{=} 0, \quad (46)$$

where step (i) uses $(\mathbf{1}_{G_l^*})_{G_k^*} = 0$, (35), and (40), and step (ii) uses (37), which requires (C2) and (C3). Rearranging the terms, this computation implies:

$$\mathbf{B}_{G_k^*,G_l^*}^S \mathbf{1}_{|G_l^*|} = \frac{1}{2} \left(\sum_{i \in G_l^*} \alpha_i^S \right) \mathbf{1}_{|G_k^*|} + \frac{|G_l^*|}{2} \alpha_{G_k^*}^S - |G_l^*| \mathbf{X}_{S,G_k^*}^\top \bar{\mathbf{X}}_{S,l}^*. \quad (47)$$

Using (45) with k replaced by l , and (43), which require (C2) and (C3), this computation implies:

$$\begin{aligned} \mathbf{B}_{G_k^*, G_l^*}^S \mathbf{1}_{|G_l^*|} &= \left\{ \frac{|G_l^*|}{2} \|\bar{\mathbf{X}}_{S,l}^*\|_2^2 - \frac{|G_k^*| + |G_l^*|}{2|G_k^*|} \lambda^S - \frac{|G_l^*|}{2} \|\bar{\mathbf{X}}_{S,k}^*\|_2^2 \right\} \mathbf{1}_{|G_k^*|} \\ &\quad + |G_l^*| \mathbf{X}_{S,G_k^*}^\top \bar{\mathbf{X}}_{S,k}^* - |G_l^*| \mathbf{X}_{S,G_k^*}^\top \bar{\mathbf{X}}_{S,l}^*. \end{aligned}$$

This implies that, for $i \in G_k^*$, the i th row sum of $\mathbf{B}_{G_k^*, G_l^*}^S$ is expressed as:

$$\begin{aligned} (\mathbf{B}_{G_k^*, G_l^*}^S \mathbf{1}_{|G_l^*|})_i &\stackrel{(i)}{=} - \frac{|G_k^*| + |G_l^*|}{2|G_k^*|} \lambda^S + \frac{|G_l^*|}{2} (\|\bar{\mathbf{X}}_{S,l}^*\|_2^2 - 2\mathbf{X}_{S,i}^\top \bar{\mathbf{X}}_{S,l}^* + \|\mathbf{X}_{S,i}\|_2^2) \\ &\quad - \frac{|G_l^*|}{2} (\|\bar{\mathbf{X}}_{S,k}^*\|_2^2 - 2\mathbf{X}_{S,i}^\top \bar{\mathbf{X}}_{S,k}^* + \|\mathbf{X}_{S,i}\|_2^2), \\ &= - \frac{|G_k^*| + |G_l^*|}{2|G_k^*|} \lambda^S + \frac{|G_l^*|}{2} (\|\bar{\mathbf{X}}_{S,l}^* - \mathbf{X}_{S,i}\|_2^2 - \|\bar{\mathbf{X}}_{S,k}^* - \mathbf{X}_{S,i}\|_2^2), \end{aligned}$$

where step (i) adds and subtracts $\frac{|G_l^*|}{2} \|\mathbf{X}_{S,i}\|_2^2$. This concludes the derivation of condition (22) and completes the proof of Lemma 18. $\square$

Remark 32. If we construct α^S and $\mathbf{B}_{G_k^*, G_l^*}^S$ according to conditions (21) and (22), then step (ii) in (42) and (46) hold without requiring conditions (C2) and (C3), as the construction is explicitly designed to satisfy these equalities. This observation is useful in the proof of Lemma 21, presented in Section 11.5.4.

11.5.3 Proof of Lemma 20

Let $S \subset [p]$ be a support set, and consider the corresponding primal SDP problem (20). We show that constructing $\dot{\alpha}^S(\lambda^S)$, $\dot{\mathbf{B}}^S(\lambda^S)$ and $\dot{\mathbf{W}}^S(\lambda^S)$ as in Lemma 19 guarantees the following:

1. Conditions (C3) and (C4) in Lemma 17 are satisfied for any λ^S ,
2. Conditions (C1) and (C5) are satisfied for $\lambda^S < U^S(\mathbf{X})$.

Proof. For notational simplicity, we omit the notation λ^S from $\dot{\alpha}^S(\lambda^S)$, $\dot{\mathbf{B}}^S(\lambda^S)$, and $\dot{\mathbf{W}}^S(\lambda^S)$ and denote them $\dot{\alpha}^S$, $\dot{\mathbf{B}}^S$, and $\dot{\mathbf{W}}^S$, respectively.

11.5.3.1 1. Conditions (C3) and (C4). Condition (C4) follows directly from the block-diagonal structure of both $\mathbf{Z}^* = \sum_{k=1}^K |G_k^*|^{-1} \mathbf{1}_{G_k^*} \mathbf{1}_{G_k^*}^\top$ and $\dot{\mathbf{B}}^S$. To verify condition (C3), which requires that $\langle \dot{\mathbf{W}}^S, \mathbf{Z}^* \rangle = 0$, we use the definition $\mathbf{Z}^* = \sum_{k=1}^K |G_k^*|^{-1} \mathbf{1}_{G_k^*} \mathbf{1}_{G_k^*}^\top$ to expand the inner product as

$$\langle \dot{\mathbf{W}}^S, \mathbf{Z}^* \rangle = \frac{1}{|G_k^*|} \sum_{k=1}^K \mathbf{1}_{G_k^*}^\top \dot{\mathbf{W}}^S \mathbf{1}_{G_k^*}.$$

Thus, it suffices to verify that $\mathbf{1}_{G_k^*}^\top \dot{\mathbf{W}}^S \mathbf{1}_{G_k^*} = 0$ for all $k = 1, \dots, K$. Since our construction sets $\dot{\mathbf{B}}_{G_k^*, G_k^*}^S = \mathbf{0}$, we can invoke (44) to state that

$$\mathbf{1}_{G_k^*}^\top \dot{\mathbf{W}}^S \mathbf{1}_{G_k^*} = \lambda^S |G_k^*| + |G_k^*| \sum_{i \in G_k^*} (\dot{\alpha}^S(\lambda_S))_i - |G_k^*|^2 \|\bar{\mathbf{X}}_{S,k}^*\|_2^2.$$

Now using our construction of $\dot{\alpha}^S(\lambda_S)$, recalled below:

$$\dot{\alpha}_{G_k^*}^S = 2\mathbf{X}_{S,G_k^*}^\top \bar{\mathbf{X}}_{S,k}^* - \frac{\lambda^S}{|G_k^*|} \mathbf{1}_{|G_k^*|} - \|\bar{\mathbf{X}}_{S,k}^*\|_2^2 \mathbf{1}_{|G_k^*|},$$

we can conclude:

$$\mathbf{1}_{G_k^*}^\top \dot{\mathbf{W}}^S \mathbf{1}_{G_k^*} = \lambda^S |G_k^*| + 2|G_k^*|^2 \|\bar{\mathbf{X}}_{S,k}^*\|_2^2 - \lambda^S |G_k^*| - |G_k^*|^2 \|\bar{\mathbf{X}}_{S,k}^*\|_2^2 - |G_k^*|^2 \|\bar{\mathbf{X}}_{S,k}^*\|_2^2 = 0.$$

This concludes the verification of condition (C3).

11.5.3.2 2. Conditions (C1) and (C5). We now show that the condition $\lambda^S < U^S(\mathbf{X})$ ensures conditions (C1) and (C5). Since our construction sets the diagonal block $\dot{\mathbf{B}}_{G_k^*, G_k^*}^S$ as a zero matrix, it suffices to show that if $\lambda^S < U^S(\mathbf{X})$, then $\dot{\mathbf{B}}_{G_k^*, G_l^*}^S > 0$ for all

$1 \leq k \neq l \leq K$. By our construction in (24), for $i \in G_k^*$ and $j \in G_l^*$, the (i, j) th element of $\dot{\mathbf{B}}_{G_k^*, G_l^*}^S$ is

$$\frac{r_i^{(k,l,S)} r_j^{(l,k,S)}}{\sum_{j \in G_l^*} r_j^{(l,k,S)}}. \quad (48)$$

where

$$r_i^{(k,l,S)} = -\frac{|G_k^*| + |G_l^*|}{2|G_k^*|} \lambda^S + \frac{|G_l^*|}{2} (\|\bar{\mathbf{X}}_{S,l}^* - \mathbf{X}_{S,i}\|_2^2 - \|\bar{\mathbf{X}}_{S,k}^* - \mathbf{X}_{S,i}\|_2^2),$$

and

$$r_j^{(l,k,S)} = -\frac{|G_l^*| + |G_k^*|}{2|G_l^*|} \lambda^S + \frac{|G_k^*|}{2} (\|\bar{\mathbf{X}}_{S,k}^* - \mathbf{X}_{S,j}\|_2^2 - \|\bar{\mathbf{X}}_{S,l}^* - \mathbf{X}_{S,j}\|_2^2).$$

Therefore, all elements of $\dot{\mathbf{B}}_{G_k^*, G_l^*}^S$ for all $1 \leq k \neq l \leq K$ are strictly positive if, for all $1 \leq k \neq l \leq K$ and $i \in G_k^*$, the inequality $r_i^{(k,l,S)} > 0$ holds, or equivalently,

$$\lambda^S < \frac{|G_l^*||G_k^*|}{|G_k^*| + |G_l^*|} (\|\bar{\mathbf{X}}_{S,l}^* - \mathbf{X}_i\|_2^2 - \|\bar{\mathbf{X}}_{S,k}^* - \mathbf{X}_i\|_2^2).$$

This condition is precisely $\lambda^S < U^S(\mathbf{X})$, completing the proof of Lemma 20. $\square$

11.5.4 Proof of Lemma 21

The proof proceeds in four steps:

1. Reduction to uniform control of a quadratic form over a subspace (equation (49)),
2. Derivation of $L_1^S(\mathbf{X})$ (equation (51)),
3. Derivation of $t_{lk}^S(\lambda^S)$ (equation (52)),
4. Derivation of $T_{1,kl}^S$, $T_{2,kl}^S$, and $T_{3,kl}^S$ (equation (53)).

Proof. Given λ_S , we construct $\dot{\mathbf{B}}(\lambda_S)$, $\dot{\alpha}^S(\lambda_S)$, and $\dot{\mathbf{W}}^S(\lambda_S)$ according to Lemma 19 and denote them by $\dot{\alpha}^S$, $\dot{\mathbf{B}}^S$, and $\dot{\mathbf{W}}^S$, respectively, for notational simplicity.

11.5.4.1 1. Reduction to uniform control of a quadratic form over a subspace.

By Lemma 20, these constructed dual variables satisfy condition (C4) of Lemma 17. Therefore, by Remark 32, for any $k = 1, \dots, K$, we have $(\dot{\mathbf{W}}^S \mathbf{1}_{G_k^*})_{G_k^*} = \mathbf{0}$, from (21), and for all $l \neq k$, we have $(\dot{\mathbf{W}}^S \mathbf{1}_{G_k^*})_{G_l^*} = \mathbf{0}$ by (22). Therefore, we have $\dot{\mathbf{W}}^S \mathbf{1}_{G_k^*} = \mathbf{0}$ for $k = 1, \dots, K$. This implies that $\mathbf{1}_{G_1^*}, \dots, \mathbf{1}_{G_K^*}$ are eigenvectors of $\dot{\mathbf{W}}^S$ corresponding to zero eigenvalues. Since any vector in $\mathbb{R}^n$ can be decomposed into the sum of its projections onto $\text{span}(\mathbf{1}_{G_k^*} : k \in [K])$ and its orthogonal complement, denoted as Γ_K , to show that $\dot{\mathbf{W}}^S \succeq 0$, it suffices to prove that

$$\mathbf{v}^\top \dot{\mathbf{W}}^S \mathbf{v} \geq 0 \quad \text{for all } \mathbf{v} \in \Gamma_K \text{ such that } \|\mathbf{v}\|_2 = 1. \quad (49)$$

11.5.4.2 2. Derivation of $L_1^S(\mathbf{X})$. By the definition of Γ_k , $\mathbf{v} \in \Gamma_k$ implies $\mathbf{v}^\top \mathbf{1}_{G_1^*} = \mathbf{v}^\top \mathbf{1}_{G_2^*} = \dots = \mathbf{v}^\top \mathbf{1}_{G_K^*} = 0$, which implies

$$\sum_{j \in G_k^*} v_j = \sum_{i=1}^n v_i = 0 \quad \text{for all } k = 1, \dots, K. \quad (50)$$

Now we compute the quadratic form $\mathbf{v}^\top \dot{\mathbf{W}}^S \mathbf{v}$ for $\mathbf{v} \in \Gamma_K$ such that $\|\mathbf{v}\|_2 = 1$:

$$\begin{aligned} \mathbf{v}^\top \dot{\mathbf{W}}^S \mathbf{v} &\stackrel{(i)}{=} \lambda^S \mathbf{v}^\top \mathbf{I}_n \mathbf{v} + \frac{1}{2} \mathbf{v}^\top \mathbf{1}_n (\alpha^S)^\top \mathbf{v} + \frac{1}{2} \mathbf{v}^\top \alpha^S \mathbf{1}_n^\top \mathbf{v} - \mathbf{v}^\top \mathbf{X}_{S,\cdot}^\top \mathbf{X}_{S,\cdot} \mathbf{v} - \mathbf{v}^\top \dot{\mathbf{B}}^S \mathbf{v} \\ &\stackrel{(ii)}{=} \lambda^S + (\mathbf{v}^\top \alpha_S) \sum_{i=1}^n v_i - \|\mathbf{X}_{S,\cdot} \mathbf{v}\|_2^2 - \mathbf{v}^\top \dot{\mathbf{B}}^S \mathbf{v} \\ &\stackrel{(ii)}{=} \lambda^S - \|\mathbf{X}_{S,\cdot} \mathbf{v}\|_2^2 - \mathbf{v}^\top \dot{\mathbf{B}}^S \mathbf{v}, \end{aligned}$$

where step (i) uses the construction $\dot{\mathbf{W}}^S = \lambda^S \mathbf{I}_n + \frac{1}{2} \mathbf{1}_n (\dot{\alpha}^S)^\top + \frac{1}{2} \dot{\alpha}^S \mathbf{1}_n^\top - \mathbf{X}_{S,\cdot}^\top \mathbf{X}_{S,\cdot} - \dot{\mathbf{B}}^S$, step (ii) uses $\mathbf{v}^\top \mathbf{I}_n \mathbf{v} = \|\mathbf{v}\|^2 = 1$, and step (ii) uses (50). Using our model and notation in

Definition 16,

$$\mathbf{X}_{S,\mathbf{v}} = \sum_{i=1}^n v_i \mathbf{X}_{S,i} = \sum_{k=1}^K \left(\sum_{i \in G_k^*} v_i (\mathbf{c}_k)_S + \sum_{i=1}^n v_i \mathcal{E}_{S,i} \right) \stackrel{(i)}{=} \sum_{i=1}^n v_i \mathcal{E}_{S,i} = \mathcal{E}_{S,\mathbf{v}},$$

where step (i) uses (50). Therefore, a sufficient condition for (49) is:

$$\lambda^S \geq \underbrace{\sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} \|\mathcal{E}_{S,\mathbf{v}}\|_2^2}_{L_1^S(\mathbf{X})} + \sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} \mathbf{v}^\top \dot{\mathbf{B}}^S \mathbf{v}. \quad (51)$$

This completes the derivation of $L_1^S(\mathbf{X})$.

11.5.4.3 3. Derivation of $t_{lk}^S(\lambda^S)$. Now we start to derive $L_2^S(\mathbf{X}, \lambda^S)$. Recall from Lemma 19 that given λ^S , we construct $\dot{\mathbf{B}}$ as a block-diagonal matrix, where the diagonal blocks are zero matrices and for $1 \leq k \neq l \leq K$, the (k, l) th off-diagonal block is defined as:

$$\dot{\mathbf{B}}_{G_k^*, G_l^*}^S = \frac{1}{\mathbf{1}_{|G_l^*|}^\top \mathbf{r}^{(l,k,S)}} \mathbf{r}^{(k,l,S)} (\mathbf{r}^{(l,k,S)})^\top,$$

with $\mathbf{r}^{(k,l,S)} \in \mathbb{R}^{|G_k^*|}$ defined componentwise by

$$r_i^{(k,l,S)} = -\frac{|G_k^*| + |G_l^*|}{2|G_k^*|} \lambda^S + \frac{|G_l^*|}{2} (\|\bar{\mathbf{X}}_{S,l}^* - \mathbf{X}_{S,i}\|_2^2 - \|\bar{\mathbf{X}}_{S,k}^* - \mathbf{X}_{S,i}\|_2^2).$$

Using this definition, we compute $\mathbf{v}^\top \dot{\mathbf{B}}^S \mathbf{v}$ in (51):

$$\mathbf{v}^\top \dot{\mathbf{B}}^S \mathbf{v} \stackrel{(i)}{=} \sum_{1 \leq k \neq l \leq K} \mathbf{v}_{G_k^*}^\top \dot{\mathbf{B}}_{G_k^*, G_l^*}^S \mathbf{v}_{G_l^*} \stackrel{(ii)}{=} \sum_{1 \leq k \neq l \leq K} \sum_{i \in G_k^*} \sum_{j \in G_l^*} \frac{(\mathbf{v}_{G_k^*}^\top \mathbf{r}^{(k,l,S)}) (\mathbf{v}_{G_l^*}^\top \mathbf{r}^{(l,k,S)})}{\underbrace{\mathbf{1}_{|G_l^*|}^\top \mathbf{r}^{(l,k,S)}}_{=t_{lk}^S(\lambda^S)}}, \quad (52)$$

where step (i) uses $\dot{\mathbf{B}}_{G_k^*, G_k^*}^S = \mathbf{0}$ for $k = 1, \dots, K$, and step (ii) uses our construction of

$$\dot{\mathbf{B}}_{G_k^*, G_l^*}^S.$$

11.5.4.4 4. Derivation of $T_{1,kl}^S$, $T_{2,kl}^S$, and $T_{3,kl}^S$. From (52), we analyze one term in the numerator:

$$\begin{aligned}
& \mathbf{v}_{G_k^*}^\top \mathbf{r}^{(k,l,S)} \\
&= \sum_{i \in G_k^*} v_i r_i^{(k,l,S)} \\
&= \sum_{i \in G_k^*} v_i \left\{ -\frac{|G_k^*| + |G_l^*|}{2|G_k^*|} \lambda^S + \frac{|G_l^*|}{2} (\|\bar{\mathbf{X}}_{S,l}^* - \mathbf{X}_{S,i}\|_2^2 - \|\bar{\mathbf{X}}_{S,k}^* - \mathbf{X}_{S,i}\|_2^2) \right\} \\
&\stackrel{(i)}{=} \frac{|G_l^*|}{2} \sum_{i \in G_k^*} v_i (\|\bar{\mathbf{X}}_{S,l}^* - \mathbf{X}_{S,i}\|_2^2 - \|\bar{\mathbf{X}}_{S,k}^* - \mathbf{X}_{S,i}\|_2^2) \\
&= \frac{|G_l^*|}{2} \sum_{i \in G_k^*} v_i (\|\bar{\mathbf{X}}_{S,l}^*\|_2^2 + \|\mathbf{X}_{S,i}\|_2^2 - 2\langle \bar{\mathbf{X}}_{S,l}^*, \mathbf{X}_{S,i} \rangle - \|\bar{\mathbf{X}}_{S,k}^*\|_2^2 - \|\mathbf{X}_{S,i}\|_2^2 + 2\langle \bar{\mathbf{X}}_{S,k}^*, \mathbf{X}_{S,i} \rangle) \\
&\stackrel{(ii)}{=} |G_l^*| \sum_{i \in G_k^*} v_i \langle \bar{\mathbf{X}}_{S,k}^* - \bar{\mathbf{X}}_{S,l}^*, \mathbf{X}_{S,i} \rangle \\
&\stackrel{(iii)}{=} |G_l^*| \sum_{i \in G_k^*} v_i \langle (\mathbf{c}_k^* - \mathbf{c}_l^*)_S + \bar{\mathcal{E}}_{S,k}^* - \bar{\mathcal{E}}_{S,l}^*, \mathcal{E}_{S,i} \rangle \\
&= |G_l^*| \sum_{i \in G_k^*} v_i \langle (\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}, \mathcal{E}_{S \cap S^*,i} \rangle + |G_l^*| \sum_{i \in G_k^*} v_i \langle \bar{\mathcal{E}}_{S,k}^* - \bar{\mathcal{E}}_{S,l}^*, \mathcal{E}_{S,i} \rangle \\
&= I_{kl}^S(\mathbf{v}, \mathbf{X}) + N_{kl}^S(\mathbf{v}, \mathbf{X})
\end{aligned}$$

where step (i) and (ii) use $\sum_{i \in G_k^*} v_i = 0$ for $k = 1, \dots, K$, and step (iii) comes from our model and notation in Definition 16. Similarly, we have

$$\mathbf{v}_{G_l^*}^\top \mathbf{r}^{(l,k,S)} = I_{lk}^S(\mathbf{v}, \mathbf{X}) + N_{lk}^S(\mathbf{v}, \mathbf{X}).$$

Therefore, we have

$$\begin{aligned}
\mathbf{v}^\top \dot{\mathbf{B}}^S \mathbf{v} &= \sum_{1 \leq k \neq l \leq K} \sum_{i \in G_k^*} \sum_{j \in G_l^*} \frac{1}{t_{lk}^S(\lambda^S)} (I_{kl}^S(\mathbf{v}, \mathbf{X}) + N_{kl}^S(\mathbf{v}, \mathbf{X})) (I_{lk}^S(\mathbf{v}, \mathbf{X}) + N_{lk}^S(\mathbf{v}, \mathbf{X})) \\
&= \sum_{1 \leq k \neq l \leq K} \sum_{i \in G_k^*} \sum_{j \in G_l^*} \frac{1}{t_{lk}^S(\lambda^S)} (T_{1,kl}^S(\mathbf{v}) + T_{2,kl}^S(\mathbf{v}) + T_{3,kl}^S(\mathbf{v})). \tag{53}
\end{aligned}$$

Plugging this into (51) completes the proof of Lemma 21.

□

11.5.5 Proof of Lemma 23

For notational simplicity, we assume $\sigma^2 = 1$ here. The proof proceeds in the following steps:

1. Decomposition of $D_{kli}^S(\mathbf{X})$ into Chi-squares, Gaussian and inner product of Gaussians,
2. Bounding the Gaussian and Chi-squares using the tail bounds,
3. Bounding the Gaussian inner product using conditioning, Gaussian tail bounds and separation condition,
4. Defining the uniform bound event,
5. Bounding the probability of the uniform bound event.

Proof. We proceed by following the steps outlined above in sequence.

11.5.5.1 1. Decomposition of $D_{kli}^S(\mathbf{X})$ into Chi-squares, Gaussian and inner product of Gaussians.

Fix indices $1 \leq k \neq l \leq K$ and choose $i \in G_k^*$. Based on our model and notation in Definition 16, we consider the following decomposition of $D_{kli}^S(\mathbf{X})$:

$$\begin{aligned}
D_{kli}^S(\mathbf{X}) &= \|\mathbf{X}_{S,i} - \bar{\mathbf{X}}_{S,l}^*\|_2^2 - \|\mathbf{X}_{S,i} - \bar{\mathbf{X}}_{S,k}^*\|_2^2 \\
&= \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_S + \mathcal{E}_{S,i} - \bar{\mathcal{E}}_{S,l}^*\|_2^2 - \|\mathcal{E}_{S,i} - \bar{\mathcal{E}}_{S,k}^*\|_2^2 \\
&= \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_S + \mathcal{E}_{S,i} - \bar{\mathcal{E}}_{S,l}^*\|_2^2 - \left(\frac{|G_k^*| - 1}{|G_k^*|}\right)^2 \|\mathcal{E}_{S,i} - \bar{\mathcal{E}}_{S,k \setminus \{i\}}^*\|_2^2 \\
&\stackrel{(i)}{=} \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + \|\mathcal{E}_{S,i}\|_2^2 + \|\bar{\mathcal{E}}_{S,l}^*\|_2^2 + 2\langle (\mathbf{c}_k^* - \mathbf{c}_l^*)_S, \mathcal{E}_{S,i} \rangle \\
&\quad - 2\langle (\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}, \bar{\mathcal{E}}_{S \cap S^*, l}^* \rangle - 2\langle \bar{\mathcal{E}}_{S,l}^*, \mathcal{E}_{S,i} \rangle \\
&\quad - \left(\frac{|G_k^*| - 1}{|G_k^*|}\right)^2 (\|\mathcal{E}_{S,i}\|_2^2 + \|\bar{\mathcal{E}}_{S,k \setminus \{i\}}^*\|_2^2 - 2\langle \mathcal{E}_{S,i}, \bar{\mathcal{E}}_{S,k \setminus \{i\}}^* \rangle) \\
&= \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + \|\bar{\mathcal{E}}_{S,l}^*\|_2^2 + \frac{2|G_k^*| - 1}{|G_k^*|^2} \|\mathcal{E}_{S,i}\|_2^2 \\
&\quad - \left(\frac{|G_k^*| - 1}{|G_k^*|}\right)^2 \|\bar{\mathcal{E}}_{S,k \setminus \{i\}}^*\|_2^2 - 2\langle (\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}, \bar{\mathcal{E}}_{S \cap S^*, l}^* \rangle \\
&\quad + 2\left\langle (\mathbf{c}_k^* - \mathbf{c}_l^*)_S - \bar{\mathcal{E}}_{S,l}^* + \left(\frac{|G_k^*| - 1}{|G_k^*|}\right)^2 \bar{\mathcal{E}}_{S,k \setminus \{i\}}^*, \mathcal{E}_{S,i} \right\rangle, \tag{54}
\end{aligned}$$

where step (i) uses the sparsity assumption on cluster center difference.

The six terms in this decomposition are classified into the following categories:

- Signal: $\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2$
- Chi-square noise:

$$\begin{aligned}
\text{CN}_l^S &:= \|\bar{\mathcal{E}}_{S,l}^*\|_2^2, \\
\text{CN}_i^S &:= \frac{2|G_k^*| - 1}{|G_k^*|^2} \|\mathcal{E}_{S,i}\|_2^2, \text{ and} \\
\text{CN}_{ki}^S &:= -\left(\frac{|G_k^*| - 1}{|G_k^*|}\right)^2 \|\bar{\mathcal{E}}_{S,k \setminus \{i\}}^*\|_2^2
\end{aligned}$$

- Gaussian signal-noise interaction:

$$\text{GSN}_{kl}^S := -2\langle (\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}, \bar{\mathcal{E}}_{S \cap S^*, l}^* \rangle$$

- Inner product of Gaussians:

$$\text{IP}_{kli}^S := 2 \left\langle (\mathbf{c}_k^* - \mathbf{c}_l^*)_S - \bar{\mathcal{E}}_{S, l}^* + \left(\frac{|G_k^*| - 1}{|G_k^*|} \right)^2 \bar{\mathcal{E}}_{S, k \setminus \{i\}}^*, \mathcal{E}_{S, i} \right\rangle,$$

In summary, we have:

$$D_{kli}^S(\mathbf{X}) = \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + \text{CN}_l^S + \text{CN}_i^S + \text{CN}_{ki}^S + \text{GSN}_{kl}^S + \text{IP}_{kli}^S. \quad (55)$$

11.5.5.2 2. Bounding the Gaussian and Chi-squares using the tail inequalities. Chi-square noise and Gaussian signal-noise interaction terms ($\text{CN}_l^S, \text{CN}_i^S, \text{CN}_{ki}^S$, and GSN_{kl}^S) in (55) are straightforward to bound from below. Define the following lower bound events:

$$\mathcal{B}_{kl, S}^{(i, 1)} := \left\{ \text{CN}_l^S \geq \frac{|S| - 2\sqrt{|S|\zeta^S}}{|G_l^*|} \right\}, \quad (56)$$

$$\mathcal{B}_{kl, S}^{(i, 2)} := \left\{ \text{CN}_i^S \geq \frac{2|G_k^*| - 1}{|G_k^*|^2} (|S| - 2\sqrt{|S|\zeta^S}) \right\}, \quad (57)$$

$$\mathcal{B}_{kl, S}^{(i, 3)} := \left\{ \text{CN}_{ki}^S \geq \frac{1 - |G_k^*|}{|G_k^*|^2} (|S| + 2\sqrt{|S|\zeta^S} + 2\zeta^S) \right\}, \quad (58)$$

$$\mathcal{B}_{kl, S}^{(i, 4)} := \left\{ \text{GSN}_{kl}^S \geq -2\sqrt{\frac{2\zeta^S}{|G_l^*|}} \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2 \right\}, \quad (59)$$

By setting $\zeta^S = \log(n^2|S|^2 \binom{p}{|S|})$, Lemma 7 implies that each event has a probability of at least $1 - 1/(n^2 K^2 |S|^2 \binom{p}{|S|})$, for any $S \in \mathcal{S}$.

11.5.5.3 3. Bounding the Gaussian inner product using conditioning, Gaussian tail bounds and separation condition. Here, we use the separation condition to

bound the term. If we condition on $\bar{\mathcal{E}}_{S,l}^*$ and $\bar{\mathcal{E}}_{S,k \setminus \{i\}}^*$, then $-\text{IP}_{kli}^S$ has a univariate Gaussian distribution with mean zero and variance as the following:

$$\begin{aligned} \text{Var}(-\text{IP}_{kli}^S \mid \bar{\mathcal{E}}_{S,l}^*, \bar{\mathcal{E}}_{S,k \setminus \{i\}}^*) &= \left\| (\mathbf{c}_k^* - \mathbf{c}_l^*)_S - \bar{\mathcal{E}}_{S,l}^* + (1 - |G_k^*|^{-1})^2 \bar{\mathcal{E}}_{S,k \setminus \{i\}}^* \right\|_2^2 \\ &\stackrel{(i)}{=} \underbrace{\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + \left\| \bar{\mathcal{E}}_{S,l}^* - (1 - |G_k^*|^{-1})^2 \bar{\mathcal{E}}_{S,k \setminus \{i\}}^* \right\|_2^2}_{:= \text{CN}_{kli}^S} \\ &\quad + 2 \underbrace{\langle (\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}, (1 - |G_k^*|^{-1})^2 \bar{\mathcal{E}}_{S \cap S^*, k \setminus \{i\}}^* - \bar{\mathcal{E}}_{S \cap S^*, l}^* \rangle}_{:= \text{GN}_{kli}^S}, \end{aligned}$$

where step (i) uses our sparsity assumption on cluster center difference.

Next, we define events that bound this variance from above:

$$\begin{aligned} \mathcal{B}_{kl,S}^{(i,5)} &:= \left\{ \text{CN}_{kli}^S \leq (|G_l^*|^{-1} + |G_k^*|^{-1})(|S| + 2\sqrt{|S|\zeta^S} + 2\zeta^S) \right\}, \\ \mathcal{B}_{kl,S}^{(i,6)} &:= \left\{ \text{GN}_{kli}^S \leq 2\sqrt{2(|G_l^*|^{-1} + |G_k^*|^{-1})\zeta^S} \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2 \right\}. \end{aligned}$$

By setting $\zeta^S = \log(n^2|S|^2 \binom{p}{|S|})$, Lemma 7 implies that each event has a probability of at least $1 - 1/(n^2|S|^2 \binom{p}{|S|})$, uniformly overall all $S \in \mathcal{S}$.

Thus, conditioned on events $\mathcal{B}_{kl,S}^{(i,5)}$ and $\mathcal{B}_{kl,S}^{(i,6)}$, the variance can be bounded as

$$\begin{aligned} \text{Var}(-\text{IP}_{kli}^S \mid \bar{\mathcal{E}}_{S,l}^*, \bar{\mathcal{E}}_{S,k \setminus \{i\}}^*) &\leq \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + (|G_l^*|^{-1} + |G_k^*|^{-1})(|S| + 2\sqrt{|S|\zeta^S} + 2\zeta^S) \\ &\quad + 2\sigma^2 \sqrt{2(|G_l^*|^{-1} + |G_k^*|^{-1})\zeta^S} \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2. \end{aligned}$$

To further bound this variance, it is convenient to remove the first-order signal term $\|(\mathbf{c}_k^* -$

$\mathbf{c}_l^*)_{S \cap S^*}\|_2$. To this end, we collect some conditions. First, recall from Theorem 1 that

$$m = 2 \min_{1 \leq k \neq l \leq K} (|G_l^*|^{-1} + |G_k^*|^{-1})^{-1}.$$

From (10), recall that for $S \in \mathcal{S}$, we have

$$\min_{1 \leq k \neq l \leq K} \|(\mathbf{c}_l^* - \mathbf{c}_k^*)_{S \cap S^*}\|_2^2 \gtrsim \sigma^2 \left(\log n + \frac{|S| \log p}{m} + \sqrt{\frac{|S| \log p}{m}} \right), \quad (60)$$

and $|S| \leq \sqrt{p}$. Under these conditions, by treating K as a constant, we have

$$\zeta^S = \log(n^2 |S|^2 \binom{p}{|S|}) \lesssim |S| \log p + \log n,$$

which implies

$$2\sqrt{2(|G_l^*|^{-1} + |G_k^*|^{-1})\zeta^S} \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2 \leq 2\sqrt{\frac{\zeta^S}{m}} \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2 \lesssim \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2.$$

Therefore, we can further bound the variance as

$$\text{Var}(-\text{IP}_{kli}^S) \lesssim \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + (|G_l^*|^{-1} + |G_k^*|^{-1})(|S| + 2\sqrt{|S|\zeta^S} + 2\zeta^S).$$

Therefore, by the Gaussian tail inequality, we have

$$\begin{aligned} \text{CP}_{kli}^S &:= \mathbb{P} \left(-\text{IP}_{kli}^S > \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 \mid \bar{\mathcal{E}}_{S,l}^*, \bar{\mathcal{E}}_{S,k \setminus \{i\}}^* \right) \\ &\leq \exp \left(\frac{-\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^4}{2\text{Var}(-\text{IP}_{kli}^S \mid \bar{\mathcal{E}}_{S,l}^*, \bar{\mathcal{E}}_{S,k \setminus \{i\}}^*)} \right) \\ &\lesssim \exp \left(\frac{-\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^4}{\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + 2(|G_l^*|^{-1} + |G_k^*|^{-1})(|S| + 2\sqrt{|S|\zeta^S} + 2\zeta^S)} \right), \quad (61) \end{aligned}$$

where the last step uses the variance bound. We want to bound the last term by C/n^2 ,

with C a constant. To this end, it suffices to show

$$\frac{\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^4}{\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|^2 + 2(|G_l^*|^{-1} + |G_k^*|^{-1})(|S| + 2\sqrt{|S|\zeta^S} + 2\zeta^S)} \gtrsim \log n,$$

which is indeed satisfied under our separation condition in (60).

11.5.5.4 4. Defining the uniform bound event. We collect the bounds in (56), (57), (58), (59), and (61). From (55), recalled below,

$$D_{kli}^S(\mathbf{X}) = \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + \text{CN}_l^S + \text{CN}_i^S + \text{CN}_{ki}^S + \text{GSN}_{kl}^S + \text{IP}_{kli}^S.$$

we can say

$$\begin{aligned} D_{kli}^S(\mathbf{X}) &\gtrsim \|(\mathbf{c}_l^* - \mathbf{c}_k^*)_{S \cap S^*}\|_2^2 + \frac{|S| - 2\sqrt{|S|\zeta^S}}{|G_l^*|} + \frac{2|G_k^*| - 1}{|G_k^*|^2}(|S| - 2\sqrt{|S|\zeta^S}) \\ &\quad + \frac{1 - |G_k^*|}{|G_k^*|^2}(|S| + 2\sqrt{|S|\zeta^S} + 2\zeta^S) - 2\sqrt{\frac{2\zeta^S}{|G_l^*|}}\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2 \\ &\quad + \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2, \\ &\gtrsim \|(\mathbf{c}_l^* - \mathbf{c}_k^*)_{S \cap S^*}\|_2^2 - \sqrt{\frac{\zeta^S}{|G_l^*|}}\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2 \\ &\quad - \left(\frac{1}{|G_k^*|} + \frac{1}{|G_l^*|}\right)|S| - \left(\frac{1}{|G_k^*|} + \frac{1}{|G_l^*|}\right)\sqrt{|S|\zeta^S} - \frac{1}{|G_k^*|}\zeta^S \\ &= \|(\mathbf{c}_l^* - \mathbf{c}_k^*)_{S \cap S^*}\|_2^2 - \left(\frac{1}{|G_k^*|} + \frac{1}{|G_l^*|}\right)|S| - R_{kl}^S, \end{aligned}$$

where

$$R_{kl}^S = \sqrt{\frac{\zeta^S}{|G_l^*|}}\|(\mathbf{c}_l^* - \mathbf{c}_k^*)_{S \cap S^*}\|_2 + \left(\frac{1}{|G_k^*|} + \frac{1}{|G_l^*|}\right)\sqrt{|S|\zeta^S} + \frac{1}{|G_k^*|}\zeta^S.$$

As a result, we define an event

$$\mathcal{A}_{kli}^S := \left\{ D_{kli}^S(\mathbf{X}) \gtrsim \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + \left(\frac{1}{|G_k^*|} + \frac{1}{|G_l^*|} \right) |S| - R_{kl}^S \right\}, \quad (62)$$

with the index (k, l, i) ranging $1 \leq k \neq l \leq K$ and all $i \in G_k^*$, and S ranging over $\mathcal{S}$. The proof is complete once we show that the intersection event

$$\bigcap_{1 \leq k \neq l \leq K} \bigcap_{i \in G_k^*} \bigcap_{S \in \mathcal{S}} \mathcal{A}_{kli}^S$$

holds with high probability.

11.5.5.5 5. Bounding the probability of the uniform bound event. To prove the lemma, it suffices to show that

$$\mathbb{P} \left(\bigcup_{1 \leq k \neq l \leq K} \bigcup_{i \in G_k^*} \bigcup_{S \in \mathcal{S}} (\mathcal{A}_{kli}^S)^C \right) \lesssim \frac{1}{n}.$$

To show this, first denote

$$\mathcal{B}_{kl,S}^{(i)} := \bigcap_{\ell=1}^6 \mathcal{B}_{kl,S}^{(i,\ell)}.$$

Then, we have

$$\begin{aligned}
& \mathbb{P}\left(\bigcup_{1 \leq k \neq l \leq K} \bigcup_{i \in G_k^*} \bigcup_{S \in \mathcal{S}} (\mathcal{A}_{kli}^S)^C\right) \\
& \leq \sum_{1 \leq k \neq l \leq K} \sum_{i \in G_k^*} \sum_{S \in \mathcal{S}} \mathbb{P}((\mathcal{A}_{kli}^S)^C) \\
& = \sum_{1 \leq k \neq l \leq K} \sum_{i \in G_k^*} \sum_{S \in \mathcal{S}} \mathbb{E}[\mathbb{P}((\mathcal{A}_{kli}^S)^C \mid \bar{\mathcal{E}}_{S,l}^*, \mathcal{E}_{S,i}, \bar{\mathcal{E}}_{S,k \setminus \{i\}})] \\
& \leq \sum_{1 \leq k \neq l \leq K} \sum_{i \in G_k^*} \sum_{S \in \mathcal{S}} \mathbb{E}[\mathbb{P}((\mathcal{A}_{kli}^S)^C \mid \bar{\mathcal{E}}_{S,l}^*, \mathcal{E}_{S,i}, \bar{\mathcal{E}}_{S,k \setminus \{i\}}) \mathbb{1}(\mathcal{B}_{kl,S}^{(i)}) + \mathbb{1}(\mathcal{B}_{kl,S}^{(i)})^C] \\
& \stackrel{(i)}{\leq} \left(\sum_{1 \leq k \neq l \leq K} \sum_{i \in G_k^*} \sum_{q=1}^{\lfloor \sqrt{p} \rfloor} \sum_{S \in \mathcal{S}, |S|=q} \mathbb{E}[\text{CP}_{kli}^S \mathbb{1}(\mathcal{B}_{kl,S}^{(i)})] \right) + \frac{\pi^2}{n} \\
& \stackrel{(ii)}{\lesssim} \frac{K^2 \pi^2}{6n} + \frac{K^2 \pi^2}{n},
\end{aligned}$$

where step (i) uses $\sum_{q=1}^{\infty} (1/q^2) = \pi^2/6$, and step (ii) uses (61). Treating K as a constant, we conclude that

$$D_{kli}^S(\mathbf{X}) \gtrsim \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + \left(\frac{1}{|G_k^*|} + \frac{1}{|G_l^*|} \right) |S| - R_{kl}^S.$$

uniformly over $S \in \mathcal{S}$, with probability at least $1 - C/n$, where $C > 0$ is a constant. This completes the proof of Lemma 23.

□

11.5.6 Proof of Lemma 24

Proof. Given a support set $S \subset [p]$ with $|S| \leq \sqrt{p}$, we have

$$L_1^S(\mathbf{X}) = \sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} \|\mathcal{E}_{S,\cdot} \mathbf{v}\|_2^2 \leq \sup_{\mathbf{v} \in \mathbb{R}^n, \|\mathbf{v}\|_2=1} \|\mathcal{E}_{S,\cdot} \mathbf{v}\|_2^2 = \|\mathcal{E}_{S,\cdot}\|_{\text{op}}^2.$$

Here, $\mathcal{E}_{S,\cdot}$ is a $|S| \times n$ random matrix with i.i.d. Gaussian entries distributed as $\mathcal{N}(0, \sigma^2)$.

Therefore, by Lemma 11, if we define an event

$$\mathcal{C}_L^S := \{L_1^S(\mathbf{X}) \leq \sigma^2(n + |S| + \xi^S)\}, \quad (63)$$

where $\xi^S = \log(n|S|^2 \binom{p}{|S|})$, then we have $\mathbb{P}(\mathcal{C}_L^S) \geq 1 - 1/(n|S|^2 \binom{p}{|S|})$. Now we show that the event $\mathcal{C}_L^S$ holds uniformly across all $S \subset [p]$ such that $|S| \leq \sqrt{p}$, with probability at least $1 - C_7/n$, where $C_7 > 0$ is a constant.

$$\mathbb{P}\left(\bigcup_{S \subset [p], |S| \leq \sqrt{p}} (\mathcal{C}_L^S)^c\right) \leq \sum_{q=1}^{\lfloor \sqrt{p} \rfloor} \sum_{|S|=q} \mathbb{P}((\mathcal{C}_L^S)^c) \leq \sum_{q=1}^{\lfloor \sqrt{p} \rfloor} \sum_{|S|=q} \frac{1}{n|S|^2 \binom{p}{|S|}} = \sum_{q=1}^{\lfloor \sqrt{p} \rfloor} \frac{1}{nq^2} \leq \frac{\pi^2}{6n},$$

where the last step uses $\sum_{q=1}^{\infty} \frac{1}{q^2} = \pi^2/6$. Since $|S| \leq \sqrt{p}$, $|S| \log p$ dominates $\log |S|$. Therefore, we conclude that uniformly across $S \in [p]$ such that $|S| \leq \sqrt{p}$, with probability at least $1 - C_7/n$, it holds that

$$L_1^S(\mathbf{X}) \lesssim \sigma^2(n + \log n + |S| \log p).$$

This completes the proof of Lemma 24. □

11.5.7 Proof of Lemma 25

The proof proceeds in the following steps:

1. The first inequality $L_1^S(\mathbf{X}) \lesssim \dot{\lambda}^S$.
2. The second inequality $\dot{\lambda}^S \lesssim U^S(\mathbf{X})$.

Proof. For simplicity, we assume $\sigma^2 = 1$ here.

11.5.7.1 1. The first inequality. We aim to show $L_1^S(\mathbf{X}) \lesssim \dot{\lambda}^S$. From (10), recall that for $S \in \mathcal{S}$, we have

$$\Delta_{S \cap S^*}^2 = \min_{1 \leq k \neq l \leq K} \|(\mathbf{c}_l^* - \mathbf{c}_k^*)_{S \cap S^*}\|_2^2 \gtrsim \sigma^2 \left(\log n + \frac{|S| \log p}{m} + \sqrt{\frac{|S| \log p}{m}} \right), \quad (64)$$

where

$$m = 2 \min_{1 \leq k \neq l \leq K} \left\{ \left(\frac{1}{|G_k^*|} + \frac{1}{|G_l^*|} \right)^{-1} \right\}.$$

Therefore, with probability at least $1 - C_7/n$ uniformly across all $S \in \mathcal{S}$, where $C_7 > 0$ is a constant from Lemma 24, it holds that

$$\begin{aligned} L_1^S(\mathbf{X}) &\lesssim (n + \log n + |S| \log p) \\ &\stackrel{(i)}{\lesssim} |S| + \frac{1}{4} (m \log n + |S| \log p + \sqrt{m|S| \log p}) \\ &\lesssim |S| + \frac{m}{4} \Delta_{S \cap S^*}^2 \\ &= \dot{\lambda}^S, \end{aligned}$$

where step (i) uses Lemma 24. This completes the derivation of the first inequality.

11.5.7.2 2. The second inequality. We aim to show $\dot{\lambda}^S \lesssim U^S(\mathbf{X})$. Recall from (26) that

$$U^S(\mathbf{X}) = \min_{1 \leq k \neq l \leq K} \left\{ \left(\frac{1}{|G_k^*|} + \frac{1}{|G_l^*|} \right)^{-1} \min_{i \in G_k^*} D_{kli}^S(\mathbf{X}) \right\}.$$

By Lemma 23, with probability at least $1 - C_6/n$ uniformly across $S \in \mathcal{S}$ and uniformly over (k, l, i) where $1 \leq k \neq l \leq K$ and $i \in G_k^*$, we have

$$\left(\frac{1}{|G_k^*|} + \frac{1}{|G_l^*|} \right)^{-1} D_{kli}^S(\mathbf{X}) \gtrsim \underbrace{\left(\frac{1}{|G_k^*|} + \frac{1}{|G_l^*|} \right)^{-1} \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2}_{\text{Part A}} + \underbrace{|S| - \left(\frac{1}{|G_k^*|} + \frac{1}{|G_l^*|} \right)^{-1} R_{kl}^S}_{\text{Part B}},$$

where

$$R_{kl}^S = \sqrt{\frac{\xi^S}{|G_l^*|}} \|(\mathbf{c}_l^* - \mathbf{c}_k^*)_{S \cap S^*}\|_2 + \left(\frac{1}{|G_k^*|} + \frac{1}{|G_l^*|} \right) \sqrt{|S| \xi^S} + \frac{1}{|G_k^*|} \xi^S,$$

and $\xi^S = \log(n^2 K^2 |S|^2 \binom{p}{|S|})$. Under the separation condition for $\mathcal{S}$, recalled in (64), and noting that $|S| \leq \sqrt{p}$ for all $S \in \mathcal{S}$, Part A dominates Part B in the inequality above.

Therefore, we have

$$\begin{aligned} \min_{1 \leq k \neq l \leq K} \left\{ \left(\frac{1}{|G_k^*|} + \frac{1}{|G_l^*|} \right)^{-1} D_{kli}^S(\mathbf{X}) \right\} &\gtrsim \min_{1 \leq k \neq l \leq K} \left(\frac{1}{|G_k^*|} + \frac{1}{|G_l^*|} \right)^{-1} \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + |S| \\ &\gtrsim |S| + \frac{m}{4} \Delta_{S \cap S^*}^2 \\ &= \dot{\lambda}^S. \end{aligned}$$

This concludes the derivation of the second inequality and thereby completes the proof. $\square$

11.5.8 Proof of Lemma 26

Proof. For notational simplicity, let $\sigma^2 = 1$. Fix $S \in \mathcal{S}$ and (l, k) with $1 \leq l \neq k \leq K$.

Condition on the high-probability event in Lemma 23. Then we have

$$\begin{aligned}
t_{lk}^S(\dot{\lambda}^S) &\stackrel{(i)}{=} \sum_{j \in G_l^*} \left\{ \frac{|G_k^*|}{2} D_{lkj}^S(\mathbf{X}) - \frac{|G_k^*| + |G_l^*|}{2|G_l^*|} \dot{\lambda}^S \right\} \\
&\stackrel{(ii)}{=} \sum_{j \in G_l^*} \left\{ \frac{|G_k^*|}{2} D_{lkj}^S(\mathbf{X}) - \frac{|G_k^*| + |G_l^*|}{2|G_l^*|} (|S| + \frac{m}{4} \Delta_{S \cap S^*}^2) \right\} \\
&\stackrel{(iii)}{\gtrsim} \sum_{j \in G_l^*} \left\{ \frac{|G_k^*|}{2} (|S| + \|(\mathbf{c}_l^* - \mathbf{c}_k^*)_{S \cap S^*}\|_2^2) - \frac{|G_k^*| + |G_l^*|}{2|G_l^*|} (|S| + \frac{m}{4} \Delta_{S \cap S^*}^2) \right\} \\
&\stackrel{(iii)}{\geq} \sum_{j \in G_l^*} \left(\frac{|G_k^*|}{2} |S| + \frac{|G_k^*|}{2} \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 \right. \\
&\quad \left. - \frac{|G_k^*| + |G_l^*|}{2|G_l^*|} |S| - \frac{|G_k^*| + |G_l^*|}{2|G_l^*|} \frac{m}{4} \|(\mathbf{c}_l^* - \mathbf{c}_k^*)_{S \cap S^*}\|_2^2 \right) \\
&\stackrel{(iv)}{\geq} \sum_{j \in G_l^*} \left\{ \left(\frac{|G_k^*|}{2} - \frac{|G_k^*| + |G_l^*|}{2|G_l^*|} \right) |S| + \frac{|G_k^*|}{4} \|(\mathbf{c}_l^* - \mathbf{c}_k^*)_{S \cap S^*}\|_2^2 \right\} \\
&= \frac{|G_l^*| |G_k^*|}{4} \|\mathbf{c}_k^* - \mathbf{c}_l^*\|_2^2 + \frac{|G_l^*| |G_k^*|}{2} |S| - \frac{|G_k^*| + |G_l^*|}{2} |S|. \\
&\stackrel{(v)}{\gtrsim} |G_l^*| |G_k^*| (\|\mathbf{c}_k^* - \mathbf{c}_l^*\|_2^2 + |S|)
\end{aligned}$$

where step (i) follows from the definition of $t_{lk}^S(\lambda^S)$ in (33), step (ii) follows from the definition of $\dot{\lambda}^S$ in Lemma 25, step (iii) uses $-\Delta_{S \cap S^*}^2 > -\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2$, step (iv) uses

$$-m \geq -2 \left\{ \frac{|G_k^*| \cdot |G_l^*|}{|G_k^*| + |G_l^*|} \right\},$$

and step (v) uses $|G_k^*| \geq 2$ and $|G_l^*| \geq 2$. This completes the proof of Lemma 26. $\square$

11.5.9 Proof of Lemma 27

the proof proceeds in the following steps.

1. Individual bound by Chi-square tail bound,

2. Uniform bound.

Proof. For notational simplicity, let $\sigma^2 = 1$.

11.5.9.1 1. Individual bound by Chi-square tail bound. Fix (k, l) such that $1 \leq k \neq l \leq K$ and $S \in \mathcal{S}$. Then we have

$$\begin{aligned}
\sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} I_{kl}^S(\mathbf{v}, \mathbf{X}) &= \sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} |G_l^*| \sum_{i \in G_k^*} v_i \langle (\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}, \mathcal{E}_{S \cap S^*, i} \rangle \\
&\leq \sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} |G_l^*| \left| \sum_{i \in G_k^*} v_i \langle (\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}, \mathcal{E}_{S \cap S^*, i} \rangle \right| \\
&\stackrel{(i)}{\leq} \sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} |G_l^*| \left(\sum_{i \in G_k^*} v_i^2 \right)^{1/2} \left(\sum_{i \in G_k^*} \langle (\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}, \mathcal{E}_{S \cap S^*, i} \rangle^2 \right)^{1/2} \\
&\leq |G_l^*| \left(\sum_{i \in G_k^*} \langle (\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}, \mathcal{E}_{S \cap S^*, i} \rangle^2 \right)^{1/2},
\end{aligned}$$

where step (i) uses the Cauchy-Schwarz inequality.

Now we derive an inequality for $\sum_{i \in G_k^*} \langle (\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}, \mathcal{E}_{S \cap S^*, i} \rangle^2$. Since

$$\frac{1}{\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2} \langle (\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}, \mathcal{E}_{S \cap S^*, i} \rangle \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1), \quad i = 1, \dots, |G_k^*|,$$

the square of this quantity follows a chi-square distribution with $|G_k^*|$ degrees of freedom, provided that S is nonempty. Applying the chi-square tail inequality (Lemma 7), if we define

$$\mathcal{C}_{Ikl}^S := \left\{ \sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} I_{kl}^S(\mathbf{v}, \mathbf{X}) \leq \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2 (|G_k^*| + 2\sqrt{|G_k^*| \zeta_I^S} + 2\zeta_I^S)^{1/2} \right\}, \quad (65)$$

where $\zeta_I^S = \log(nK^2|S|^2 \binom{p}{|S|})$, then we have $\mathbb{P}(\mathcal{C}_{Ikl}^S) \geq 1 - 1/(n^2 K^2 |S|^2 \binom{p}{|S|})$.

11.5.9.2 2. Uniform bound. Now we show that the event $\mathcal{C}_{Ikl}^S$ holds uniformly across all $S \subset \mathcal{S}$ and $1 \leq k \neq l \leq K$, with probability at least $1 - C_9/n$, where $C_9 > 0$ is a constant.

$$\mathbb{P}\left(\bigcup_{1 \leq k \neq l \leq K} \bigcup_{S \in \mathcal{S}} (\mathcal{C}_{Ikl}^S)^C\right) \leq \sum_{1 \leq k \neq l \leq K} \sum_{q=1}^{\lfloor \sqrt{p} \rfloor} \sum_{|S|=q} \mathbb{P}((\mathcal{C}_{Ikl}^S)^C) \stackrel{(i)}{\leq} \sum_{q=1}^{\lfloor \sqrt{p} \rfloor} \sum_{|S|=q} \frac{1}{n|S|^2 \binom{p}{|S|}} = \sum_{q=1}^{\lfloor \sqrt{p} \rfloor} \frac{1}{nq^2} \leq \frac{\pi^2}{6n},$$

where the last step uses $\sum_{q=1}^{\infty} \frac{1}{q^2} = \pi^2/6$. Since $|S| \leq \sqrt{p}$, $|S| \log p$ dominates $\log |S|$. Therefore, we conclude that uniformly across $S \in \mathcal{S}$ and (k, l) such that $1 \leq k \neq l \leq K$, with probability at least $1 - C_9/n$, it holds that

$$\begin{aligned} & \sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} I_{kl}^S(\mathbf{v}, \mathbf{X}) \\ & \lesssim |G_l^*| \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2 \left(|G_k^*| + \sqrt{|G_k^*| \log n} + \sqrt{|G_k^*| |S| \log p} + \log n + |S| \log p \right)^{1/2}. \end{aligned}$$

This completes the proof of Lemma 27. $\square$

11.5.10 Proof of Lemma 28

Proof. For notational simplicity, let $\sigma^2 = 1$ and for $k = 1, \dots, K$, denote

$$\text{IB}_k := \left(|G_k^*| + \sqrt{|G_k^*| \log n} + \sqrt{|G_k^*| |S| \log p} + \log n + |S| \log p \right)^{1/2}.$$

Then we have

$$\begin{aligned}
\sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} \sum_{1 \leq k \neq l \leq K} \frac{T_{1,kl}^S(\mathbf{v})}{t_{lk}^S(\dot{\lambda}^S)} &\stackrel{(i)}{\lesssim} \sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} \sum_{1 \leq k \neq l \leq K} \frac{I_{kl}^S(\mathbf{v}, \mathbf{X}) \times I_{lk}^S(\mathbf{v}, \mathbf{X})}{|G_l^*| \cdot |G_k^*| \cdot \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + |S|} \\
&\leq \sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} \sum_{1 \leq k \neq l \leq K} \frac{I_{kl}^S(\mathbf{v}, \mathbf{X}) \times I_{lk}^S(\mathbf{v}, \mathbf{X})}{|G_l^*| \cdot |G_k^*| \cdot \|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2} \\
&\stackrel{(ii)}{\lesssim} \sum_{1 \leq k \neq l \leq K} \text{IB}_k^S \times \text{IB}_l^S \\
&\leq \sum_{k=1}^K \text{IB}_k^S \sum_{l=1}^K \text{IB}_l^S \\
&\stackrel{(iii)}{\lesssim} n + \sqrt{n \log n} + \sqrt{n|S| \log p} + \log n + |S| \log p \\
&\stackrel{(iv)}{\lesssim} |S| + m \log n + \sqrt{m|S| \log p} + |S| \log p \\
&\stackrel{(v)}{\lesssim} |S| + \frac{m}{4} \Delta_{S \cap S^*}^2 = \dot{\lambda}^S,
\end{aligned}$$

where step (i) uses Lemma 26 and definition $T_{1,kl}^S(\mathbf{v}) = I_{kl}^S(\mathbf{v}, \mathbf{X}) \times I_{lk}^S(\mathbf{v}, \mathbf{X})$ presented in (30), step (ii) uses Lemma 27, step (iii) uses the identity $\sum_{k=1}^K |G_k^*| = n$ and inequality $\sum_{k=1}^K \sqrt{|G_k^*|} \leq \sqrt{K \sum_{k=1}^K |G_k^*|} \lesssim \sqrt{n}$, step (iv) uses the condition $m \gtrsim n/\log n$ of Theorem 1, and step (v) uses the separation condition of $\mathcal{S}$, presented in (10). This completes the proof of Lemma 28. $\square$

11.5.11 Proof of Lemma 29

With appropriate algebraic manipulation, the problem reduces to an application of the concentration inequality for an unbounded empirical process developed by Adamczak (2008) (Lemma 12). This requires estimating the expectation, variance, and sub-exponential norm of the empirical process. Accordingly, the proof proceeds in the following steps.

1. Reformulation into a quadratic form,
2. Bounding the operator norm dependent terms,

3. Bounding the expectation of the supremum,
4. Individual bound using Hanson-wright inequality,
5. Uniform bound.

Proof. For notational simplicity, we set $\sigma^2 = 1$ and assume, without loss of generality, that $G_k^* = 1, \dots, |G_k^*|$ throughout the proof.

11.5.11.1 1. Transformation into a quadratic form. Fix (k, l) such that $1 \leq k \neq l \leq K$ and $S \subset [p]$ such that $|S| \leq \sqrt{p}$. We can express $G_{3,kl}^S(\mathbf{v})$ as a scaled quadratic form:

$$\begin{aligned}
G_{1,kl}^S(\mathbf{v}) &= \frac{|G_l^*|}{|G_k^*|} \sum_{\{(i,j) \in G_k^*\}} v_i \langle \mathcal{E}_{S,i}, \mathcal{E}_{S,j} \rangle \\
&= \frac{|G_l^*|}{|G_k^*|} \sum_{\{(i,j) \in G_k^*\}} \frac{1}{2} (v_i + v_j) \langle \mathcal{E}_{S,i}, \mathcal{E}_{S,j} \rangle \\
&= \frac{|G_l^*|}{|G_k^*|} \underbrace{\text{vec}(\mathcal{E}_{S,G_k^*})^\top (\mathbf{A}^{\mathbf{v}} \otimes \mathbf{I}_{|S|}) \text{vec}(\mathcal{E}_{S,G_k^*})}_{\text{QF}_{kl}^S(\mathbf{A}^{\mathbf{v}} \otimes \mathbf{I}_{|S|}, \mathbf{X})},
\end{aligned} \tag{66}$$

Here, the (i, j) th element of the matrix $\mathbf{A}^{\mathbf{v}} \in \mathbb{R}^{|G_k^*| \times |G_k^*|}$ is defined as $A_{i,j}^{\mathbf{v}} := (v_i + v_j)/2$, and $\text{vec}(\mathcal{E}_{S,G_k^*})$ denotes the $|S| \cdot |G_k^*|$ -dimensional vector obtained by stacking the columns of $\mathcal{E}_{S,G_k^*}$ in order, with the first column at the top and the last column at the bottom, and $\mathbf{A}^{\mathbf{v}} \otimes \mathbf{I}_{|S|}$ denotes the tensor product:

$$\mathbf{A}^{\mathbf{v}} \otimes \mathbf{I}_{|S|} = \begin{bmatrix} A_{1,1}^{\mathbf{v}} \mathbf{I}_{|S|} & \cdots & A_{1,|G_k^*|}^{\mathbf{v}} \mathbf{I}_{|S|} \\ \vdots & \ddots & \vdots \\ A_{|G_k^*|,1}^{\mathbf{v}} \mathbf{I}_{|S|} & \cdots & A_{|G_k^*|,|G_k^*|}^{\mathbf{v}} \mathbf{I}_{|S|} \end{bmatrix}.$$

The quadratic form $\text{QF}_{kl}^S(\mathbf{A}^\mathbf{v} \otimes \mathbf{I}_{|S|}, \mathbf{X})$ is centered because

$$\mathbb{E}[\text{QF}_{kl}^S(\mathbf{A}^\mathbf{v} \otimes \mathbf{I}_{|S|}, \mathbf{X})] = \mathbb{E} \left[\sum_{\{(i,j) \in G_k^*\}} v_i \mathbb{E}[\langle \mathcal{E}_{S,i}, \mathcal{E}_{S,j} \rangle] \right] = \sum_{i \in G_k^*} v_i \mathbb{E}[\|\mathcal{E}_{S,i}\|_2^2] \stackrel{(i)}{=} |S| \sum_{i \in G_k^*} v_i = 0,$$

where step (i) uses $\sum_{i=1}^n v_i = 0$, shown in (50), and indexed by

$$\mathcal{M}_{kl}^S = \{\mathbf{A}^\mathbf{v} \otimes \mathbf{I}_{|S|} \mid \mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2 = 1\}.$$

To derive the uniform bound, we apply the uniform Hanson-Wright inequality (Lemma 15) to obtain the concentration inequality for

$$\sup_{\mathbf{M} \in \mathcal{M}_{kl}^S} \text{QF}_{kl}^S(\mathbf{M}, \mathbf{X}) - \mathbb{E} \left[\sup_{\mathbf{M} \in \mathcal{M}_{kl}^S} \text{QF}_{kl}^S(\mathbf{M}, \mathbf{X}) \right].$$

To apply Lemma 15, we need to bound the following two quantities:

$$\sup_{\mathbf{M} \in \mathcal{A}_{kl}^S} \|\mathbf{M}\|_{\text{op}} \quad \text{and} \quad \|\text{vec}(\mathcal{E}_{S,G_k^*})\|_{\mathcal{A}_{kl}^S} := \mathbb{E} \left[\sup_{\mathbf{M} \in \mathcal{A}_{kl}^S} \|(\mathbf{M} + \mathbf{M}^\top) \text{vec}(\mathcal{E}_{S,G_k^*})\|_2 \right],$$

where bounding both terms critically depends on $\|\mathbf{A}^\mathbf{v}\|_{\text{op}}$, and the expectation of supremum

$$\mathbb{E} \left[\sup_{\mathbf{M} \in \mathcal{M}_{kl}^S} \text{QF}_{kl}^S(\mathbf{M}, \mathbf{X}) \right].$$

We present the bounds in order.

11.5.11.2 2. Bounding the operator norm dependent terms. We start by bounding $\sup_{\mathbf{M} \in \mathcal{A}_{kl}^S} \|\mathbf{M}\|_{\text{op}}$. First note that for $\mathbf{M} \in \mathcal{A}_{kl}^S$, its operator norm can be simplified into:

$$\|\mathbf{M}\|_{\text{op}} = \|\mathbf{A}^\mathbf{v} \otimes \mathbf{I}_{|S|}\|_{\text{op}} = \|\mathbf{A}^\mathbf{v}\|_{\text{op}} \|\mathbf{I}_{|S|}\|_{\text{op}} = \|\mathbf{A}^\mathbf{v}\|_{\text{op}}.$$

Therefore it suffices to bound the supremum of $\|\mathbf{A}^\mathbf{v}\|_{\text{op}}$:

$$\begin{aligned}
\|\mathbf{A}^\mathbf{v}\|_{\text{op}} &= \max_{\|\mathbf{u}\|_2=1, \mathbf{u} \in \mathbb{R}^{|G_k^*|}} \mathbf{u}^T \mathbf{A}^\mathbf{v} \mathbf{u} \\
&= \max_{\|\mathbf{u}\|_2=1, \mathbf{u} \in \mathbb{R}^{|G_k^*|}} \sum_{\{(i,j) \in G_k^*\}} u_i u_j \frac{v_i + v_j}{2} \\
&= \max_{\|\mathbf{u}\|_2=1, \mathbf{u} \in \mathbb{R}^{|G_k^*|}} \left(\sum_{i=1}^{|G_k^*|} u_i v_i \right) \left(\sum_{j=1}^{|G_k^*|} u_j \right) \\
&\stackrel{(i)}{\leq} \max_{\|\mathbf{u}\|_2=1, \mathbf{u} \in \mathbb{R}^{|G_k^*|}} \left(\sum_{i=1}^{|G_k^*|} u_i^2 \right)^{1/2} \left(\sum_{i=1}^{|G_k^*|} v_i^2 \right)^{1/2} \left(\sum_{j=1}^{|G_k^*|} u_j^2 \right)^{1/2} \sqrt{|G_k^*|} \\
&\stackrel{(ii)}{\leq} \sqrt{|G_k^*|},
\end{aligned}$$

where step (i) uses Cauchy-Schwarz inequality, and step (ii) uses $\|\mathbf{v}\|_2 \leq 1$ for $\mathbf{v} \in \mathbb{V}_k$.

Since the last term do not depend on $\mathbf{v}$, we have

$$\sup_{\mathbf{M} \in \mathcal{A}_{kl}^S} \|\mathbf{M}\|_{\text{op}} \leq \sqrt{|G_k^*|}. \quad (67)$$

Next, we bound $\|\text{vec}(\mathcal{E}_{S, G_k^*})\|_{\mathcal{A}_{kl}^S}^2$:

$$\begin{aligned}
\|\text{vec}(\mathcal{E}_{S, G_k^*})\|_{\mathcal{A}_{kl}^S}^2 &= (\mathbb{E}[\sup_{\mathbf{M} \in \mathcal{A}_{kl}^S} \|(\mathbf{M} + \mathbf{M}^\top) \text{vec}(\mathcal{E}_{S, G_k^*})\|_2])^2 \\
&\leq (\mathbb{E}[\sup_{\mathbf{M} \in \mathcal{A}_{kl}^S} \|(\mathbf{M} + \mathbf{M}^\top)\|_{\text{op}} \|\text{vec}(\mathcal{E}_{S, G_k^*})\|_2])^2 \\
&\stackrel{(i)}{\leq} (\mathbb{E}[2|G_k^*|^{1/2} \|\text{vec}(\mathcal{E}_{S, G_k^*})\|_2])^2 \\
&\stackrel{(ii)}{\leq} 4|G_k^*| \mathbb{E}[\|\text{vec}(\mathcal{E}_{S, G_k^*})\|_2^2] \\
&= 4|G_k^*| \sum_{i \in G_k^*} \mathbb{E}[\|\mathcal{E}_{S, i}\|_2^2] \\
&= 4|G_k^*|^2 |S|,
\end{aligned}$$

where step (i) uses (67), step (ii) applies Jensen's inequality to a convex function $x \mapsto x^2$.

11.5.11.3 3. Bounding the expectation of the supremum. It is more convenient to start from (66) and decompose the quadratic form $\text{QF}_{kl}^S(\mathbf{A}^{\mathbf{v}}, \mathbf{X})$ into the following two terms:

$$\text{QF}_{kl}^S(\mathbf{A}^{\mathbf{v}}, \mathbf{X}) = \sum_{\{(i,j) \in G_k^* : i \neq j\}} v_j \langle \mathcal{E}_{S,i}, \mathcal{E}_{S,j} \rangle + \sum_{i \in G_k^*} v_i \|\mathcal{E}_{S,i}\|_2^2.$$

For these two terms, the index set $\mathcal{A}_{kl}^S$, which indexes the quadratic form, is more suitably represented by

$$\mathbb{V}_k := \left\{ \mathbf{v}_{G_k^*} : \mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2 = 1 \right\}.$$

We provide the bound for each of these two terms, starting with the first one. First note that given $\mathbf{v} \in \mathbb{V}_k$, we have

$$\begin{aligned} \sup_{\mathbf{v} \in \mathbb{V}_k} \sum_{\{(i,j) \in G_k^* : i \neq j\}} v_j \langle \mathcal{E}_{S,i}, \mathcal{E}_{S,j} \rangle &= \sup_{\mathbf{v} \in \mathbb{V}_k} \left| \sum_{\{(i,j) \in G_k^* : i \neq j\}} v_j \langle \mathcal{E}_{S,i}, \mathcal{E}_{S,j} \rangle \right| \\ &= \sup_{\mathbf{v} \in \mathbb{V}_k} \left| \sum_{j \in G_k^*} v_j \langle \mathcal{E}_{S,j}, \sum_{i \in G_k^*, i \neq j} \mathcal{E}_{S,i} \rangle \right| \\ &\stackrel{(i)}{\leq} \sup_{\mathbf{v} \in \mathbb{V}_k} \left(\sum_{j \in G_k^*} v_j^2 \right)^{1/2} \left(\sum_{j \in G_k^*} \langle \mathcal{E}_{S,j}, \sum_{i \in G_k^*, i \neq j} \mathcal{E}_{S,i} \rangle^2 \right)^{1/2} \\ &\stackrel{(ii)}{\leq} \left(\sum_{j \in G_k^*} \langle \mathcal{E}_{S,j}, \sum_{i \in G_k^*, i \neq j} \mathcal{E}_{S,i} \rangle^2 \right)^{1/2}, \end{aligned}$$

where step (i) uses Cauchy-Schwarz inequality and step (ii) uses $\|\mathbf{v}\|_2 \leq 1$ for $\mathbf{v} \in \mathbb{V}_k$.

Using this bound, we can upper bound the expectation as follows:

$$\begin{aligned}
\mathbb{E} \left[\sup_{\mathbf{v} \in \mathbb{V}_k} \sum_{\{(i,j) \in G_k^*, i \neq j\}} v_j \langle \mathcal{E}_{S,i}, \mathcal{E}_{S,j} \rangle \right] &\leq \mathbb{E} \left[\left(\sum_{j \in G_k^*} \langle \mathcal{E}_{S,j}, \sum_{i \in G_k^*, i \neq j} \mathcal{E}_{S,i} \rangle \right)^2 \right]^{1/2} \\
&\stackrel{(i)}{\leq} \left(\sum_{j \in G_k^*} \mathbb{E} [\langle (\varepsilon_j)_S, \sum_{i \neq j, i \in [n_k]} \mathcal{E}_{S,i} \rangle^2] \right)^{1/2} \\
&\stackrel{(ii)}{=} \left(\sum_{j \in G_k^*} \mathbb{E} \left\| \sum_{i \in G_k^*, i \neq j} \mathcal{E}_{S,i} \right\|_2^2 \right)^{1/2} \\
&\stackrel{(iii)}{\leq} (|G_k^*|(|G_k^*| - 1)|S|)^{1/2} \\
&\leq |G_k^*| \sqrt{|S|}, \tag{68}
\end{aligned}$$

where step (i) uses Jensen's inequality to a concave function $x \mapsto x^{1/2}$, step (ii) the independence and zero-mean assumption, and step (iii) uses isotropic Gaussian assumption.

Next, we bound the expectation of the supremum of $\sum_{i \in G_k^*} v_i \|\mathcal{E}_{S,i}\|_2^2$.

$$\begin{aligned}
\mathbb{E} \sup_{\mathbf{v} \in \mathbb{V}_k} \left| \sum_{i \in G_k^*} v_i \|\mathcal{E}_{S,i}\|_2^2 \right| &\stackrel{(i)}{=} \mathbb{E} \sup_{\mathbf{v} \in \mathbb{V}_k} \left| \sum_{i \in G_k^*} v_i (\|\mathcal{E}_{S,i}\|_2^2 - |S|) \right| \\
&\stackrel{(ii)}{\leq} \mathbb{E} \sup_{\mathbf{v} \in \mathbb{V}_k} \left(\sum_{i \in G_k^*} v_i^2 \right)^{1/2} \left\{ \sum_{i \in G_k^*} (\|\mathcal{E}_{S,i}\|_2^2 - |S|)^2 \right\}^{1/2} \\
&\stackrel{(iii)}{\leq} \mathbb{E} \left\{ \sum_{i \in G_k^*} (\|\mathcal{E}_{S,i}\|_2^2 - |S|)^2 \right\}^{1/2} \\
&\stackrel{(iv)}{\leq} \left\{ \sum_{i \in G_k^*} \mathbb{E} (\|\mathcal{E}_{S,i}\|_2^2 - |S|)^2 \right\}^{1/2} \\
&= \left(\sum_{i \in G_k^*} \text{Var} \|\mathcal{E}_{S,i}\|_2^2 \right)^{1/2} \\
&\stackrel{(v)}{=} \sqrt{2|G_k^*||S|}, \tag{69}
\end{aligned}$$

where step (i) uses $\sum_{i \in G_k^*} v_i = 0$, which follows from the fact that Γ_k is orthogonal to $\mathbb{1}_{G_k^*}$,

step (ii) uses Cauchy-Schwarz inequality, step (iii) uses $\|\mathbf{v}\|_2 \leq 1$, step (iv) uses Jensen's inequality to a concave function $x \mapsto x^{1/2}$, and step (v) uses our Gaussian noise assumption and the variance formula for Chi-square random variables.

Combining (68) and (69), we conclude that

$$\mathbb{E} \left[\sup_{\mathbf{M} \in \mathcal{M}_{kl}^S} \text{QF}_{kl}^S(\mathbf{M}, \mathbf{X}) \right] \lesssim |G_k^*| \sqrt{|S|}.$$

11.5.11.4 4. Individual bound using Hanson-wright inequality. Applying Lemma 15 with the bounds established in the previous steps and using the formulation (66) of $G_{1,kl}^S(\mathbf{v})$, we define an event

$$\mathcal{C}_{N,kl}^S := \left\{ \sup_{\mathbf{v} \in \mathbb{V}_k} \frac{|G_l^*|}{|G_k^*|} \sum_{\{(i,j) \in G_k^*\}} v_i \langle \mathcal{E}_{S,i}, \mathcal{E}_{S,j} \rangle \lesssim \frac{|G_l^*|}{|G_k^*|} \underbrace{|G_k^*| \sqrt{|S|}}_{\mathbb{E}[\sup_{\mathbf{M} \in \mathcal{M}_{kl}^S} \text{QF}_{kl}^S(\mathbf{M}, \mathbf{X})] \lesssim} + \frac{|G_l^*|}{|G_k^*|} \zeta_{N,kl}^S \right\},$$

where $\zeta_{N,kl}^S := |G_k^*| \sqrt{|S|(\log n + 2 \log |S| + \log \binom{p}{|S|})}$, then we have

$$\begin{aligned} \mathbb{P} \left((\mathcal{C}_{N,kl}^S)^C \right) &\lesssim \exp \left[- \min \left(\underbrace{\frac{(\zeta_{N,kl}^S)^2}{|G_k^*|^2 |S|}}_{\|\text{vec}(\mathcal{E}_{S,G_k^*})\|_{\mathcal{A}_{kl}^S}^2 \lesssim \sup_{\mathbf{M} \in \mathcal{A}_{kl}^S} \|\mathbf{M}\|_{\text{op}} \lesssim}, \frac{\zeta_{N,kl}^S}{\sqrt{|G_k^*|}} \right) \right] \\ &\lesssim \frac{1}{n |S|^2 \binom{p}{|S|}}. \end{aligned}$$

The proof is complete once we show that the intersection event

$$\bigcap_{S \in [p], |S| \leq \sqrt{p}} \bigcap_{1 \leq k \neq l \leq K} \mathcal{C}_{N,kl}^S$$

holds with high probability.

11.5.11.5 5. Uniform bound. Now we show that the event $\mathcal{C}_{N,kl}^S$ holds uniformly across all $S \in [p]$ such that $|S| \leq \sqrt{p}$ and $1 \leq k \neq l \leq K$, with probability at least $1 - C_{11}/n$, where $C_{11} > 0$ is a constant.

$$\begin{aligned}
\mathbb{P}\left(\bigcup_{S \in [p], |S| \leq \sqrt{p}} \bigcup_{1 \leq k \neq l \leq K} (\mathcal{C}_{N,kl}^S)^C\right) &\leq \sum_{1 \leq k \neq l \leq K} \sum_{q=1}^{\lfloor \sqrt{p} \rfloor} \sum_{|S|=q} \mathbb{P}((\mathcal{C}_{N,kl}^S)^C) \\
&\lesssim \sum_{q=1}^{\lfloor \sqrt{p} \rfloor} \sum_{|S|=q} \frac{K^2}{n|S|^2 \binom{p}{|S|}} \\
&= \sum_{q=1}^{\lfloor \sqrt{p} \rfloor} \frac{K^2}{nq^2} \\
&\leq \frac{K^2 \pi^2}{6n},
\end{aligned}$$

where the last step uses $\sum_{q=1}^{\infty} \frac{1}{q^2} = \pi^2/6$. Since $|S| \leq \sqrt{p}$, $|S| \log p$ dominates $\log |S|$. Also, we treat K as a constant. Therefore, we conclude that uniformly across $S \in \mathcal{S}$ and (k, l) such that $1 \leq k \neq l \leq K$, with probability at least $1 - C_{11}/n$, it holds that

$$\sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} G_{1,kl}^S(\mathbf{v}) \lesssim |G_l^*| \sqrt{|S|(\log n + |S| \log p)}$$

This completes the proof of Lemma 29.

□

11.5.12 Proof of Lemma 30

The proof proceeds in the following steps.

1. Conditional analysis using a concentration inequality for the supremum of a Gaussian process,
2. Individual bound by Chi-square tail bound,

3. Uniform bound

Proof. For notational simplicity, let $\sigma^2 = 1$.

11.5.12.1 1. Conditional analysis. Since the sum of inner products of Gaussian random vectors is challenging to analyze directly, we first condition on $\bar{\mathcal{E}}_{S,l}^*$ and then apply the concentration inequality for the supremum of a Gaussian process (Lemma 13). Fix (k, l) such that $1 \leq k \neq l \leq K$ and $S \subset [p]$ such that $|S| \leq \sqrt{p}$. Conditioned on $\bar{\mathcal{E}}_{S,l}^*$, $G_{2,kl}^S(\mathbf{v})$ can be viewed as a canonical Gaussian process $\langle G, \mathbf{w} \rangle$, where $G \sim \mathcal{N}(0, \mathbf{I}_{|G_k^*|})$ and $\mathbf{w} \in \mathbb{W}_k$, with

$$\mathbb{W}_k := \left\{ \|\bar{\mathcal{E}}_{S,l}^*\|_2 \mathbf{v}_{\upharpoonright G_k^*} : \mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2 = 1 \right\},$$

where $\mathbf{v}_{\upharpoonright G_k^*}$ is the restriction of $\mathbf{v} \in \Gamma_K$ onto G_k^* , and the canonical metric on $\mathbb{W}_k$ is the ℓ_2 -metric.

To apply Lemma 13, we have to bound

$$\begin{aligned} \text{EC}_{kl}^S(\bar{\mathcal{E}}_{S,l}^*) &:= \mathbb{E} \left[\sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} \sum_{i \in G_k^*} v_i \langle \bar{\mathcal{E}}_{S,l}^*, \mathcal{E}_{S,i} \rangle \mid \bar{\mathcal{E}}_{S,l}^* \right] = \mathbb{E} \left[\sup_{\mathbf{w} \in \mathbb{W}_k} \langle G, \mathbf{w} \rangle \mid \bar{\mathcal{E}}_{S,l}^* \right], \text{ and} \\ \text{SVC}_{kl}^S(\bar{\mathcal{E}}_{S,l}^*) &:= \sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} \text{Var} \left(\sum_{i \in G_k^*} v_i \langle \bar{\mathcal{E}}_{S,l}^*, \mathcal{E}_{S,i} \rangle \mid \bar{\mathcal{E}}_{S,l}^* \right) = \sup_{\mathbf{w} \in \mathbb{W}_k} \text{Var}(\langle G, \mathbf{w} \rangle \mid \bar{\mathcal{E}}_{S,l}^*). \end{aligned}$$

We bound $\text{EC}_{kl}^S(\bar{\mathcal{E}}_{S,l}^*)$ using Dudley's integral inequality (Lemma 14):

$$\text{EC}_{kl}^S(\bar{\mathcal{E}}_{S,l}^*) \lesssim \int_0^\infty \sqrt{\log N(\mathbb{W}_k, \epsilon)} d\epsilon \stackrel{(i)}{\lesssim} \|\bar{\mathcal{E}}_{S,l}^*\|_2 \int_0^\infty \sqrt{|G_k^*| \log(1/\epsilon)} d\epsilon = \|\bar{\mathcal{E}}_{S,l}^*\|_2 \sqrt{|G_k^*|},$$

where step (i) uses the fact that the ϵ -covering entropy of the unit sphere in $\mathbb{R}^{|G_k^*|}$ is at most $|G_k^*| \log(1/\epsilon)$, up to a constant factor, for any $\epsilon \in (0, 1)$.

Next, we bound $\text{SVC}_{kl}^S(\bar{\mathcal{E}}_{S,l}^*)$:

$$\text{SVC}_{kl}^S(\bar{\mathcal{E}}_{S,l}^*) = \sup_{\mathbf{w} \in \mathbb{W}_k} \text{Var}(\langle G, \mathbf{w} \rangle \mid \bar{\mathcal{E}}_{S,l}^*) = \sup_{\mathbf{w} \in \mathbb{W}_k} \|\bar{\mathcal{E}}_{S,l}^*\|_2^2 \sum_{i \in G_k^*} v_i^2 \leq \|\bar{\mathcal{E}}_{S,l}^*\|_2^2,$$

where the last inequality follows from $\|\mathbf{v}\|_2 = 1$ for $\mathbf{v} \in \Gamma_K$. Let us denote $\xi^S = \log(n|S|^2 \binom{p}{|S|})$. Define the following conditional probability as follows:

$$\text{CP}_{N,kl}^S := \mathbb{P} \left(\sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} |G_{2,kl}^S(\mathbf{v})| \gtrsim |G_l^*| \underbrace{\|\bar{\mathcal{E}}_{S,l}^*\|_2 \sqrt{|G_k^*|}}_{\gtrsim \text{EC}_{kl}^S(\bar{\mathcal{E}}_{S,l}^*)} + |G_l^*| \underbrace{\|\bar{\mathcal{E}}_{S,l}^*\|_2}_{\gtrsim \sqrt{\text{SVC}_{kl}^S(\bar{\mathcal{E}}_{S,l}^*)}} \sqrt{\xi^S} \mid \bar{\mathcal{E}}_{S,l}^* \right),$$

Then by Lemma 13, we have

$$\text{CP}_{N,kl}^S \leq \frac{1}{n|S|^2 \binom{p}{|S|}}. \quad (70)$$

11.5.12.2 2. Chi-square tail bound Now we use the Chi-square tail inequality to further bound $\|\bar{\mathcal{E}}_{S,l}^*\|_2$ and $\|\bar{\mathcal{E}}_{S,l}^*\|_2^2$. If we define the following event:

$$\mathcal{B}_{N,kl}^S := \left\{ \|\bar{\mathcal{E}}_{S,l}^*\|_2^2 \leq \frac{1}{|G_l^*|} \left(|S| + 2\sqrt{|S|\xi^S} + 2\xi^S \right) \right\}.$$

By Lemma 7, we have

$$\mathbb{P}((\mathcal{B}_{N,kl}^S)^C) \leq \frac{1}{n|S|^2 \binom{p}{|S|}}.$$

Combining the bounds in $\text{CP}_{N,kl}^S$ and $\mathcal{B}_{N,kl}^S$, we define an event:

$$\mathcal{A}_{N,kl}^S := \left\{ \sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} |G_{2,kl}^S(\mathbf{v})| \lesssim \sqrt{|G_l^*|} \sqrt{|S| + 2\sqrt{|S|\xi^S} + 2\xi^S} (\sqrt{|G_k^*|} + \sqrt{\xi^S}) \right\}.$$

The proof is complete once we show that the intersection event

$$\bigcap_{1 \leq k \neq l \leq K} \bigcap_{S \in [p], |S| \leq \sqrt{p}} \mathcal{A}_{N,kl}^S$$

holds with high probability.

11.5.12.3 3. Uniform bound. We have

$$\begin{aligned}
& \mathbb{P}\left(\bigcup_{1 \leq k \neq l \leq K} \bigcup_{S \in [p], |S| \leq \sqrt{p}} (\mathcal{A}_{N,kl}^S)^C\right) \\
& \leq \sum_{1 \leq k \neq l \leq K} \sum_{q=1}^{\lfloor \sqrt{p} \rfloor} \sum_{|S|=q} \mathbb{P}((\mathcal{A}_{N,kl}^S)^C) \\
& = \sum_{1 \leq k \neq l \leq K} \sum_{q=1}^{\lfloor \sqrt{p} \rfloor} \sum_{|S|=q} \mathbb{E}[\mathbb{P}((\mathcal{A}_{N,kl}^S)^C \mid \bar{\mathcal{E}}_{S,l}^*)] \\
& \leq \sum_{1 \leq k \neq l \leq K} \sum_{q=1}^{\lfloor \sqrt{p} \rfloor} \sum_{|S|=q} \mathbb{E}[\mathbb{P}((\mathcal{A}_{N,kl}^S)^C \mid \bar{\mathcal{E}}_{S,l}^*) \mathbb{1}(\mathcal{B}_{N,kl}^S) + \mathbb{1}((\mathcal{B}_{N,kl}^S)^C)] \\
& \stackrel{(i)}{\leq} \left(\sum_{1 \leq k \neq l \leq K} \sum_{q=1}^{\lfloor \sqrt{p} \rfloor} \sum_{|S|=q} \mathbb{E}[\text{CP}_{N,kl}^S \mathbb{1}(\mathcal{B}_{N,kl}^S)] \right) + \frac{K^2 \pi^2}{6n} \\
& \stackrel{(ii)}{\lesssim} \frac{K^2 \pi^2}{6n} + \frac{K^2 \pi^2}{6n},
\end{aligned}$$

where step (i) uses $\sum_{q=1}^{\infty} (1/q^2) = \pi^2/6$, and step (ii) also uses it and additionally uses (70).

Treating K as a constant, we conclude that uniformly across $S \in \mathcal{S}$ and (k, l) such that $1 \leq k \neq l \leq K$, with probability at least $1 - C_{12}/n$, where $C_{12} > 0$ is a constant, it holds that

$$\sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} |G_{2,kl}^S(\mathbf{v})| \lesssim \sqrt{|G_l^*|} \sqrt{|S| + 2\sqrt{|S|\xi^S} + 2\xi^S} \sqrt{|G_k^*| + \sqrt{\xi^S}}.$$

This completes the proof of Lemma 30. $\square$

11.5.13 Proof of Lemma 31

Leveraging the decomposition of $T_{2,kl}^S$ into $G_{1,kl}^S$ and $G_{2,kl}^S$ given in Section 11.4.3, the proof proceeds in the following intermediate steps:

1. Problem decomposition
2. Inequality for the $G_1 \times G_1$ term
3. Inequality for the $G_2 \times G_2$ term
4. Conclusion

Proof. For notational simplicity, let $\sigma^2 = 1$.

11.5.13.1 1. Problem decomposition. Recall that we defined $\Delta_{S \cap S^*}^2 = \min_{1 \leq k \neq l \leq K} \|(\mathbf{c}_l^* - \mathbf{c}_k^*)_{S \cap S^*}\|_2^2$ in Theorem 1, and $\dot{\lambda}^S = |S| + m\Delta_{S \cap S^*}^2/4$ in Lemma 25.

$$\begin{aligned}
& \sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} \sum_{1 \leq k \neq l \leq K} \frac{T_{2,kl}^S(\mathbf{v})}{t_{lk}^S(\dot{\lambda}^S)} \\
& \stackrel{(i)}{=} \sum_{1 \leq k \neq l \leq K} \sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} \frac{N_{kl}^S(\mathbf{v}, \mathbf{X}) N_{lk}^S(\mathbf{v}, \mathbf{X})}{t_{lk}^S(\dot{\lambda}^S)} \\
& \stackrel{(ii)}{\lesssim} \sum_{1 \leq k \neq l \leq K} \sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} \frac{N_{kl}^S(\mathbf{v}, \mathbf{X}) N_{lk}^S(\mathbf{v}, \mathbf{X})}{|G_l^*| |G_k^*| (\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + |S|)} \\
& = \sum_{1 \leq k \neq l \leq K} \sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} \frac{(G_{1,kl}^S(\mathbf{v}) + G_{2,kl}^S(\mathbf{v}))(G_{1,lk}^S(\mathbf{v}) + G_{2,lk}^S(\mathbf{v}))}{|G_l^*| |G_k^*| (\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + |S|)}
\end{aligned}$$

where step (i) uses the definition of $T_{2,kl}^S(\mathbf{v})$ stated in (31) and step (ii) uses Lemma 26. Expanding the numerator yields four terms. We establish the inequality for two of them, the $G_1 \times G_1$ and $G_2 \times G_2$ terms, as the cross terms can be easily handled by leveraging the results from these two cases. Before proceeding, we recall from (10) that $\mathcal{S}$ is defined as

$$\mathcal{S} = \left\{ S : S \subset [p], |S| \leq \sqrt{p}, \Delta_{S \cap S^*}^2 \gtrsim \left(\log n + \frac{|S| \log p}{m} + \sqrt{\frac{|S| \log p}{m}} \right) \right\}.$$

11.5.13.2 2. Inequality for the $G_1 \times G_1$ term. By Lemma 29, uniformly across $S \in \mathcal{S}$ and (k, l) such that $1 \leq k \neq l \leq K$, with probability at least $1 - C_{11}/n$, it holds

that

$$\begin{aligned}
& \sum_{1 \leq k \neq l \leq K} \sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} \frac{G_{1,kl}^S(\mathbf{v}) G_{1,lk}^S(\mathbf{v})}{|G_l^*| |G_k^*| (\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + |S|)} \\
& \stackrel{(i)}{\lesssim} \sum_{1 \leq k \neq l \leq K} \frac{|S|(\log n + |S| \log p)}{(\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + |S|)} \\
& \lesssim \frac{|S| \log n + |S|^2 \log p}{(\Delta_{S \cap S^*}^2 + |S|)}.
\end{aligned}$$

Therefore it suffices to show that

$$|S| \log n + |S|^2 \log p \lesssim (|S| + m \Delta_{S \cap S^*}^2)(\Delta_{S \cap S^*}^2 + |S|),$$

which is true since $|S| \Delta_{S \cap S^*}^2 \gtrsim |S| \log n$ and $m|S| \Delta_{S \cap S^*}^2 \gtrsim |S|^2 \log p$ under the separation condition of $\mathcal{S}$.

11.5.13.3 3. Inequality for the $G_2 \times G_2$ term. By Lemma 30, uniformly across $S \in \mathcal{S}$ and (k, l) such that $1 \leq k \neq l \leq K$, with probability at least $1 - C_{12}/n$, it holds that

$$\begin{aligned}
& \sum_{1 \leq k \neq l \leq K} \sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} \frac{G_{2,kl}^S(\mathbf{v}) G_{2,lk}^S(\mathbf{v})}{|G_l^*| |G_k^*| (\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + |S|)} \\
& \stackrel{(i)}{\lesssim} \sum_{1 \leq k \neq l \leq K} \frac{\sqrt{|G_l^*|} \sqrt{|S| + 2\sqrt{|S|} \xi^S + 2\xi^S} \sqrt{|G_k^*| + \sqrt{\xi^S}} \sqrt{|G_k^*|} \sqrt{|S| + 2\sqrt{|S|} \xi^S + 2\xi^S} \sqrt{|G_l^*| + \sqrt{\xi^S}}}{|G_l^*| |G_k^*| (\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + |S|)} \\
& = \sum_{1 \leq k \neq l \leq K} \frac{(|S| + 2\sqrt{|S|} \xi^S + 2\xi^S) \sqrt{|G_k^*| |G_l^*| + \sqrt{\xi^S}(|G_k^*| + |G_l^*|)} + \xi^S}{\sqrt{|G_l^*| |G_k^*|} (\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + |S|)} \\
& = \sum_{1 \leq k \neq l \leq K} \frac{(|S| + 2\sqrt{|S|} \xi^S + 2\xi^S) (\sqrt{|G_k^*| |G_l^*|} + \sqrt{\sqrt{\xi^S}(|G_k^*| + |G_l^*|)} + \sqrt{\xi^S})}{\sqrt{|G_l^*| |G_k^*|} (\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + |S|)},
\end{aligned}$$

where $\xi^S = \log(n|S|^2(\frac{p}{|S|}))$. The last term can be decomposed into three parts, and we establish an inequality for each one in turn, starting with the first term:

$$\begin{aligned}
& \sum_{1 \leq k \neq l \leq K} \frac{(|S| + 2\sqrt{|S|\xi^S} + 2\xi^S)(\sqrt{|G_k^*||G_l^*|})}{\sqrt{|G_l^*||G_k^*|}(\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + |S|)} \\
&= \sum_{1 \leq k \neq l \leq K} \frac{(|S| + 2\sqrt{|S|\xi^S} + 2\xi^S)}{(\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + |S|)} \\
&\lesssim \frac{(|S| + |S|\sqrt{\log p} + \sqrt{|S|\log n} + \log n + |S|\log p)}{\Delta_{S \cap S^*}^2 + |S|} \\
&\lesssim |S| + m\Delta_{S \cap S^*}^2,
\end{aligned}$$

where the last inequality holds due to the separation condition of $\mathcal{S}$.

Now, for the second term, we first recall from Theroem 1 that $m = 2 \min_{1 \leq k \neq l \leq K} \left\{ \frac{|G_k^*||G_l^*|}{|G_k^*| + |G_l^*|} \right\}$.

Then we have:

$$\begin{aligned}
& \sum_{1 \leq k \neq l \leq K} \frac{(|S| + 2\sqrt{|S|\xi^S} + 2\xi^S)(\sqrt{\xi^S(|G_k^*| + |G_l^*|)})}{\sqrt{|G_l^*||G_k^*|}(\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + |S|)}, \\
&\lesssim \frac{(|S| + 2\sqrt{|S|\xi^S} + 2\xi^S)(\xi^S)^{1/4}}{\sqrt{m}(\Delta_{S \cap S^*}^2 + |S|)} \\
&\lesssim |S| + m\Delta_{S \cap S^*}^2,
\end{aligned}$$

where the last inequality holds because under the separation condition of $\mathcal{S}$, we have $(|S| + 2\sqrt{|S|\xi^S} + 2\xi^S) \lesssim |S| + m\Delta_{S \cap S^*}^2$ and $(\xi^S)^{1/4} \lesssim \sqrt{m}(\Delta_{S \cap S^*}^2 + |S|)$.

Finally, for the third term, we have

$$\begin{aligned}
& \sum_{1 \leq k \neq l \leq K} \frac{(|S| + 2\sqrt{|S|\xi^S} + 2\xi^S)\sqrt{\xi^S}}{\sqrt{|G_l^*||G_k^*|}(\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + |S|)} \\
& \lesssim \sum_{1 \leq k \neq l \leq K} \frac{(|S| + 2\sqrt{|S|\xi^S} + 2\xi^S)\sqrt{\xi^S}}{\sqrt{m}(\|(\mathbf{c}_k^* - \mathbf{c}_l^*)_{S \cap S^*}\|_2^2 + |S|)} \\
& \lesssim |S| + m\Delta_{S \cap S^*}^2,
\end{aligned}$$

where the last inequality holds because under the separation condition of $\mathcal{S}$, we have $(|S| + 2\sqrt{|S|\xi^S} + 2\xi^S) \lesssim |S| + m\Delta_{S \cap S^*}^2$ and $(\xi^S)^{1/2} \lesssim \sqrt{m}(\Delta_{S \cap S^*}^2 + |S|)$.

11.5.13.4 4. Conclusion. The inequalities for the cross terms can be verified in a straightforward manner. Combining all the inequalities established above, we conclude that, uniformly over $S \in \mathcal{S}$ and all (k, l) with $1 \leq k \neq l \leq K$, the following holds with probability at least $1 - C_{13}/n$, where $C_{13} > 0$ is a constant, it holds that:

$$\sup_{\mathbf{v} \in \Gamma_K, \|\mathbf{v}\|_2=1} \sum_{1 \leq k \neq l \leq K} \frac{T_{2,kl}^S(\mathbf{v})}{t_{lk}^S(\dot{\lambda}^S)}.$$

This completes the proof of Lemma 31. □

11.6 Proof of the Minimax Lower Bound

The proof proceeds in the following intermediate steps:

1. Reduction to symmetric two-cluster problem (71),
2. Reduction to average case risk (72),
3. Reduction to Bayes risk (73),
4. Bounding the Bayes risk (75).

Proof. We proceed by following the steps outlined above in sequence.

1. Reduction to symmetric two-cluster problem. We denote the separation by $\rho > 0$ and leave its value unspecified for now. First, we define a reduced class of distributions consisting of two symmetric, sparse cluster centers.

$$\tilde{\theta}(s, \rho, 2) := \tilde{\mathcal{M}}(s, \rho, 2) \times \mathcal{H}(2)$$

where

$$\begin{aligned} \tilde{\mathcal{M}}(s, \rho, 2) := \{(\mathbf{c}_1, \mathbf{c}_2) : \mathbf{c}_1 = \mathbf{c}, \mathbf{c}_2 = -\mathbf{c}, \mathbf{c} \in \mathbb{R}^p, \|\mathbf{c}_1 - \mathbf{c}_2\|_2^2 \geq \rho^2, \\ \text{supp}(\mathbf{c}) = S^* \subset [p], |S^*| \leq s\}, \end{aligned}$$

Then we have (by augmenting the matrices in $\mathcal{H}(2)$ with zero columns to match the dimensions):

$$\tilde{\theta}(s, \rho, 2) \subset \Psi(s, \rho, K).$$

This inclusion immediately implies that

$$\inf_{\hat{\mathbf{H}}} \sup_{\psi \in \Psi(s, \rho, K)} \mathbb{P}_\psi(\hat{\mathbf{H}}(\mathbf{X}) \neq \mathbf{H}) \geq \inf_{\hat{\mathbf{H}}} \sup_{\psi \in \tilde{\theta}(s, \rho, 2)} \mathbb{P}_\psi(\hat{\mathbf{H}}(\mathbf{X}) \neq \mathbf{H}), \quad (71)$$

since the supremum over a smaller set is less than or equal to that over a larger set.

2. Reduction to average case risk. Next, we lower bound the worst-case risk in (71) by the average risk.

$$\begin{aligned} \inf_{\hat{\mathbf{H}}} \sup_{\psi \in \tilde{\mathcal{H}}(s, \rho, 2)} \mathbb{P}_{\psi}(\hat{\mathbf{H}}(\mathbf{X}) \neq \mathbf{H}) &= \inf_{\hat{\mathbf{H}}} \sup_{(\mathbf{c}_1, \mathbf{c}_2) \in \tilde{\mathcal{M}}(s, \rho, 2)} \sup_{\mathbf{H} \in \mathcal{H}(2)} \mathbb{P}_{\psi}(\hat{\mathbf{H}}(\mathbf{X}) \neq \mathbf{H}) \\ &\geq \inf_{\hat{\mathbf{H}}} \sup_{(\mathbf{c}_1, \mathbf{c}_2) \in \tilde{\mathcal{M}}(s, \rho, 2)} \mathbb{E}_{\mathbf{H}(\mathbf{X}) \sim \pi} \mathbb{P}_{\psi}(\hat{\mathbf{H}} \neq \mathbf{H}), \end{aligned} \quad (72)$$

where π is a uniform prior on $\mathcal{H}(2)$, formally given by, for $i = 1, \dots, n$,

$$H_{i,1} \stackrel{\text{i.i.d.}}{\sim} \text{Ber}(1/2), \quad H_{i,2} = 1 - H_{i,1}.$$

3. Reduction to Bayes risk. The risk (72) is equivalent to the minimax risk under zero-one loss, where the data is generated from a uniform hierarchical Gaussian mixture model defined as follows:

$$\begin{aligned} H_{i,1} &\sim \text{Ber}(1/2), \\ \mathbf{X}_i \mid H_{i,1} &\sim \mathcal{N}((-1)^{H_{i,1}} \mathbf{c}, \mathbf{I}_p). \end{aligned}$$

In this setting, the algorithm that minimizes $\mathbb{E}_{\mathbf{H} \sim \pi} \mathbb{P}_{\psi}(\hat{\mathbf{H}}(\mathbf{X}) \neq \mathbf{H})$ is the Bayes classifier $\mathbf{H}^{\mathbf{c}}$, which assumes knowledge of the true cluster centers. It operates by thresholding the Gaussian likelihood ratio, and under symmetric cluster centers with isotropic covariance, this yields the decision rule $H_{i,1}^{\mathbf{c}} = \mathbb{1}(\mathbf{c}^{\top} \mathbf{X}_i > 0)$ for $i = 1, \dots, n$. Its error probability for one observation, known as the Bayes rate, is explicitly given by (Cai & Zhang 2019):

$$\mathcal{R}_{\mathbf{c}_1, \mathbf{c}_2}^* := \Phi\left(-\frac{\|\mathbf{c}_1 - \mathbf{c}_2\|_2}{2\sigma}\right), \quad (73)$$

where Φ denotes the cumulative distribution function of standard normal distribution. Therefore we can bound (72) further from below:

$$\begin{aligned}
\inf_{\hat{\mathbf{H}}} \sup_{(\mathbf{c}_1, \mathbf{c}_2) \in \tilde{\mathcal{M}}(s, \rho, 2)} \mathbb{E}_{\mathbf{H}(\mathbf{X}) \sim \pi} \mathbb{P}_{\psi}(\hat{\mathbf{H}} \neq \mathbf{H}) &\stackrel{(i)}{\geq} \sup_{(\mathbf{c}_1, \mathbf{c}_2) \in \tilde{\mathcal{M}}(s, \rho, 2)} \inf_{\hat{\mathbf{H}}} \mathbb{E}_{\mathbf{H}(\mathbf{X}) \sim \pi} \mathbb{P}_{\psi}(\hat{\mathbf{H}} \neq \mathbf{H}) \\
&\stackrel{(ii)}{=} \sup_{(\mathbf{c}_1, \mathbf{c}_2) \in \tilde{\mathcal{M}}(s, \rho, 2)} 1 - (1 - \mathcal{R}_{\mathbf{c}_1, \mathbf{c}_2}^*)^n \\
&\stackrel{(iii)}{=} 1 - (1 - \Phi(-\rho/\sigma))^n, \tag{74}
\end{aligned}$$

where step (i) applies the max-min inequality (see, for example, Section 5.4.1 of Boyd and Vandenberghe, 2004), step (ii) follows from the optimality of Bayes classifier, and step (iii) uses (73).

4. Bounding the Bayes risk. If we set $\rho^2 = C_4 \sigma^2 \log n$, we have

$$\Phi(-\rho/\sigma) = \Phi(-\sqrt{C_4 \log n}) = 1 - \mathbb{P}(Z > -\sqrt{C_4 \log n}) \stackrel{(i)}{\geq} 1 - \exp(-\frac{C_4}{2} \log n) = 1 - n^{-C_4/2},$$

where Z denotes a standard normal random variable and step (i) uses Gaussian tail bound. Leveraging this, for any $n \geq 2$, we further bound (74) as:

$$1 - \{1 - \Phi(-\sqrt{C_4 \log n})\}^n \geq 1 - n^{-C_4 n/2} > C_5, \tag{75}$$

where $C_5 > 0$ depends on C_4 . Collecting the reductions (71), (72), (73), and the bound (75), we conclude that

$$\inf_{\hat{\mathbf{H}}} \sup_{\psi \in \Psi(s, \sigma^2 \log n, K)} \mathbb{P}_{\psi}(\hat{\mathbf{H}}(\mathbf{X}) \neq \mathbf{H}) > C_5.$$

This completes the proof of the lower bound part of Theorem 1. $\square$

Remark 33. *The reduction in step (i) of (74) is not tight in some regime. A sharper analysis can be obtained by decomposing the average risk $\mathbb{E}_{\mathbf{H}(\mathbf{X}) \sim \pi} \mathbb{P}_\psi(\hat{\mathbf{H}} \neq \mathbf{H})$ into the Bayes risk $\mathcal{R}_{\mathbf{c}_1, \mathbf{c}_2}^*$ and the excess risk $\mathbb{E}_{\mathbf{H}(\mathbf{X}) \sim \pi} \mathbb{P}_\psi(\hat{\mathbf{H}} \neq \mathbf{H}) - \mathcal{R}_{\mathbf{c}_1, \mathbf{c}_2}^*$. Cai et al. (2019) shows that the excess risk is of order $(s \log p)/n$ in a setting slightly different from ours. Rather than modifying their analysis, we assume a standard high-dimensional sparse regime $s \log p/n = o(1)$, so that the Bayes risk term $\log n$ dominates. In this regime, this lower bound matches the separation bound established in first claim of Theorem 1 in the main text.*

12 Details for Robust MMD Statistic

This section provides the detailed algebraic derivations, omitted from Section 3.2.3, that connect the population and empirical Maximum Mean Discrepancy (MMD) to our population clustering parameters and proposed test statistic. First, we show that the population squared MMD simplifies to the squared difference of the true transformed means using a linear kernel:

$$\begin{aligned} \text{MMD}_k^2(P_j, Q_j) &:= \mathbb{E}_{P_j}[k(\tilde{X}, \tilde{X}')] - 2\mathbb{E}_{P_j, Q_j}[k(\tilde{X}, \tilde{X}')] + \mathbb{E}_{Q_j}[k(\tilde{X}', \tilde{X}'')] \\ &= (\mathbb{E}_{P_j}[\tilde{X}])^2 - 2\mathbb{E}_{P_j}[\tilde{X}]\mathbb{E}_{Q_j}[\tilde{X}'] + (\mathbb{E}_{Q_j}[\tilde{X}'])^2 \\ &= (\mathbb{E}_{P_j}[\tilde{X}] - \mathbb{E}_{Q_j}[\tilde{X}'])^2 = ((\Omega^* \mathbf{c}_1^*)_j - (\Omega^* \mathbf{c}_2^*)_j)^2. \end{aligned}$$

Next, we show that if the true, uncorrupted partition G^* is known, the V-statistic empirical MMD plug-in estimator $\widehat{\text{MMD}}_k^2(G^*, \mathbf{X}_j)$ exactly reduces to the square of our test statistic

$\Lambda(G^*, \mathbf{X}_j)$ defined in (12).

$$\begin{aligned} \widehat{\text{MMD}}_k^2(G^*, \mathbf{X}_j) &:= \frac{1}{|G_1^*|^2} \sum_{\ell, m \in G_1^*} k(\tilde{X}_{\ell, j}, \tilde{X}_{m, j}) + \frac{1}{|G_2^*|^2} \sum_{\ell, m \in G_2^*} k(\tilde{X}_{\ell, j}, \tilde{X}_{m, j}) - \frac{2}{|G_1^*||G_2^*|} \sum_{\ell \in G_1^*} \sum_{m \in G_2^*} k(\tilde{X}_{\ell, j}, \tilde{X}_{m, j}) \\ &= \left(\frac{1}{|G_1^*|} \sum_{\ell \in G_1^*} \tilde{X}_{\ell, j} - \frac{1}{|G_2^*|} \sum_{m \in G_2^*} \tilde{X}_{m, j} \right)^2 = \Lambda(G^*, \mathbf{X}_j)^2. \end{aligned}$$

Finally, we present a theorem adapted from [Schrab & Kim \(2025\)](#) and rewritten in our notation, which provides a theoretical guarantee for Algorithm 4.

Theorem 34 (Power guarantee of Algorithm 4, Theorem 1 of [Schrab & Kim 2025](#)). *If the estimated cluster sizes are proportional ($n_1 \asymp n_2$ with $n_1 \leq n_2$) and the number of permutations B satisfies $B > 3(\tau_\epsilon - \alpha)^{-2} \log\left(\frac{8}{1-\tau_\epsilon-\alpha}\right) + (\tau_\epsilon - \alpha)(1 - (\tau_\epsilon - \alpha))$, then for any true data-generating distributions that satisfy*

$$|(\Omega^* \mathbf{c}_1^*)_j - (\Omega^* \mathbf{c}_2^*)_j| \geq C_K \max \left\{ \sqrt{\frac{\max\{\log(\frac{e}{\tau_\epsilon - \alpha}), \log(\frac{e}{1-\tau_\epsilon - \alpha})\}}{n}}, \frac{r}{n} \right\},$$

where $C_K > 0$ is a constant, the probability of rMMD correctly outputting 1 is bounded as:

$$\mathbb{P}_{P_j, Q_j}(\text{reject } H_0 \mid \text{at most } r \text{ misclustered samples}) \geq \tau_\epsilon + \alpha. \quad (76)$$

Because our TVS algorithm conditions on the observed data $\tilde{\mathbf{X}}$, the marginal probability bound in (76), taken with respect to P_j and Q_j , does not directly apply and should not be used as rigorous justification of the oracle margin assumption (Assumption 2). We present Theorem 34 to conceptually motivate using the robust MMD test for the reward step.

13 Control of Noise Variable Selection by TVS Algorithm

This section shows that the TVS algorithm selects noise variables only with low probability, under a mild assumption. We first note that under the model defined in (1), any non-signal variable $j \notin S^*$ represents pure random noise and is independent of the underlying cluster structure. Consequently, the permutation null hypothesis holds strictly for any current cluster estimate $\hat{G}^{t-1}$, meaning that mean reward $\theta_j(\hat{G}^{t-1})$ precisely represents the Type I error rate of the test $\varphi_j(\hat{G}^{t-1}; U^t)$.

Assumption 4 (Type I Error Calibration). *For a given threshold $\tau_\epsilon > 0$ and margin $\alpha \in (0, \tau_\epsilon)$, the permutation test is calibrated to a significance level $\tau_\epsilon - \alpha$. Consequently, the test satisfies the uniform noise rejection condition:*

$$\max_G \theta_j(G) = \tau_\epsilon - \alpha, \forall j \notin S^*$$

where the maximum is taken over all possible partitions G of $[n]$.

Under Assumption 4, the following theorem states that TVS algorithm tightly controls the inclusion of noise variables. In a more formal way, with high probability, the number of times any noise variable is selected grows at most logarithmically with the time horizon T (proof provided in Section 13.1).

Theorem 35 (Noise exclusion guarantee). *Suppose Assumption 4 holds. For fixed integer prior parameters $a_j^0 \geq 1$ and $b_j^0 \geq 1$ for all $j \in [p]$, conditioned on the data $\tilde{\mathbf{X}}$, we have*

$$\mathbb{P} \left(\max_{j \notin S^*} \sum_{t=1}^T \mathbb{1}\{j \in \hat{S}^t\} \leq \left\lceil \max \left\{ \frac{8}{\alpha^2} \log(2npT), \max_{j \notin S^*} \frac{4(a_j^0 - 1)}{\alpha} \right\} \right\rceil \right) \geq 1 - \frac{1}{n},$$

where the probability is taken over random seeds $U^1, \dots, U^T$.

Notably, Theorem 35 establishes this noise exclusion guarantee without relying on any assumptions regarding the strength of the underlying clustering signal.

13.1 Proof of Theorem 35

Proof. Let $\mathcal{F}_{t-1} = \sigma(U^1, \dots, U^{t-1})$ denote the filtration generated by the random permutation seeds up to the end of iteration $t - 1$. Conditioned on the data $\widetilde{\mathbf{X}}$, all algorithmic randomness is captured by this filtration. To track the state of each arm j at iteration t , we denote its selection count and average observed reward prior to iteration t as:

$$N_j^t := \sum_{k < t} \mathbb{1}(j \in \hat{S}^k) = a_j^{t-1} + b_j^{t-1} - a_j^0 - b_j^0 \quad \text{and} \quad \hat{\mu}_j^t := \frac{a_j^{t-1} - a_j^0}{\sum_{k < t} \mathbb{1}(j \in \hat{S}^k)} = \frac{a_j^{t-1} - a_j^0}{N_j^t}. \quad (77)$$

Note that both N_j^t and $\hat{\mu}_j^t$ are $\mathcal{F}_{t-1}$ -measurable.

Fix a noise variable $j \notin S^*$. Leveraging Assumption 4, we first show that for any $M \geq \frac{4(a_j^0 - 1)}{\alpha}$ and any finite horizon $T \geq 1$,

$$\mathbb{P}(N_j^{T+1} > M) \leq 2T \exp(-M\alpha^2/8). \quad (78)$$

The proof is as follows. If $N_j^{T+1} > M$, the algorithm must have selected variable j for the $(M + 1)$ -th time at some specific iteration $t \leq T$. Therefore we can write as

$$\{N_j^{T+1} > M\} \subseteq \bigcup_{t=1}^T \{j \in \hat{S}^t, N_j^t = M\}, \quad \text{thus} \quad \mathbb{P}(N_j^{T+1} > M) \leq \sum_{t=1}^T \mathbb{P}(j \in \hat{S}^t, N_j^t = M).$$

Fix an arbitrary time index t . The algorithm selects variable j at iteration t only if the sampled Beta random variable satisfies $\vartheta_j^t \geq \tau_\epsilon$. We decompose the probability of this

selection event around the empirical mean $\hat{\mu}_j^t$:

$$\left\{ \vartheta_j^t \geq \tau_\epsilon \right\} = \left\{ (\vartheta_j^t - \hat{\mu}_j^t) + \hat{\mu}_j^t \geq \tau_\epsilon \right\} \subseteq \left\{ \vartheta_j^t - \hat{\mu}_j^t \geq \frac{\alpha}{2} \right\} \cup \left\{ \hat{\mu}_j^t \geq \tau_\epsilon - \frac{\alpha}{2} \right\}.$$

We bound the probability of these two events separately, conditioning explicitly on the history.

We start with the event related to posterior sampling in the lemma below (proof in [Section 13.2](#)).

Lemma 36 (Event 1: posterior sampling deviation.). *For $M \geq \frac{4(a_j^0 - 1)}{\alpha}$, we have*

$$\mathbb{P} \left(\vartheta_j^t \geq \hat{\mu}_j^t + \frac{\alpha}{2} \mid \mathcal{F}_{t-1} \right) \mathbb{1}\{N_j^t = M\} \leq \exp \left(-M \frac{\alpha^2}{8} \right). \quad (79)$$

The next lemma bounds the reward deviation probability (proof in [Section 13.3](#)).

Lemma 37 (Event 2: Reward deviation). *For a positive integer M , we have:*

$$\mathbb{P} \left(\left\{ \hat{\mu}_j^t \geq \tau_\epsilon - \frac{\alpha}{2} \right\} \cap \{N_j^t = M\} \right) \leq \exp \left(-M \frac{\alpha^2}{2} \right).$$

Combining the bounds for Event 1 and Event 2 requires formally intersecting them with the historical event $\{N_j^t = M\}$. Recall our earlier decomposition of the selection event $\{j \in \hat{S}^t\} = \{\vartheta_j^t \geq \tau_\epsilon\}$:

$$\left\{ j \in \hat{S}^t \right\} \cap \left\{ N_j^t = M \right\} \subseteq \left(\left\{ \vartheta_j^t - \hat{\mu}_j^t \geq \frac{\alpha}{2} \right\} \cup \left\{ \hat{\mu}_j^t \geq \tau_\epsilon - \frac{\alpha}{2} \right\} \right) \cap \left\{ N_j^t = M \right\}.$$

Applying the union bound to this intersection yields:

$$\mathbb{P}\left(j \in \hat{S}^t, N_j^t = M\right) \leq \mathbb{P}\left(\vartheta_j^t - \hat{\mu}_j^t \geq \frac{\alpha}{2}, N_j^t = M\right) + \mathbb{P}\left(\hat{\mu}_j^t \geq \tau_\epsilon - \frac{\alpha}{2}, N_j^t = M\right).$$

For the first term (Event 1), because $\{N_j^t = M\}$ is strictly $\mathcal{F}_{t-1}$ -measurable, we can evaluate the joint probability by applying the Law of Total Expectation (the Tower Property) to our conditional bound:

$$\begin{aligned} \mathbb{P}\left(\vartheta_j^t - \hat{\mu}_j^t \geq \frac{\alpha}{2}, N_j^t = M\right) &= \mathbb{E}\left[\mathbb{1}_{\{N_j^t = M\}} \mathbb{P}\left(\vartheta_j^t - \hat{\mu}_j^t \geq \frac{\alpha}{2} \mid \mathcal{F}_{t-1}\right)\right] \\ &\leq \mathbb{E}\left[\mathbb{1}_{\{N_j^t = M\}} \exp\left(-M \frac{\alpha^2}{8}\right)\right] \\ &\leq \exp\left(-M \frac{\alpha^2}{8}\right). \end{aligned}$$

Summing these two probabilities and noting that $\exp(-M\alpha^2/2) \leq \exp(-M\alpha^2/8)$ since $\alpha > 0$ and $M \geq 1$, we obtain the joint probability bound for a single iteration t :

$$\mathbb{P}\left(j \in \hat{S}^t, N_j^t = M\right) \leq \exp\left(-M \frac{\alpha^2}{8}\right) + \exp\left(-M \frac{\alpha^2}{2}\right) \leq 2 \exp\left(-M \frac{\alpha^2}{8}\right).$$

Finally, because the event of pulling the arm more than M times implies that the $(M+1)$ -th pull must occur at some specific iteration $t \in \{1, \dots, T\}$, we apply a union bound over all T possible time steps. This yields the desired bound for a single noise arm across the entire horizon:

$$\mathbb{P}(N_j^{T+1} > M) \leq \sum_{t=1}^T \mathbb{P}\left(j \in \hat{S}^t, N_j^t = M\right) \leq \sum_{t=1}^T 2 \exp\left(-M \frac{\alpha^2}{8}\right) = 2T \exp\left(-M \frac{\alpha^2}{8}\right).$$

Uniform bound. If $M \geq \max_{j \notin S^*} \frac{4(a_j^0 - 1)}{\alpha}$, applying (78) and the union bound over all

$(p - |S^*|) \leq p$ noise variables :

$$\mathbb{P}(\exists j \notin S^* : N_j^{T+1} > M) \leq \sum_{j \notin S^*} \mathbb{P}(N_j^{T+1} > M) \leq 2pT \exp(-M\alpha^2/8).$$

For the right-hand side to be smaller than δ , we need

$$M > \frac{8}{\alpha^2} \log\left(\frac{2pT}{\delta}\right).$$

Therefore, if

$$M \geq M(\delta, T) = \left\lceil \frac{8}{\alpha^2} \log\left(\frac{2pT}{\delta}\right) \vee \max_{j \notin S^*} \frac{4(a_j^0 - 1)}{\alpha} \right\rceil$$

we have

$$\mathbb{P}\left(\max_{j \notin S^*} [N_j^{T+1}] \leq M(\delta, T)\right) \geq 1 - \delta.$$

This completes the proof of Theorem 35. □

13.2 Proof of Lemma 36

Proof. Conditioned on the filtration $\mathcal{F}_{t-1}$, the historical pull count N_j^t and the empirical mean $\hat{\mu}_j^t$ are deterministic. We conduct our analysis conditionally on the event $\{N_j^t = M\}$, which signifies that arm j has been pulled exactly M times prior to iteration t . On this event, the generative distribution of the Thompson sampling variable ϑ_j^t is formed using exactly M historical reward observations.

Given $\{N_j^t = M\}$, the variable ϑ_j^t is drawn from the following Beta distribution:

$$\text{Beta}(a_j^0 + M\hat{\mu}_j^t, b_j^0 + M(1 - \hat{\mu}_j^t)) = \text{Beta}(a_j^{t-1}, b_j^{t-1}).$$

For notational simplicity, let $s = M\hat{\mu}_j^t$ denote the sum of rewards observed from arm

j . Because the rewards are binary, s is an integer. Consequently, the updated prior parameters $a_j^{t-1} = a_j^0 + s$ and $b_j^{t-1} = b_j^0 + M - s$ are strictly positive integers. This integer property allows us to leverage the exact identity between the Beta cumulative distribution function (CDF) and the Binomial CDF. Specifically, for integer parameters $\alpha, \beta > 0$, $1 - F_{\text{beta}}(x; \alpha, \beta) = F_{\text{binom}}(\alpha - 1; \alpha + \beta - 1, x)$. Applying this mapping, the corresponding Binomial parameters, namely the number of trials n^t and the success threshold k^t , are:

$$n^t := M + a_j^0 + b_j^0 - 1, \quad k^t := s + a_j^0 - 1.$$

We now evaluate the conditional probability of a posterior sampling deviation:

$$\begin{aligned} \mathbb{P}\left(\vartheta_j^t \geq \hat{\mu}_j^t + \frac{\alpha}{2} \mid \mathcal{F}_{t-1}, N_j^t = M\right) &= 1 - F_{\text{beta}}(\hat{\mu}_j^t + \alpha/2; a_j^{t-1}, b_j^{t-1}) \\ &= 1 - \left(1 - F_{\text{binom}}(k^t; n^t, \hat{\mu}_j^t + \alpha/2)\right) \\ &= F_{\text{binom}}(k^t; n^t, \hat{\mu}_j^t + \alpha/2). \end{aligned} \tag{80}$$

To rigorously apply the lower-tail Chernoff-Hoeffding bound to this Binomial CDF (80), we must first verify that the threshold ratio $\frac{k^t}{n^t}$ is strictly less than the success probability parameter $(\hat{\mu}_j^t + \frac{\alpha}{2})$:

$$\left(\hat{\mu}_j^t + \frac{\alpha}{2}\right) > \frac{k^t}{n^t}.$$

Let's expand the difference:

$$\begin{aligned}
\left(\hat{\mu}_j^t + \frac{\alpha}{2}\right) - \frac{k^t}{n^t} &= \frac{\alpha}{2} + \hat{\mu}_j^t - \frac{M\hat{\mu}_j^t + a_j^0 - 1}{M + a_j^0 + b_j^0 - 1} \\
&= \frac{\alpha}{2} + \frac{\hat{\mu}_j^t(M + a_j^0 + b_j^0 - 1) - (M\hat{\mu}_j^t + a_j^0 - 1)}{n^t} \\
&= \frac{\alpha}{2} + \frac{(a_j^0 + b_j^0 - 1)\hat{\mu}_j^t - a_j^0 + 1}{n^t}.
\end{aligned}$$

Let us focus on the numerator term:

$$f(\hat{\mu}_j^t) := (a_j^0 + b_j^0 - 1)\hat{\mu}_j^t - a_j^0 + 1.$$

Because $a_j^0 \geq 1$ and $b_j^0 \geq 1$, the slope $(a_j^0 + b_j^0 - 1)$ is strictly positive. Therefore, this is an increasing linear function of $\hat{\mu}_j^t$. To find the universal lower bound, we simply evaluate this function at its absolute minimum possible value, which occurs when $\hat{\mu}_j^t = 0$:

$$f(0) = (a_j^0 + b_j^0 - 1)(0) - a_j^0 + 1 = 1 - a_j^0.$$

Now, substitute this worst-case numerator back into the full expression:

$$\left(\hat{\mu}_j^t + \frac{\alpha}{2}\right) - \frac{k^t}{n^t} \geq \frac{\alpha}{2} + \frac{1 - a_j^0}{n^t} = \frac{\alpha}{2} - \frac{a_j^0 - 1}{n^t} \stackrel{(i)}{\geq} \frac{\alpha}{2} - \frac{a_j^0 - 1}{M} \stackrel{(ii)}{\geq} \frac{\alpha}{4} > 0,$$

where each step uses

$$(i) \quad n^t = M + a_j^0 + b_j^0 - 1 \geq M$$

$$(ii) \quad M \geq \frac{4(a_j^0 - 1)}{\alpha} \implies \frac{a_j^0 - 1}{M} \leq \frac{\alpha}{4}.$$

Therefore we can use the lower-tail Chernoff-Hoeffding bound to (80):

$$\mathbb{P}\left(\vartheta_j^t \geq \hat{\mu}_j^t + \frac{\alpha}{2} \mid \mathcal{F}_{t-1}, N_j^t = M\right) = F_{\text{binom}}(k^t; n^t, \hat{\mu}_j^t + \alpha/2) \leq \exp\left(-n^t \cdot d\left(\frac{k^t}{n^t}, \hat{\mu}_j^t + \frac{\alpha}{2}\right)\right),$$

where $d(p, q)$ denotes the KL Divergence between two Bernoulli random variables with success probabilities p and q . By Pinsker's inequality, the KL divergence is bounded by the squared difference: $d(p, q) \geq 2(p - q)^2$. Applying this to our gap:

$$d\left(\frac{k^t}{n^t}, \hat{\mu}_j^t + \frac{\alpha}{2}\right) \geq 2\left\{\left(\hat{\mu}_j^t + \frac{\alpha}{2}\right) - \frac{k^t}{n^t}\right\}^2 \geq 2\left(\frac{\alpha}{4}\right)^2 = \frac{\alpha^2}{8}.$$

Finally, since $a_j^0 \geq 1$ and $b_j^0 \geq 1$, we know the number of binomial trials satisfies $n^t = M + a_j^0 + b_j^0 - 1 \geq M + 1 > M$. Therefore:

$$\mathbb{P}\left(\vartheta_j^t \geq \hat{\mu}_j^t + \frac{\alpha}{2} \mid \mathcal{F}_{t-1}, N_j^t = M\right) \leq \exp\left(-M \frac{\alpha^2}{8}\right).$$

This completes the proof of Lemma 36. □

13.3 Proof of Lemma 37

Proof. We must account for the fact that the sequence of rewards is not i.i.d. We construct a supermartingale by tracking the specific iterations where arm j is selected. Following Definition 44, let τ_j^w denote the random time step at which arm j is pulled for the w -th time. Let $\mathcal{G}_w := \mathcal{F}_{\tau_j^w}$ denote the stopped filtration capturing all algorithmic history up to and including the w -th pull, with $\mathcal{G}_0$ representing the initial state. Let $Y_j^{\tau_j^w} \in \{0, 1\}$ denote the binary permutation test reward obtained at this w -th pull. By Assumption 4, the permutation test controls the Type I error uniformly over *all* possible cluster partitions. Therefore, conditioning on the history up to the moment just before the w -th pull, the expected reward is strictly bounded:

$$\mathbb{E}[Y_j^{\tau_j^w} \mid \mathcal{G}_{w-1}] \leq \tau_\epsilon - \alpha.$$

We define the centered difference sequence:

$$Z_w = Y_j^{\tau_j^w} - (\tau_\epsilon - \alpha).$$

By our Type I error bound, this sequence satisfies $\mathbb{E}[Z_w \mid \mathcal{G}_{w-1}] \leq 0$.

Let $S_M = \sum_{w=1}^M Z_w$ denote the sequence of partial sums of these deviations, initialized at $S_0 = 0$. We formally establish that the stochastic process $\{S_M\}_{M \geq 0}$ forms a discrete-time supermartingale with respect to the stopped filtration $\{\mathcal{G}_M\}_{M \geq 0}$ by verifying the requisite properties:

1. **Integrability:** because $Y_j^{\tau_j^w} \in \{0, 1\}$, each difference Z_w is strictly bounded within an interval of length 1:

$$Z_w \in [-(\tau_\epsilon - \alpha), 1 - (\tau_\epsilon - \alpha)].$$

Consequently, the partial sum S_M is bounded and absolutely integrable, i.e., $\mathbb{E}[|S_M|] < \infty$ for any given M .

2. **Measurability:** The sum S_M depends only on the algorithmic choices and rewards up to the M -th pull, meaning S_M is adapted to $\mathcal{G}_M$.
3. **Conditional Expectation:** Because S_1 is fully determined by the history up to the (1)-th pull, it is $\mathcal{G}_1$ -measurable. By the linearity of conditional expectation, we can extract it:

$$\mathbb{E}[S_M \mid \mathcal{G}_1] = \mathbb{E}[S_1 + Z_M \mid \mathcal{G}_1] = S_1 + \mathbb{E}[Z_M \mid \mathcal{G}_1].$$

Applying our established bound $\mathbb{E}[Z_M \mid \mathcal{G}_1] \leq 0$ to the final term yields the defining property of a supermartingale:

$$\mathbb{E}[S_M \mid \mathcal{G}_1] \leq S_1.$$

On the event $\{N_j^t = M\}$, exactly M pulls of arm j have occurred prior to iteration t , meaning the empirical mean is exactly $\hat{\mu}_j^t = \frac{1}{M} \sum_{w=1}^M Y_j^{\tau_j^w}$. We can therefore bound the joint probability by bounding the supermartingale sequence at step M . Applying the Azuma-Hoeffding inequality for supermartingales with bounded differences (where the interval length $c_w = 1$), we obtain:

$$\begin{aligned}
\mathbb{P} \left(\left\{ \hat{\mu}_j^t \geq \tau_\epsilon - \frac{\alpha}{2} \right\} \cap \{N_j^t = M\} \right) &\leq \mathbb{P} \left(\frac{1}{M} \sum_{w=1}^M Y_j^{\tau_j^w} - (\tau_\epsilon - \alpha) \geq \frac{\alpha}{2} \right). \\
&= \mathbb{P} \left(S_M \geq M \frac{\alpha}{2} \right) \\
&\leq \exp \left(- \frac{2 \left(M \frac{\alpha}{2} \right)^2}{\sum_{w=1}^M 1^2} \right) \\
&= \exp \left(- \frac{M^2 \alpha^2 / 2}{M} \right) = \exp \left(- M \frac{\alpha^2}{2} \right).
\end{aligned}$$

This completes the proof of Lemma 37. $\square$

14 Proofs of Preliminary Results in Section 3.3

This section presents preliminary results that support the theoretical guarantees of the proposed TVS algorithm.

14.1 Proof of Lemma 4

Proof. On $\mathcal{E}_{\text{SDP}}$, we have $\widehat{\mathbf{Z}}(S) = \mathbf{Z}^*$ for every $S \in \mathcal{S}$. Therefore if $\widehat{S}^t \in \mathcal{S}$ then $\widehat{\mathbf{Z}}(\widehat{S}^t) = \mathbf{Z}^*$. The clustering step in Algorithm 1 yields $(\widehat{G}_1^t, \widehat{G}_2^t)$ equal to (G_1^*, G_2^*) up to label permutation. Since $Y_j(\cdot)$ in Definition 2 depends on data only on the induced grouping of indices, the resulting rewards agree with oracle rewards. $\square$

14.2 Justification of Assumption 2

We provide an example of a reward mechanism that satisfies Assumption 2 using a randomized test. Consider the (known covariance) Gaussian mixture model (1) with $K = 2$ and let $n_k := |G_k^*|$, and m as in (10) so that $(1/n_1 + 1/n_2) = 2/m$. Recall $\tilde{\mathbf{X}} = \Omega^* \mathbf{X}$.

Let $u \in (0, 1)$ be the target nominal level for the noise features. Choose a strictly smaller threshold level $u_0 \in (0, u)$. Consider the following randomized test mechanism. For each feature j , we define the standardized statistic:

$$D_j(\tilde{\mathbf{X}}) := \frac{(\bar{\tilde{\mathbf{X}}}_{G_1^*} - \bar{\tilde{\mathbf{X}}}_{G_2^*})_j}{\sqrt{\Omega_{jj}^*(1/n_1 + 1/n_2)}}.$$

Given the data $\tilde{\mathbf{X}}$, the decision $\Lambda_j(G^*, \tilde{\mathbf{X}}) \in \{0, 1\}$ is drawn conditionally independently as:

$$\Lambda_j(G^*, \tilde{\mathbf{X}}) \mid \tilde{\mathbf{X}} \sim \text{Bernoulli}(\phi_j(D_j(\tilde{\mathbf{X}}))),$$

where the randomized test function ϕ_j is defined as:

$$\phi_j(d) = \begin{cases} 1 & \text{if } |d| > z_{1-u_0/2}, \\ p & \text{if } |d| \leq z_{1-u_0/2}, \end{cases}$$

with the randomization probability $p := \frac{u-u_0}{1-u_0} \in (0, 1)$, and $z_q := \Phi^{-1}(q)$. The following proposition justifies Assumption 2.

Proposition 1. *If $j \notin S^*$ (equivalently $(\Omega^*(\mathbf{c}_1^* - \mathbf{c}_2^*))_j = 0$), then the oracle success probability is exactly $\theta_j^{\text{orc}} = u$.*

If $j \in S^$ and we define the standardized effect size $\gamma_j := \sqrt{\frac{m}{2}} \cdot \frac{|(\Omega^*(\mathbf{c}_1^* - \mathbf{c}_2^*))_j|}{\sqrt{\Omega_{jj}^*}}$, then the*

oracle success probability satisfies the explicit lower bound

$$\theta_j^{\text{orc}} = \mathbb{P}(|N(\gamma_j, 1)| > z_{1-u_0/2}) + p \mathbb{P}(|N(\gamma_j, 1)| \leq z_{1-u_0/2}) > 1 - \Phi(z_{1-u_0/2} - \gamma_j).$$

Consequently, if we choose $u \leq \tau_\epsilon - \alpha$ and assume

$$\min_{j \in S^*} \gamma_j \geq z_{1-u_0/2} + \Phi^{-1}(\tau_\epsilon + \alpha), \quad (81)$$

then Assumption 2 holds.

Proof. Under the described Gaussian model, conditional on G^* , the standardized test statistic $D_j(\widetilde{\mathbf{X}})$ is Gaussian with mean γ_j and variance 1. The unconditional oracle success probability is $\theta_j^{\text{orc}} = \mathbb{E}[\phi_j(D_j)]$. For $j \notin S^*$, we have $\gamma_j = 0$, so $D_j \sim N(0, 1)$. The oracle probability evaluates to:

$$\begin{aligned} \theta_j^{\text{orc}} &= \mathbb{P}(|N(0, 1)| > z_{1-u_0/2}) + p \mathbb{P}(|N(0, 1)| \leq z_{1-u_0/2}) \\ &= u_0 + \left(\frac{u - u_0}{1 - u_0} \right) (1 - u_0) = u. \end{aligned}$$

For $j \in S^*$, we bound the success probability from below by discarding the strictly positive randomized component:

$$\begin{aligned} \theta_j^{\text{orc}} &> \mathbb{P}(|N(\gamma_j, 1)| > z_{1-u_0/2}) \\ &\geq \mathbb{P}(N(\gamma_j, 1) > z_{1-u_0/2}) = 1 - \Phi(z_{1-u_0/2} - \gamma_j). \end{aligned}$$

If $u \leq \tau_\epsilon - \alpha$, then all noise features satisfy $\theta_j^{\text{orc}} = u \leq \tau_\epsilon - \alpha$. If (81) holds, then for every signal feature $j \in S^*$,

$$1 - \Phi(z_{1-u_0/2} - \gamma_j) \geq 1 - \Phi(-\Phi^{-1}(\tau_\epsilon + \alpha)) = \tau_\epsilon + \alpha,$$

establishing the oracle margin (15). □

15 Regret Bound for TVS Algorithm

This section presents the regret bound of the TVS algorithm introduced in Section 3.2.1. This result partially demonstrates that the TVS algorithm is capable of recovering the oracle variable subset S_ϵ^* defined in (14), under the oracle margin condition. Moreover, elements of its proof are instrumental in establishing exact variable selection consistency in Theorem 5. For any $S \subseteq [p]$, define the oracle gap as:

$$\Delta_S := r_\epsilon^{\text{orc}}(S_\epsilon^*) - r_\epsilon^{\text{orc}}(S), \quad \Delta_{\max} := \max_{S \subseteq [p]} \Delta_S. \quad (82)$$

The pseudo-oracle regret over T iterations is

$$\text{Reg}^{\text{orc}}(T) = \mathbb{E} \left[\sum_{t=1}^T \{r_\epsilon^{\text{orc}}(S_\epsilon^*) - r_\epsilon^{\text{orc}}(\hat{S}^t)\} \right] = \sum_{t=1}^T \mathbb{E} [\Delta_{\hat{S}^t}],$$

where expectation is over posterior sampling $\vartheta_j^t \sim \text{Beta}(a_j^t, b_j^t)$ and the rewards mechanism involving Bernoulli feedbacks Y_j^{orc} . The following theorem 38 (proof in Section 16) bounds the oracle TVS regret.

Theorem 38 (Cumulative regret bound for oracle TVS). *Consider the oracle TVS problem where if a variable j is selected at time t , the algorithm observes an i.i.d. oracle Bernoulli reward $Y_j^{\text{orc},t} \sim \text{Bernoulli}(\theta_j^{\text{orc}})$. Under Assumption 2, we have for all $T \geq \max_{j \in [p]} \exp(\alpha(a_j^0 - 1)/2)$,*

$$\text{Reg}^{\text{orc}}(T) \leq \Delta_{\max} \min \left\{ T, p \left(3 + \frac{4}{\alpha^2} + \frac{8 \log T}{\alpha^2} \right) \right\}.$$

Since the TVS algorithm and its analysis are conditioned on the data matrix $\mathbf{X}$, we treat

the ambient dimension p as fixed. Under this setting, the cumulative regret grows on the order of $\log T$; consequently, the average regret is of order $(\log T)/T$, which converges to zero as we keep running the algorithm ($T \rightarrow \infty$).

16 Proof of Theorem 38

Proof. The proof of Theorem 38 is under the global assumption of oracle TVS. We generalize the proof of Liu & Ročková (2023) to include general priors $\{a_j^0, b_j^0\}_{j=1}^p$ and a cutoff $\tau_\epsilon = \log(1/\epsilon)/\log(\frac{\epsilon+1}{\epsilon})$. Recall the notation defined in (77):

$$N_j^t = \sum_{k < t} \mathbb{1}(j \in \hat{S}^k) = a_j^{t-1} + b_j^{t-1} - a_j^0 - b_j^0 \quad \text{and} \quad \hat{\mu}_j^t = \frac{a_j^{t-1} - a_j^0}{\sum_{k < t} \mathbb{1}(j \in \hat{S}^k)} = \frac{a_j^{t-1} - a_j^0}{N_j^t}.$$

As in Liu & Ročková (2023), our proof is structured around the hierarchical relationship between the two primary sources of randomness in TVS: Bernoulli and Beta. Specifically, the empirical mean $\hat{\mu}_j^t$, which is itself a random variable, parameterizes the Beta posterior distribution of the Thompson sample ϑ_j^t . We first decompose the regret with respect to these sources of randomness, and then bound each of these regret components separately.

Let us start the regret decomposition. Since $\Delta_{\hat{S}^t} = r_\epsilon^{\text{orc}}(S_\epsilon^*) - r_\epsilon^{\text{orc}}(\hat{S}^t)$ is zero when $\hat{S}^t = S_\epsilon^*$, we can write the cumulative expected regret as

$$\text{Reg}^{\text{orc}}(T) = \mathbb{E}\left[\sum_{t=1}^T \Delta_{\hat{S}^t}\right] = \sum_{t=1}^T \mathbb{E}[\Delta_{\hat{S}^t} \mathbb{1}\{\hat{S}^t \neq S_\epsilon^*\}].$$

To decompose this regret, we partition the t -th iteration error event

$$A^t := \{\hat{S}^t \neq S_\epsilon^*\} \tag{83}$$

into the following sub-events that categorize the source of the selection failure:

- **Rare event on reward (B^t):** a signal arm has been under-rewarded or a selected noise arm has been over-rewarded relative to the threshold τ_ϵ .

$$B^t := \left\{ \exists j \in S_\epsilon^* : \hat{\mu}_j^t < \tau_\epsilon + \frac{\alpha}{2} \right\} \cup \left\{ \exists j \in \hat{S}^t \setminus S_\epsilon^* : \hat{\mu}_j^t > \tau_\epsilon - \frac{\alpha}{2} \right\}$$

- **Rare event on posterior sampling (C^t):** The Thompson sample $\vartheta_j^t \sim \text{Beta}(a_j^t, b_j^t)$,

is drawn too small for a signal arm, or drawn too large for a selected noise arm, relative to the empirical mean $\hat{\mu}_j^t$.

$$C^t := \left\{ \exists j \in S_\epsilon^* : \vartheta_j^t < \hat{\mu}_j^t - \frac{\alpha}{2} \right\} \cup \left\{ \exists j \in \hat{S}^t \setminus S_\epsilon^* : \vartheta_j^t > \hat{\mu}_j^t + \frac{\alpha}{2} \right\}$$

- **Insufficient exploration (D^t):** An under-explored arm has been selected:

$$D^t := \bigcup_{j \in \hat{S}^t} \left\{ N_j^t < \frac{8 \log T}{\alpha^2} \right\}. \quad (84)$$

Using these events, the following lemma (Proof in Section 16.1) decomposes the regret into three terms (explained in Table 2).

Lemma 39 (Regret decomposition). *The oracle cumulative regret $\text{Reg}^{\text{orc}}(T)$ is decomposed as follows:*

$$\begin{aligned} \text{Reg}^{\text{orc}}(T) &= \underbrace{\sum_{t=1}^T \mathbb{E} \left[\Delta_{\hat{S}^t} \mathbf{1}(A^t \cap B^t) \right]}_{:= \text{Reg}_1^{\text{orc}}(T)} + \underbrace{\sum_{t=1}^T \mathbb{E} \left[\Delta_{\hat{S}^t} \mathbf{1}(A^t \cap (B^t)^C \cap C^t \cap (D^t)^C) \right]}_{:= \text{Reg}_2^{\text{orc}}(T)} \\ &\quad + \underbrace{\sum_{t=1}^T \mathbb{E} \left[\Delta_{\hat{S}^t} \mathbf{1}(A^t \cap (B^t)^C \cap C^t \cap D^t) \right]}_{\text{Reg}_3^{\text{orc}}(T)}. \end{aligned}$$

Subsequent sections bound each of these terms separately.

Table 2: Summary of regret components

Regret	Main Source of Error	$\hat{S}^t$ State
$\text{Reg}_1^{\text{orc}}(T)$	B^t : Unlucky testing so far (Bernoulli distribution); requires Assumption 2 to bound	Any state
$\text{Reg}_2^{\text{orc}}(T)$	C^t : Unlucky Thompson sample now (Beta distribution)	$(D^t)^C$: only the sufficiently explored are selected
$\text{Reg}_3^{\text{orc}}(T)$	C^t : Unlucky Thompson sample now (Beta distribution)	D^t : an underexplored arm is selected, leading to large Beta posterior variance

The following lemma (proof in Section 16.2) generalizes of Lemma 2, 4, and 8 and of Liu & Ročková 2023, bounding the three components in Lemma 39. Recall that $\Delta_{\max} := \max_{S \subset [p]} \Delta_S$. We claim the following lemma:

Lemma 40 (Generalization of Lemma 2, 4, and 8 of Liu & Ročková 2023). *For a given constant $C \in (0, 1)$, its corresponding selection threshold τ_ϵ , and a margin $\alpha \in (0, \tau_\epsilon)$, under Assumption 2, the regret components satisfy:*

$$\text{Reg}_1^{\text{orc}}(T) \leq \min\left\{p \left(1 + \frac{4}{\alpha^2}\right), T\right\} \Delta_{\max}, \quad (85)$$

$$\text{Reg}_2^{\text{orc}}(T) \leq \min\{p, T\} \Delta_{\max} \text{ for } T \geq \max_{j \in [p]} \exp(\alpha(a_j^0 - 1)/2), \quad (86)$$

$$\text{Reg}_3^{\text{orc}}(T) \leq \min\left\{p \left(1 + \frac{8 \log T}{\alpha^2}\right), T\right\} \Delta_{\max} \quad (87)$$

Combining the three bounds in Lemma 40, we conclude that

$$\begin{aligned}
\text{Reg}^{\text{orc}}(T) &< \text{Reg}_1^{\text{orc}}(T) + \text{Reg}_2^{\text{orc}}(T) + \text{Reg}_3^{\text{orc}}(T) \\
&\leq \Delta_{\max} \min \left\{ T, p \left(1 + \frac{4}{\alpha^2} \right) + p + p \left(1 + \frac{8 \log T}{\alpha^2} \right) \right\} \\
&= \Delta_{\max} \min \left\{ T, p \left(3 + \frac{4}{\alpha^2} + \frac{8 \log T}{\alpha^2} \right) \right\}
\end{aligned}$$

This completes the proof of Theorem 38. □

16.1 Proof of Lemma 39

Proof. We observe that the intersection $(B^t)^C \cap (C^t)^C$ implies $(A^t)^C$. This follows because, under $(B^t)^C$ (typical reward history) recalled below:

$$(B^t)^C = \left\{ \forall j \in S_\epsilon^* : \hat{\mu}_j^t \geq \tau_\epsilon + \frac{\alpha}{2} \right\} \cap \left\{ \forall j \in \hat{S}^t \setminus S_\epsilon^* : \hat{\mu}_j^t \leq \tau_\epsilon - \frac{\alpha}{2} \right\}$$

and $(C^t)^C$ (typical posterior sampling) recalled below:

$$(C^t)^C = \left\{ \forall j \in S_\epsilon^* : \vartheta_j^t \geq \hat{\mu}_j^t - \frac{\alpha}{2} \right\} \cap \left\{ \forall j \in \hat{S}^t \setminus S_\epsilon^* : \vartheta_j^t \leq \hat{\mu}_j^t + \frac{\alpha}{2} \right\},$$

the Thompson samples, drawn from Beta distribution, satisfy $\vartheta_j^t > \tau_\epsilon$ for all signal arms $j \in S_\epsilon^*$ and $\vartheta_j^t < \tau_\epsilon$ for all noise arms $j \notin S_\epsilon^*$.

Since $(B^t)^C \cap (C^t)^C$ implies $(A^t)^C$, we have $(B^t)^C \cap (C^t)^C \cap A^t = \emptyset$. Under these conditions, the algorithm necessarily selects $\hat{S}^t = S_\epsilon^*$, meaning no error occurs $((A^t)^C)$. Consequently, we can decompose the total expected regret by partitioning the error event A^t across the

intersections of B^t, C^t , and $(D^t)^C$:

$$\begin{aligned}
\text{Reg}(T) &= \sum_{t=1}^T \mathbb{E} [\Delta_{\hat{S}^t} \mathbb{1}(A^t)] \\
&= \sum_{t=1}^T \mathbb{E} [\Delta_{\hat{S}^t} \mathbb{1}(A^t \cap B^t) + \Delta_{\hat{S}^t} \mathbb{1}(A^t \cap (B^t)^C \cap C^t)] \\
&= \sum_{t=1}^T \mathbb{E} \left[\Delta_{\hat{S}^t} \mathbb{1}(A^t \cap B^t) + \Delta_{\hat{S}^t} \mathbb{1}(A^t \cap (B^t)^C \cap C^t \cap (D^t)^C) \right. \\
&\quad \left. + \Delta_{\hat{S}^t} \mathbb{1}(A^t \cap (B^t)^C \cap C^t \cap D^t) \right].
\end{aligned}$$

This completes the proof of Lemma 39. □

16.2 Proof of Lemma 40

This section proves the bounds (85), (86), and (87) presented in Lemma 40.

16.2.1 Bounding the First Regret Component

This section proves the bound (85), recalled below:

$$\text{Reg}_1^{\text{orc}}(T) \leq \min \left\{ p \left(1 + \frac{4}{\alpha^2} \right), T \right\} \Delta_{\max}.$$

Proof. First, $\text{Reg}_1^{\text{orc}}(T)$ has a trivial upper bound:

$$\sum_{t=1}^T \mathbb{E} [\Delta_{\hat{S}^t} \mathbb{1}(A^t \cap B^t)] \leq \Delta_{\max} T.$$

We can also bound $\text{Reg}_1^{\text{orc}}(T)$ in the following way. It is decomposed into

$$\begin{aligned} \text{Reg}_1^{\text{orc}}(T) &\leq \Delta_{\max} \sum_{j \in S_\epsilon^*} \mathbb{E} \left[\sum_{t=1}^T \mathbb{1} \left\{ j \in \hat{S}^t, \hat{\mu}_j^t < \tau_\epsilon + \frac{\alpha}{2} \right\} \right] \\ &\quad + \Delta_{\max} \sum_{j \notin S_\epsilon^*} \mathbb{E} \left[\sum_{t=1}^T \mathbb{1} \left\{ j \in \hat{S}^t, \hat{\mu}_j^t > \tau_\epsilon - \frac{\alpha}{2} \right\} \right]. \end{aligned}$$

To bound the first and second terms, we invoke Assumption 2. Specifically, we use $\theta_j^*(\hat{G}^{t-1}) = \theta_j^{\text{orc}} \geq \tau_\epsilon + \alpha$ for all $j \in S_\epsilon^*$ for the first term, and $\theta_j(\hat{G}^{t-1}) = \theta_j^{\text{orc}} < \tau_\epsilon - \alpha$ for all $j \notin S_\epsilon^*$ for the second term, respectively. As the proof structure is identical for both cases, we detail the bound only for the first term in the regret upper bound above.

Without loss of generality, fix a signal arm $j \in S_\epsilon^*$. We define $\tau_0 = 0$ and denote by $\tau_1, \tau_2, \dots$ the timestamps t at which the j -th arm is selected (i.e., $j \in \hat{S}^t$). Let $d(a, b)$ denote the KL divergence of two Bernoulli random variables with success probabilities a and b . We bound the expected number of times a signal arm $j \in S_\epsilon^*$ is selected while its empirical mean is misleadingly low:

$$\begin{aligned} \sum_{t=1}^T \mathbb{1} \left\{ j \in \hat{S}^t, \hat{\mu}_j^t < \tau_\epsilon + \frac{\alpha}{2} \right\} &\leq 1 + \sum_{w=1}^{N_j^{T+1}} \mathbb{P} \left(\hat{\mu}_j^{\tau_w} < \tau_\epsilon + \frac{\alpha}{2}, N_j^t = w \right) \\ &\stackrel{(i)}{\leq} 1 + \sum_{w=1}^{N_j^{T+1}} \exp \left(-w d \left(\tau_\epsilon + \frac{\alpha}{2}, \tau_\epsilon + \alpha \right) \right) \\ &\stackrel{(ii)}{\leq} 1 + \sum_{w=1}^{N_j^{T+1}} \exp \left(-w \frac{\alpha^2}{2} \right) \\ &\leq 1 + \sum_{w=1}^{\infty} \exp \left(-w \frac{\alpha^2}{2} \right) \\ &\stackrel{(iii)}{\leq} 1 + \frac{e^{-\alpha^2/2}}{1 - e^{-\alpha^2/2}} \\ &\leq 1 + \frac{4}{\alpha^2}, \end{aligned} \tag{88}$$

where each step uses

- (i) Chernhoff-Hoeffding bound for the average of w independent Bernoulli random variables, monotonicity of the KL divergence $d(a, b)$ in its second argument for $b > a$, and $\theta_j^*(\hat{G}^{t-1}) = \theta_j^{\text{orc}} \geq \tau_\epsilon + \alpha$ for all $j \in S_\epsilon^*$ and t ,
- (ii) The Pinsker's Inequality $d(a, b) \geq 2(a - b)^2$ which cancels out τ_ϵ ,
- (iii) The geometric series.

Multiplying this with $p\Delta_{\max}$ and taking the minimum with $\Delta_{\max}T$ completes the proof the bound (85). $\square$

16.2.2 Bounding the Second Regret Component

This section proves the bound (86), which is recalled below:

$$\text{Reg}_2^{\text{orc}}(T) = \sum_{t=1}^T \mathbb{E}[\Delta_{\hat{S}^t} \times \mathbb{1}(A^t \cap (B^t)^C \cap C^t \cap (D^t)^C)] \leq \min\{p, T\} \Delta_{\max}.$$

Proof. Recall that $\text{Reg}_2^{\text{orc}}(T)$ is defined as

$$\text{Reg}_2^{\text{orc}}(T) = \sum_{t=1}^T \mathbb{E}[\Delta_{\hat{S}^t} \times \mathbb{1}(A^t \cap (B^t)^C \cap C^t \cap (D^t)^C)].$$

Recall that the event $(D^t)^C$ is defined in (84) as

$$N_j^t > \frac{8 \log T}{\alpha^2},$$

which is equivalent to

$$\frac{\alpha}{2} > \sqrt{\frac{2 \log T}{N_j^t}}.$$

The set $(B^t)^C \cap C^t \cap (D^t)^C$ thus implies:

$$\left\{ \exists j \in \hat{S}^t \setminus S_\epsilon^* : \vartheta_j^t - \hat{\mu}_j^t > \sqrt{\frac{2 \log T}{N_j^t}} \text{ or } \exists j \in \hat{S}^t \cap S_\epsilon^* : \hat{\mu}_j^t - \vartheta_j^t > \sqrt{\frac{2 \log T}{N_j^t}} \right\}. \quad (89)$$

To proceed, we use the following lemma. Its proof is a straightforward extension of Lemma 3 in [Liu & Ročková \(2023\)](#), utilizing the approach from Lemma 36, and is therefore omitted.

Lemma 41 (generalization of Lemma 3 of [Liu & Ročková 2023](#)). *If $N_j^t \geq \frac{2(a_j^0-1)^2}{\log T}$, we have*

$$\mathbb{P} \left[\vartheta_j^t - \hat{\mu}_j^t > \sqrt{\frac{2 \log T}{N_j^t}} \right] \leq \frac{1}{T}, \quad \mathbb{P} \left[\hat{\mu}_j^t - \vartheta_j^t > \sqrt{\frac{2 \log T}{N_j^t}} \right] \leq \frac{1}{T}.$$

Since Theorem 38 assumes $T \geq \max_{j \in [p]} \exp(\alpha(a_j^0-1)/2)$, and we assume the event $(D^t)^C$, we can apply Lemma 41 to (89). Then we can write

$$\begin{aligned} \text{Reg}_2^{\text{orc}}(T) &= \sum_{t=1}^T \Delta_{\hat{S}^t} \mathbb{P}(A^t \cap (B^t)^C \cap C^t \cap (D^t)^C) \leq \Delta_{\max} \sum_{t=1}^T \mathbb{P}((B^t)^C \cap C^t \cap (D^t)^C) \\ &\leq \Delta_{\max} \sum_{t=1}^T \frac{p}{T} = p \Delta_{\max}. \end{aligned}$$

Taking the minimum between this and trivial bound $T \Delta_{\max}$ completes the proof of the bound (86). $\square$

16.2.3 Bounding the Third Regret Component

This section proves the bound (87), recalled below:

$$\text{Reg}_3^{\text{orc}}(T) \leq \min \left\{ p \left(1 + \frac{8 \log T}{\alpha^2} \right), T \right\} \Delta_{\max}$$

the term $\text{Reg}_3^{\text{orc}}(T)$, which is mainly driven by

$$D^t = \bigcup_{j \in \hat{S}^t} \left\{ N_j^t \leq \frac{8 \log T}{\alpha^2} \right\} :$$

$$\begin{aligned} \text{Reg}_3^{\text{orc}}(T) &= \mathbb{E} \left[\sum_{t=1}^T \Delta_{\hat{S}^t} \mathbb{1}(A^t \cap (B^t)^C \cap (C^t)^C \cap D^t) \right] \\ &\stackrel{(i)}{\leq} \mathbb{E} \left[\sum_{t=1}^T \Delta_{\hat{S}^t} \mathbb{1}(D^t) \right] \\ &\stackrel{(ii)}{\leq} \mathbb{E} \left[\Delta_{\max} \sum_{t=1}^T \sum_{j \in \hat{S}^t} \mathbb{1}(\{N_j^t \leq (8 \log T)/\alpha^2\}) \right] \\ &\stackrel{(iii)}{=} \mathbb{E} \left[\Delta_{\max} \sum_{t=1}^T \sum_{j=1}^p \mathbb{1}(\{j \in \hat{S}^t\} \cap \{N_j^t \leq \frac{8 \log T}{\alpha^2}\}) \right] \\ &= \mathbb{E} \left[\Delta_{\max} \sum_{j=1}^p \sum_{t=1}^T \mathbb{1}(\{j \in \hat{S}^t\} \cap \{N_j^t \leq \frac{8 \log T}{\alpha^2}\}) \right] \\ &\stackrel{(iv)}{=} \mathbb{E} \left[\Delta_{\max} \sum_{j=1}^p \sum_{w=1}^{N_j^{T+1}} \mathbb{1}(\{N_j^{\tau_j^w} \leq \frac{8 \log T}{\alpha^2}\}) \right] \\ &\stackrel{(v)}{=} \Delta_{\max} \sum_{j=1}^p \mathbb{E} \left[\sum_{w=1}^{N_j^{T+1}} \mathbb{1}(\{w-1 \leq \frac{8 \log T}{\alpha^2}\}) \right] \\ &\leq \Delta_{\max} p \left[N_j^{T+1} \times \mathbb{1}\left(N_j^{T+1} \leq \frac{8 \log T}{\alpha^2} + 1\right) \right. \\ &\quad \left. + \left(\frac{8 \log T}{\alpha^2} + 1\right) \times \mathbb{1}\left(N_j^{T+1} \geq \frac{8 \log T}{\alpha^2} + 1\right) \right] \\ &\leq \Delta_{\max} p \left(\frac{8 \log T}{\alpha^2} + 1 \right). \end{aligned}$$

where each step uses

(i) Inclusion of events,

(ii) On the event D^t , there exists at least one variable $j \in \hat{S}^t$ such that $N_j^t \leq (8 \log T)/\alpha^2$

- (iii) Moving $j \in \hat{S}^t$ to the indicator,
- (iv) Removing timestamps t such that j -th variable was not selected (τ_j^w is the w -th timestamp that j -th variable is pulled; Definition 44),
- (v) $N_j^{\tau_j^w} = w - 1$ by Definition 44 and (77).

17 Proof of Theorem 5

Proof. Recall from (83) that $A^t = \{\hat{S}^t \neq S_\epsilon^*\}$ denotes the event that the TVS algorithm selects an incorrect variable set at iteration t . By the Borel-Cantelli lemma, to establish almost sure convergence, it suffices to show that the sum of these error probabilities over all time steps is finite:

$$\sum_{t=1}^{\infty} \mathbb{P}(A^t) < \infty.$$

Applying a union bound over the p total arms, we can decompose the global error into armwise error events:

$$\sum_{t=1}^{\infty} \mathbb{P}(A^t) \leq \sum_{j \notin S_\epsilon^*} \sum_{t=1}^{\infty} \mathbb{P}(\vartheta_j^t > \tau_\epsilon) + \sum_{j \in S_\epsilon^*} \sum_{t=1}^{\infty} \mathbb{P}(\vartheta_j^t < \tau_\epsilon).$$

Therefore, the proof proceeds by upper-bounding these infinite sums for both the noise arms ($j \notin S_\epsilon^*$) and the signal arms ($j \in S_\epsilon^*$).

For the noise arm ($j \notin S_\epsilon^*$), we bound the probability sum $\sum_{t=1}^{\infty} \mathbb{P}(\vartheta_j^t > \tau_\epsilon)$ by generalizing Lemma 9 of Liu & Ročková (2023) (proof in Section 17.1).

Lemma 42 (Generalization of Lemma 9 of Liu & Ročková 2023). *Assume that there exist selection threshold $\tau_\epsilon \in (0, 1)$ and margin $\alpha \in (0, 1 - \tau_\epsilon)$ that satisfy the Assumption 2. Let $\{a_j^0, b_j^0\}_{j=1}^p$ be the prior parameters. For each noise variable $j \in S \setminus S_\epsilon^*$, let*

$M_j = \max \left(0, \left\lceil \frac{4(a_j^0 - 1)}{\alpha} \right\rceil \right)$. We have:

$$\sum_{t=1}^{\infty} \mathbb{P}(\vartheta_j^t > \tau_\epsilon) < M_j + 1 + \frac{4}{\alpha^2} + \frac{1}{1 - \exp(-\alpha^2/8)}.$$

By Lemma 42, for the noise arms, we have

$$\sum_{j \notin S_\epsilon^*} \sum_{t=1}^{\infty} \mathbb{P}(\vartheta_j^t > \tau_\epsilon) < p \max_{j \notin S_\epsilon^*} M_j + p \left(1 + \frac{4}{\alpha^2} + \frac{1}{1 - \exp(-\alpha^2/8)} \right) < \infty.$$

For the signal arms, we require the following lemma (proof in Section 17.2).

Lemma 43 (Generalization of Lemma 10 of Liu & Ročková 2023). *Fix $j \in S_\epsilon^*$. Assume that there exist selection threshold $\tau_\epsilon \in (0, 1)$ and margin $\alpha \in (0, 1 - \tau_\epsilon)$ that satisfy the Assumption 2. Let $\{a_j^0, b_j^0\}_{j=1}^p$ be the prior parameters. Then we have*

$$\sum_{t=1}^{\infty} \mathbb{P}(\vartheta_j^t < \tau_\epsilon) \leq C_2(\tau_\epsilon, \alpha, a_j^0, b_j^0) < \infty,$$

where $C_2(\tau_\epsilon, \alpha, a_j^0, b_j^0)$ is a constant that do not depend on n, p, S^* or any other problem parameters.

Therefore, we have shown that

$$\sum_{t=1}^{\infty} \mathbb{P}(A^t) \leq \sum_{j \notin S_\epsilon^*} \sum_{t=1}^{\infty} \mathbb{P}(\vartheta_j^t > \tau_\epsilon) + \sum_{j \in S_\epsilon^*} \sum_{t=1}^{\infty} \mathbb{P}(\vartheta_j^t < \tau_\epsilon) = p C_3(\tau_\epsilon, \alpha, a_j^0, b_j^0, \Delta_{\max}) < \infty.$$

where $C_3(\tau_\epsilon, \alpha, a_j^0, b_j^0, \Delta_{\max})$ does not depend on p, n, S^* or any other problem parameter.

Therefore for fixed p and TVS algorithm parameters $\tau_\epsilon, \alpha, a_j^0, b_j^0$ that satisfies the Assump-

tion 2, the sum $\sum_{t=1}^{\infty} \mathbb{P}(A^t)$ is finite. By the Borel-Cantelli lemma, we have, with p fixed,

$$\mathbb{P}\left(\liminf_{t \rightarrow \infty} \hat{S}^t = S^* \mid \widetilde{\mathbf{X}}\right) = 1.$$

This completes the proof of Theorem 5. □

17.1 Proof of Lemma 42

Proof. Fix a noise variable $j \in S \setminus S_\epsilon^*$. Recall the notation defined in (77):

$$N_j^t = \sum_{k < t} \mathbb{1}(j \in \hat{S}^k) = a_j^{t-1} + b_j^{t-1} - a_j^0 - b_j^0 \quad \text{and} \quad \hat{\mu}_j^t = \frac{a_j^{t-1} - a_j^0}{\sum_{k < t} \mathbb{1}(j \in \hat{S}^k)} = \frac{a_j^{t-1} - a_j^0}{N_j^t}.$$

The proof proceeds as follows:

$$\begin{aligned} \sum_{t=1}^{\infty} \mathbb{P}(\vartheta_j^t > \tau_\epsilon) &\leq \sum_{t=1}^{\infty} \mathbb{P}(\hat{\mu}_j^t \geq \tau_\epsilon - \frac{\alpha}{2}, \vartheta_j^t > \tau_\epsilon) + \sum_{t=1}^{\infty} \mathbb{P}\left[(\vartheta_j^t \geq \hat{\mu}_j^t + \alpha/2, \vartheta_j^t > \tau_\epsilon)\right] \\ &\stackrel{(i)}{\leq} 1 + \frac{4}{\alpha^2} + \sum_{t=1}^{\infty} \mathbb{P}\left[(\vartheta_j^t \geq \hat{\mu}_j^t + \alpha/2, \vartheta_j^t > \tau_\epsilon)\right] \\ &\stackrel{(iii)}{=} 1 + \frac{4}{\alpha^2} + \sum_{w=0}^{\infty} \sum_{t=1}^{\infty} \mathbb{E}\left[\mathbb{1}\{\vartheta_j^t \geq \hat{\mu}_j^t + \alpha/2\} \mathbb{1}\{\vartheta_j^t > \tau_\epsilon\} \mathbb{1}\{N_j^t = w\}\right] \\ &\stackrel{(iv)}{=} 1 + \frac{4}{\alpha^2} + \sum_{w=0}^{\infty} \mathbb{E}\left[\mathbb{1}\{\vartheta_j^{\tau_j^w} \geq \hat{\mu}_j^{\tau_j^w} + \alpha/2\}\right] \\ &\leq 1 + \frac{4}{\alpha^2} + M_j + \sum_{w=M_j}^{\infty} \mathbb{E}\left[\mathbb{1}\{\vartheta_j^{\tau_j^w} \geq \hat{\mu}_j^{\tau_j^w} + \alpha/2\}\right] \\ &\stackrel{(v)}{\leq} 1 + \frac{4}{\alpha^2} + M_j + \sum_{w=M_j}^{\infty} \exp\left(-w \frac{\alpha^2}{8}\right) \\ &\stackrel{(vi)}{\leq} M_j + 1 + \frac{4}{\alpha^2} + \frac{1}{1 - \exp(-\alpha^2/8)}. \end{aligned}$$

where each step uses

(i) Decomposition of the selection event around the empirical mean:

$$\{\vartheta_j^t > \tau_\epsilon\} \subseteq \left\{ \hat{\mu}_j^t \geq \tau_\epsilon - \frac{\alpha}{2}, \vartheta_j^t > \tau_\epsilon \right\} \cup \left\{ \tilde{\theta}_i^t - \hat{\mu}_j^t > \frac{\alpha}{2}, \vartheta_j^t > \tau_\epsilon \right\}.$$

(ii) $\{\vartheta_j^t > \tau_\epsilon\} = \{j \in \hat{S}^t\}$ and the same logic as (88), which relies on Assumption 2 and uses Chernoff-Hoeffding bound, Pinsker's inequality, and the geometric series.

(iii) the Tonelli's Theorem,

(iv) for a fixed w , the condition $\vartheta_j^t > \tau_\epsilon, N_j^t = w$ translates to: "arm j is pulled at time t , and it has been pulled exactly w times previously," which can happen at only one t , which we denote as τ_j^w

(v) By the definition of τ_j^w , at time τ_j^w , we have $N_j^t = w$, and Lemma 36 states that for $N_j^t \geq M_j \geq \frac{4(a_j^0 - 1)}{\alpha}$, we have

$$\mathbb{P}(\vartheta_j^t \geq \hat{\mu}_j^t + \alpha/2 | N_j^t) \leq \exp\left(-N_j^t \frac{\alpha^2}{8}\right),$$

(vi) the geometric series.

This completes the proof of Lemma 42. □

17.2 Proof of Lemma 43

Proof. Note that in between $[\tau_j^w + 1, \tau_j^{w+1} - 1]$ we have $\{\vartheta_j^t < \tau_\epsilon\}$ since the arm j is not being pulled. One can deduce that

$$\sum_{t=1}^{\infty} \mathbb{P}(\vartheta_j^t < \tau_\epsilon) = \sum_{j=0}^{\infty} \mathbb{E}[\tau_j^{w+1} - \tau_j^w - 1].$$

To bound the infinite sum, we partition it at the critical horizon $W_0 = \left\lceil \frac{4(\tau_\epsilon + \alpha/2)C_0}{\alpha} \right\rceil$. To

simplify notation, let the leading prior constant be $K_\tau = \frac{1}{(1-\tau_\epsilon)^{C_0}}$. We write:

$$\sum_{w=0}^{\infty} \left(\mathbb{E}[\tau_j^{w+1} - \tau_j^w] - 1 \right) = \sum_{w=0}^{W_0-1} \left(\mathbb{E}[\tau_j^{w+1} - \tau_j^w] - 1 \right) + \sum_{w=W_0}^{\infty} \left(\mathbb{E}[\tau_j^{w+1} - \tau_j^w] - 1 \right).$$

For the first term (the early transient phase), we apply the constant bound from Lemma 46. Since this is a finite sum of bounded terms, it evaluates to a finite constant:

$$\begin{aligned} \sum_{w=0}^{W_0-1} \left(\mathbb{E}[\tau_j^{w+1} - \tau_j^w] - 1 \right) &\leq \sum_{w=0}^{W_0-1} \left[K_\tau \left(2 + \frac{1}{\alpha} \right) - 1 \right] \\ &= W_0 \left[K_\tau \left(2 + \frac{1}{\alpha} \right) - 1 \right]. \end{aligned}$$

For the second term (the asymptotic tail), we apply Lemma 47. Crucially, we subtract 1 from both sides of the inequality in Lemma 47, which perfectly cancels the +1 term on the right-hand side. This leaves only strictly decaying components:

$$\mathbb{E}[\tau_j^{w+1} - \tau_j^w] - 1 \leq K_\tau \left[2 \exp \left(-\frac{w\alpha^2}{8} \right) + 2 \exp \left(-\frac{w\alpha^2}{2} \right) \right] + \frac{1}{\exp((w + C_0)\alpha^2/2) - 1}.$$

Summing this infinite tail yields three geometric series. First, we evaluate the two exponential terms bounded by K_τ , starting from $w = W_0$:

$$\sum_{w=W_0}^{\infty} K_\tau \left[2 \exp \left(-\frac{w\alpha^2}{8} \right) + 2 \exp \left(-\frac{w\alpha^2}{2} \right) \right] \leq K_\tau \left[\frac{2 \exp(-W_0\alpha^2/8)}{1 - \exp(-\alpha^2/8)} + \frac{2 \exp(-W_0\alpha^2/2)}{1 - \exp(-\alpha^2/2)} \right].$$

Finally, we sum the remaining fractional term. Because $C_0 \geq 0$, we strictly relax the denominator by dropping C_0 to yield $\frac{1}{\exp(w\alpha^2/2) - 1}$. As established in previous lemmas, we

partition this sum at $\lceil 2/\alpha^2 \rceil$ to apply the exponential bound $\frac{1}{e^x - 1} \leq e^{-x/2}$ for $x \geq 1$:

$$\begin{aligned}
\sum_{w=W_0}^{\infty} \frac{1}{\exp(w\alpha^2/2) - 1} &\leq \sum_{w=W_0}^{\lceil 2/\alpha^2 \rceil} \frac{1}{\exp(w\alpha^2/2) - 1} + \sum_{w=\lceil 2/\alpha^2 \rceil}^{\infty} \frac{1}{\exp(w\alpha^2/2) - 1} \\
&\leq \frac{2}{\alpha^2} \left(\frac{1}{\exp(W_0\alpha^2/2) - 1} \right) + \sum_{w=\lceil 2/\alpha^2 \rceil}^{\infty} \exp\left(-\frac{w\alpha^2}{4}\right) \\
&\leq \frac{2}{\alpha^2(\exp(W_0\alpha^2/2) - 1)} + \frac{\exp(-1/2)}{1 - \exp(-\alpha^2/4)}.
\end{aligned}$$

Combining the finite transient sum with the converged infinite geometric series, we obtain an explicit, finite upper bound for the total sum:

$$\begin{aligned}
\sum_{w=0}^{\infty} \left(\mathbb{E}[\tau_j^{w+1} - \tau_j^w] - 1 \right) &\leq W_0 \left[\frac{1}{(1 - \tau_\epsilon)^{C_0}} \left(2 + \frac{1}{\alpha} \right) - 1 \right] \\
&\quad + \frac{1}{(1 - \tau_\epsilon)^{C_0}} \left[\frac{2 \exp(-W_0\alpha^2/8)}{1 - \exp(-\alpha^2/8)} + \frac{2 \exp(-W_0\alpha^2/2)}{1 - \exp(-\alpha^2/2)} \right] \\
&\quad + \frac{2}{\alpha^2(\exp(W_0\alpha^2/2) - 1)} + \frac{\exp(-1/2)}{1 - \exp(-\alpha^2/4)} \\
&:= C_2(\tau_\epsilon, \alpha, a_j^0, b_j^0).
\end{aligned}$$

Because every term on the right-hand side is a finite constant defined purely by the hyperparameters $(\tau_\epsilon, \alpha, a_j^0, b_j^0)$, the infinite sum is strictly finite. This completes the proof of Lemma 43. $\square$

18 Expected Failure Analysis for the Signal Arm

Fix a signal arm $j \in S_\epsilon^*$. This section is dedicated to bounding from above the expected number of *not pulling* the j th arm, namely

$$\sum_{t=1}^{\infty} \mathbb{P}(\vartheta_j^t < \tau_\epsilon),$$

where $\vartheta_j^t \sim \text{Beta}(a_j^t, b_j^t)$ is the Thompson sample. The key of the proof is decomposing this sum of probabilities into a sum of waiting times. To formally define the waiting time, we present the following definition:

Definition 44 (Timestamp of w -th pull). *Let τ_j^w denote the time step t at which arm j is pulled ($\vartheta_j^t \geq \tau_\epsilon$) for the w -th time. Consequently, the sequence $\{\tau_j^w\}_{w=1}^\infty$ forms a random, strictly increasing subsequence of the natural numbers (e.g., 2, 10, 35, 234, ...).*

Now, we note that if $t \in [\tau_j^w + 1, \tau_j^{w+1} - 1]$, we have $\{\vartheta_j^t < \tau_\epsilon\}$ since the arm j is *not being pulled* between the pulls. Therefore, we can rewrite the probability sum as the sum of expected waiting times:

$$\sum_{t=1}^{\infty} \mathbb{P}(\vartheta_j^t < \tau_\epsilon) = \sum_{w=0}^{\infty} \mathbb{E}[\tau_j^{w+1} - \tau_j^w - 1]. \quad (90)$$

The subsequent subsections are devoted to bounding $\mathbb{E}[\tau_j^{w+1} - \tau_j^w]$.

18.1 Waiting as a geometric distribution

Conditioned on the history at time τ_j^w , the Thompson samples ϑ_j^t are i.i.d. draws from the same Beta distribution:

$$\vartheta_j^t \sim \text{Beta}(a_j^{\tau_j^w}, b_j^{\tau_j^w}), \quad t = \tau_j^w, (\tau_j^w + 1), \dots, (\tau_j^{w+1} - 1).$$

For $t = \tau_j^w, (\tau_j^w + 1), \dots, (\tau_j^{w+1} - 1)$, the TVS algorithm selects the arm j at time t if and only if $\vartheta_j^t > \tau_\epsilon$. Consequently, the number of iterations until the next selection, namely $(\tau_j^{w+1} - \tau_j^w)$, follows a geometric distribution with success probability:

$$p_{j,C}^w := \mathbb{P}(\vartheta_j^t > \tau_\epsilon \mid a_j^{\tau_j^w}, b_j^{\tau_j^w}).$$

By the properties of the geometric distribution, the conditional expected waiting time is the reciprocal of this probability:

$$\mathbb{E}[\tau_j^{w+1} - \tau_j^w \mid a_j^{\tau_j^w}, b_j^{\tau_j^w}] = \frac{1}{p_{j,C}^w}.$$

18.2 Unconditional Expectation via Relationship Between Beta Survival and Binomial CDF

To evaluate the success probability $p_{j,C}^w$, we utilize the fundamental identity linking the Beta Cumulative Distribution Function (CDF) to the Binomial CDF. For any positive integers α and β :

$$F_{\text{beta}}(y; \alpha, \beta) = 1 - F_{\text{bin}}(\alpha - 1; \alpha + \beta - 1, y). \quad (91)$$

For clarity, let use define

$$S_j^w := \text{the number of successful rewards observed from } w \text{ pulls of arm } j. \quad (92)$$

In our context, starting from a general prior $\text{Beta}(a_j^0, b_j^0)$, the Beta posterior where the Thompson samples are drawn from is

$$\text{Beta}(a_j^0 + S_j^w, b_j^0 + w - S_j^w).$$

Evaluating the survival function (the probability of being pulled) at the threshold $y = \tau_\epsilon \in (0, 1)$:

$$p_{j,C}^w = \mathbb{P}(\vartheta_j^t > \tau_\epsilon \mid S_j^w, w) = 1 - F_{\text{beta}}(\tau_\epsilon; a_j^0 + S_j^w, b_j^0 + w - S_j^w).$$

Substituting the identity, we express $p_{j,C}^w$ purely in terms of the prior parameters, the past

successes S_j^w , and the total pull count w :

$$\begin{aligned} p_{j,C}^w &= 1 - \left[1 - F_{\text{bin}}((a_j^0 + S_j^w) - 1; (a_j^0 + S_j^w) + (b_j^0 + w - S_j^w) - 1, \tau_\epsilon) \right] \\ &= F_{\text{bin}}(a_j^0 + S_j^w - 1; a_j^0 + b_j^0 + w - 1, \tau_\epsilon). \end{aligned}$$

Note that the only randomness lies in S_j^w , and everything else is constant. That is to say, $1/p_{j,C}^w$ is a random variable which is a transformation of S_j^w . The conditional expected waiting time, $\mathbb{E}[\tau_j^{w+1} - \tau_j^w \mid S_j^w]$, is a function of the random success count S_j^w . To obtain the unconditional expected waiting time, we apply the law of total expectation by marginalizing over all possible success counts $S_j^w \in \{0, 1, \dots, w\}$:

$$\begin{aligned} \mathbb{E}[\tau_j^{w+1} - \tau_j^w] &= \mathbb{E}[\mathbb{E}[\tau_j^{w+1} - \tau_j^w \mid a_j^{\tau_j^w}, b_j^{\tau_j^w}]] \\ &= \mathbb{E}\left[\frac{1}{p_{j,C}^w}\right] \\ &= \sum_{s=0}^w \frac{f_j^*(s; w)}{F_{\text{bin}}(a_j^0 + s - 1; a_j^0 + b_j^0 + w - 1, \tau_\epsilon)}, \end{aligned} \tag{93}$$

where $f_j^*(s; w)$ denotes the probability mass function of S_j^w evaluated at $S_j^w = s$.

18.3 Simplifying the Unconditional Expectation

We further simplify the sum form of the unconditional expectation given in (93), recalled below:

$$\mathbb{E}[\tau_j^{w+1} - \tau_j^w] = \sum_{s=0}^w \frac{f_j^*(s; w)}{F_{\text{bin}}(a_j^0 + s - 1; a_j^0 + b_j^0 + w - 1, \tau_\epsilon)}.$$

To simplify the denominator, we use two properties of the Binomial CDF. First, since our prior satisfies $a_j^0 \geq 1$, we know $a_j^0 + s - 1 \geq s$. Because the CDF is non-decreasing with

respect to the number of successes, we have:

$$F_{\text{bin}}(a_j^0 + s - 1; a_j^0 + b_j^0 + w - 1, \tau_\epsilon) \geq F_{\text{bin}}(s; a_j^0 + b_j^0 + w - 1, \tau_\epsilon).$$

Second, we utilize the recurrence property of the Binomial CDF with respect to the number of trials:

$$F_{\text{bin}}(k; n + 1, p) \geq (1 - p)F_{\text{bin}}(k; n, p).$$

By applying this recurrence relation exactly $(a_j^0 + b_j^0 - 1)$ times, we can reduce the number of trials in our CDF from $(a_j^0 + b_j^0 + w - 1)$ down to exactly w :

$$F_{\text{bin}}(s; a_j^0 + b_j^0 + w - 1, \tau_\epsilon) \geq (1 - \tau_\epsilon)^{a_j^0 + b_j^0 - 1} F_{\text{bin}}(s; w, \tau_\epsilon).$$

Substituting this lower bound into our unconditional expectation, we can extract a constant pre-factor based purely on the chosen priors and threshold:

$$\mathbb{E}[\tau_j^{w+1} - \tau_j^w] \leq \frac{1}{(1 - \tau_\epsilon)^{a_j^0 + b_j^0 - 1}} \sum_{s=0}^w \frac{f_j^*(s; w)}{F_{\text{bin}}(s; w, \tau_\epsilon)}. \quad (94)$$

To address the numerator, the following lemma provides an upper bound for $f_j^*(s; w)$, generalizing Lemma 5 of [Liu & Ročková \(2023\)](#) to an arbitrary threshold τ_ϵ (proof in Section 18.5). Because s is a sum of Bernoulli random variables with varying success probabilities, this analysis relies on Assumption 2.

Lemma 45 (Generalization of Lemma 5 of [Liu & Ročková 2023](#)). *Assume that there exist selection threshold $\tau_\epsilon \in (0, 1)$ and margin $\alpha \in (0, 1 - \tau_\epsilon)$ that satisfy the Assumption 2. Let $j \in S_\epsilon^*$ be a signal arm, and let $f_j^*(s; w)$ be the probability that after w pulls of arm j ,*

exactly s result in a reward. If $s \leq \lfloor \tau_\epsilon w \rfloor$, then:

$$f_j^*(s; w) \leq \binom{w}{s} (\tau_\epsilon + \alpha)^s (1 - \tau_\epsilon - \alpha)^{w-s}.$$

Leveraging this Lemma, subsequent subsections explicitly bound the expected waiting time

$$\mathbb{E}[\tau_j^{w+1} - \tau_j^w]$$

18.4 Bounding the Expected in Two Regimes

We bound the unconditional expected waiting time given in (94) (which relies on Assumption 2) recalled below:

$$\mathbb{E}[\tau_j^{w+1} - \tau_j^w] \leq \frac{1}{(1 - \tau_\epsilon)^{a_j^0 + b_j^0 - 1}} \sum_{s=0}^w \frac{f_j^*(s; w)}{F_{\text{bin}}(s; w, \tau_\epsilon)}.$$

Let us denote

$$\Sigma(s_1, s_2) := \sum_{s=s_1}^{s_2} \frac{f_i^*(s; w)}{F_{\text{bin}}(s; w, \tau_\epsilon)}.$$

Thus bounding the expected waiting time hinges on bounding the sums $\Sigma(s_1, s_2)$.

The following lemma (proof in Section 18.6) does not rely on any assumptions about w and shows that the early sum $\Sigma(0, \lfloor \tau_\epsilon w \rfloor)$ decays exponentially in w . However, without further assumptions on w , the remainder of the sum can only be bounded by a constant.

Lemma 46 (Generalization of Lemma 6 of Liu & Ročková 2023). *Fix $j \in S_\epsilon^*$. Assume that there exist selection threshold $\tau_\epsilon \in (0, 1)$ and margin $\alpha \in (0, 1 - \tau_\epsilon)$ that satisfy the Assumption 2. Let $\{a_j^0, b_j^0\}_{j=1}^p$ be the prior parameters. Then we have*

$$\sum_{s=0}^{\lfloor \tau_\epsilon w \rfloor} \frac{f_j^*(s; w)}{F_{\text{bin}}(s; w, \tau_\epsilon)} \leq \frac{1}{\alpha} \exp(-2\alpha^2 w), \quad (95)$$

and

$$\sum_{s=\lceil \tau_\epsilon w \rceil}^w \frac{f_j^*(s; w)}{F_{\text{bin}}(s; w, \tau_\epsilon)} \leq 2,$$

which result in

$$\mathbb{E}[\tau_j^{w+1} - \tau_j^w] \leq \frac{2 + 1/\alpha}{(1 - \tau_\epsilon)^{a_j^0 + b_j^0 - 1}}.$$

Next, the following lemma (proof in Section 18.7) shows that for sufficiently large w ($w > 2/\alpha$), the expected waiting time $\mathbb{E}[\tau_j^{w+1} - \tau_j^w]$ is bounded by an exponentially decaying function of w .

Lemma 47 (Generalization of Lemma 6 of Liu & Ročková 2023). *Fix $j \in S_\epsilon^*$ such that Assume that there exist selection threshold $\tau_\epsilon \in (0, 1)$ and margin $\alpha \in (0, 1 - \tau_\epsilon)$ that satisfy the Assumption 2. Let $\{a_j^0, b_j^0\}_{j=1}^p$ be the prior parameters. Then for $w \geq \frac{4(\tau_\epsilon + \alpha/2)(a_j^0 + b_j^0 - 1)}{\alpha}$, we have*

$$\begin{aligned} \mathbb{E}[\tau_j^{w+1} - \tau_j^w] &\leq \frac{1}{(1 - \tau_\epsilon)^{a_j^0 + b_j^0 - 1}} \left[2 \exp\left(-\frac{w\alpha^2}{8}\right) + 2 \exp\left(-\frac{w\alpha^2}{2}\right) \right] + 1 \\ &\quad + \frac{1}{\exp\left(-\frac{(a_j^0 + b_j^0 - 1)\alpha^2}{2}\right) \exp\left(-\frac{w\alpha^2}{2}\right) - 1} \end{aligned}$$

18.5 Proof of Lemma 45

Proof. Let us denote the rewards from w pulls of arm j as $Y_j^{\tau_j^1}, \dots, Y_j^{\tau_j^w} \in \{0, 1\}$. By Assumption 2, their expected rewards of satisfies

$$\mathbb{E}[Y_j^{\tau_j^1}] = \dots = \mathbb{E}[Y_j^{\tau_j^w}] = \theta_j^{\text{orc}} \geq \tau_\epsilon + \alpha.$$

However, for the time being, we admit different success probabilities for each of w pulls and denote the individual success probabilities $p_1, \dots, p_w$. We eventually show that the probability mass function is upper bounded at $p_1 = p_2 = \dots, p_w = \tau_\epsilon + \alpha$.

The probability of observing exactly s successes in w independent trials is given by the Poisson binomial probability mass function, which is a multi-linear polynomial in terms of the individual success probabilities $p_1, \dots, p_w$:

$$f(p_1, \dots, p_w) := \sum_{A \subseteq [w], |A|=s} \prod_{k \in A} p_k \prod_{m \notin A} (1 - p_m).$$

Because this function is multi-linear with respect to each p_k , its extrema over the valid parameter space $p_k \in [\tau_\epsilon + \alpha, 1]$ must occur at the boundaries. Specifically, we want to show that under the condition $s \leq \lfloor \tau_\epsilon w \rfloor$, the maximum probability occurs when all p_k are at their lowest possible bound: $p_k = \tau_\epsilon + \alpha$. Suppose, for the sake of finding the maximum, we set r variables to their upper boundary of 1 (where $0 \leq r \leq s$), and the remaining $w - r$ variables to the lower boundary of $\tau_\epsilon + \alpha$. Let the probability of observing exactly s total successes in this configuration be denoted as $P(r)$:

$$P(r) := \binom{w-r}{r} (\tau_\epsilon + \alpha)^r (1 - \tau_\epsilon - \alpha)^{w-s}.$$

To show this probability is maximized at $r = 0$, we evaluate the ratio of adjacent configurations $P(r+1)$ and $P(r)$:

$$\frac{P(r+1)}{P(r)} = \frac{\binom{w-r-1}{r-1}}{\binom{w-r}{r}} \frac{(\tau_\epsilon + \alpha)^{r-1}}{(\tau_\epsilon + \alpha)^r} = \frac{r}{w-r} \frac{1}{\tau_\epsilon + \alpha}.$$

We can bound the fraction $\frac{r}{w-r}$ using the condition that $s \leq \lfloor \tau_\epsilon w \rfloor \implies s < \tau_\epsilon w$. Furthermore, since $r \geq 0$ and $\tau_\epsilon < 1$, we know that $-r \leq -\tau_\epsilon r$, which allows us to bound the numerator further:

$$\frac{r}{w-r} \leq \frac{\tau_\epsilon w - r}{w-r} \leq \frac{\tau_\epsilon w - \tau_\epsilon r}{w-r} = \tau_\epsilon$$

Substituting this upper bound into our ratio test yields:

$$\frac{P(r+1)}{P(r)} \leq \frac{\tau_\epsilon}{\tau_\epsilon + \alpha}$$

Since $\alpha > 0$, it is strictly true that $\frac{\tau_\epsilon}{\tau_\epsilon + \alpha} < 1$. Because $\frac{P(r+1)}{P(r)} < 1$ holds for all valid r , the sequence $P(r)$ is strictly decreasing. This confirms that the maximum possible probability is achieved at $r = 0$, where every pull operates at the worst-case expected reward of $\tau_\epsilon + \alpha$.

This establishes our upper bound:

$$f_j^*(s; w) \leq \binom{w}{s} (\tau_\epsilon + \alpha)^s (1 - \tau_\epsilon - \alpha)^{w-s}.$$

This completes the proof of Lemma [45](#). □

18.6 Proof of Lemma 46

Proof. The first argument is shown as follows:

$$\begin{aligned}
\sum_{s=0}^{\lfloor \tau_\epsilon w \rfloor} \frac{f_j^*(s; w)}{F_{\text{bin}}(s; w, \tau_\epsilon)} &\stackrel{(i)}{\leq} \sum_{s=0}^{\lfloor \tau_\epsilon w \rfloor} \frac{\binom{w}{s} (\tau_\epsilon + \alpha)^s (1 - \tau_\epsilon - \alpha)^{w-s}}{F_{\text{bin}}(s; w, \tau_\epsilon)} \\
&\stackrel{(ii)}{\leq} \sum_{s=0}^{\lfloor \tau_\epsilon w \rfloor} \frac{\binom{w}{s} (\tau_\epsilon + \alpha)^s (1 - \tau_\epsilon - \alpha)^{w-s}}{\binom{w}{s} \tau_\epsilon^s (1 - \tau_\epsilon)^{w-s}} \\
&= \left(\frac{1 - \tau_\epsilon - \alpha}{1 - \tau_\epsilon} \right)^w \sum_{s=0}^{\lfloor \tau_\epsilon w \rfloor} \left(\frac{(\tau_\epsilon + \alpha)(1 - \tau_\epsilon)}{\tau_\epsilon(1 - \tau_\epsilon - \alpha)} \right)^s \\
&\stackrel{(iii)}{=} \left(\frac{1 - \tau_\epsilon - \alpha}{1 - \tau_\epsilon} \right)^w \sum_{s=0}^{\lfloor \tau_\epsilon w \rfloor} r^s \\
&\stackrel{(iv)}{=} \left(\frac{1 - \tau_\epsilon - \alpha}{1 - \tau_\epsilon} \right)^w \frac{r^{\lfloor \tau_\epsilon w \rfloor + 1} - 1}{r - 1} \\
&\leq \left(\frac{1 - \tau_\epsilon - \alpha}{1 - \tau_\epsilon} \right)^w \frac{r \cdot r^{\tau_\epsilon w}}{r - 1} \\
&= \left[\left(\frac{1 - \tau_\epsilon - \alpha}{1 - \tau_\epsilon} \right) r^{\tau_\epsilon} \right]^w \left(\frac{r}{r - 1} \right) \\
&\stackrel{(v)}{\leq} \frac{1}{\alpha} \left[\left(\frac{1 - \tau_\epsilon - \alpha}{1 - \tau_\epsilon} \right) r^{\tau_\epsilon} \right]^w \\
&= \frac{1}{\alpha} \exp \left(-w \left[\tau_\epsilon \log \frac{\tau_\epsilon}{\tau_\epsilon + \alpha} + (1 - \tau_\epsilon) \log \frac{1 - \tau_\epsilon}{1 - \tau_\epsilon - \alpha} \right] \right) \\
&\stackrel{(vi)}{=} \frac{1}{\alpha} \exp(-w, d(\tau_\epsilon, \tau_\epsilon + \alpha)) \\
&\stackrel{(vii)}{\leq} \frac{1}{\alpha} \exp(-2\alpha^2 w)
\end{aligned}$$

where each step uses:

(i) Lemma 45,

(ii) The property that a single term of the Binomial PMF is a strict lower bound for its corresponding CDF: $F_{\text{bin}}(s; w, \tau_\epsilon) \geq \binom{w}{s} \tau_\epsilon^s (1 - \tau_\epsilon)^{w-s}$, allowing us to cancel the combinatorial term,

- (iii) The substitution $r := \frac{(\tau_\epsilon + \alpha)(1 - \tau_\epsilon)}{\tau_\epsilon(1 - \tau_\epsilon - \alpha)} > 1$,
- (iv) The formula for a finite geometric sum,
- (v) The algebraic simplification $\frac{r}{r-1} = \frac{(\tau_\epsilon + \alpha)(1 - \tau_\epsilon)}{\alpha} \leq \frac{1}{\alpha}$ (since $\tau_\epsilon + \alpha \leq 1$ and $1 - \tau_\epsilon \leq 1$),
- (vi) The definition of the Kullback-Leibler (KL) divergence between two Bernoulli distributions with parameters p and q :

$$d(p, q) = p \log \frac{p}{q} + (1 - p) \log \frac{1 - p}{1 - q},$$

- (vii) Pinsker's inequality, which states $d(p, q) \geq 2(p - q)^2$.

Now we move onto the second argument which bounds

$$\sum_{s=\lceil \tau_\epsilon w \rceil}^w \frac{f_j^*(s; w)}{F_{\text{bin}}(s; w, \tau_\epsilon)}.$$

without assumption on w . In this range, s is always greater than or equal to the mean of the Binomial distribution $B(w, \tau_\epsilon)$. A fundamental property of the Binomial distribution is that its median m always satisfies $\lfloor \tau_\epsilon w \rfloor \leq m \leq \lceil \tau_\epsilon w \rceil$. Consequently, for any $s \geq \lceil \tau_\epsilon w \rceil$, the cumulative probability $F_{\text{bin}}(s; w, \tau_\epsilon)$ represents the probability of having at most s successes, where s is already at or beyond the center of the distribution. This ensures that:

$$F_{\text{bin}}(s; w, \tau_\epsilon) \geq F_{\text{bin}}(\lceil \tau_\epsilon w \rceil; w, \tau_\epsilon) \geq \frac{1}{2}.$$

Substituting this lower bound into the upper tail summation, we can pull the constant out

of the sum:

$$\sum_{s=\lceil \tau_\epsilon w \rceil}^w \frac{f_j^*(s; w)}{F_{\text{bin}}(s; w, \tau_\epsilon)} \leq \sum_{s=\lceil \tau_\epsilon w \rceil}^w \frac{f_j^*(s; w)}{1/2} = 2 \sum_{s=\lceil \tau_\epsilon w \rceil}^w f_j^*(s; w) \leq 2(1) = 2,$$

where the last equality uses the fact that since $f_j^*(s; w)$ is a valid probability mass function, the sum of all its values over $s \in [0, w]$ is exactly 1, and therefore, the sum over any subset of s is strictly bounded by 1.

Now for the final argument, we follow (94) and multiply the previously established bounds by $\frac{1}{(1 - \tau_\epsilon)^{a_j^0 + b_j^0 - 1}}$. Combining two re-scaled bounds yield

$$\mathbb{E}[\tau_j^{w+1} - \tau_j^w] \leq \frac{2 + 1/\alpha}{(1 - \tau_\epsilon)^{a_j^0 + b_j^0 - 1}}.$$

This completes the proof of Lemma 46. □

18.7 Proof of Lemma 47

Proof. We start from (94), which relies on Assumption 2:

$$\mathbb{E}[\tau_j^{w+1} - \tau_j^w] \leq \frac{1}{(1 - \tau_\epsilon)^{a_j^0 + b_j^0 - 1}} \sum_{s=0}^w \frac{f_j^*(s; w)}{F_{\text{bin}}(s; w, \tau_\epsilon)}.$$

By (95), we already know that

$$\Sigma(0, \lfloor \tau_\epsilon w \rfloor) \leq \frac{1}{\alpha} \exp(-2\alpha^2 w).$$

Therefore we focus on bounding $\Sigma(\lfloor \tau_\epsilon w \rfloor, w)$. We partition the sum into three parts and bound each part as:

1. $\Sigma(\lceil \tau_\epsilon w \rceil, \lceil \tau_\epsilon w \rceil) \leq 2 \exp(-2w\alpha^2)$ (Section 18.7.1),

$$2. \Sigma(\lceil \tau_\epsilon w \rceil + 1, \lfloor (\tau_\epsilon + \alpha/2)(w + a_j^0 + b_j^0 - 1) \rfloor) \leq 2 \exp\left(-\frac{w\alpha^2}{8}\right) \text{ (Section 18.7.2)}$$

3. Section 18.7.3:

$$\frac{1}{(1 - \tau_\epsilon)^{a_j^0 + b_j^0 - 1}} \Sigma(\lfloor (\tau_\epsilon + \alpha/2)(w + a_j^0 + b_j^0 - 1) \rfloor + 1, w) \leq 1 + \frac{1}{\exp\left(-\frac{(a_j^0 + b_j^0 - 1)\alpha^2}{2}\right) \exp\left(-\frac{w\alpha^2}{2}\right) - 1}$$

Summing these up and multiplying by the prior pre-factor, we get:

$$\begin{aligned} \mathbb{E}[\tau_j^{w+1} - \tau_j^w] &\leq \frac{1}{(1 - \tau_\epsilon)^{a_j^0 + b_j^0 - 1}} \left[2 \exp\left(-\frac{w\alpha^2}{8}\right) + 2 \exp\left(-\frac{w\alpha^2}{2}\right) \right] + 1 \\ &\quad + \frac{1}{\exp\left(-\frac{(a_j^0 + b_j^0 - 1)\alpha^2}{2}\right) \exp\left(-\frac{w\alpha^2}{2}\right) - 1}. \end{aligned}$$

This completes the proof of Lemma 47. $\square$

18.7.1 Bounding the First Sum in Section 18.7

In this regime, the sum contains only one term corresponding to $s = \lceil \tau_\epsilon w \rceil$:

$$\Sigma(\lceil \tau_\epsilon w \rceil, \lceil \tau_\epsilon w \rceil) = \frac{f_i^*(\lceil \tau_\epsilon w \rceil; w)}{F_{\text{bin}}(\lceil \tau_\epsilon w \rceil; w, \tau_\epsilon)}.$$

Therefore, we lower bound the denominator and upper bound the numerator.

We start with the denominator. For binomial distribution with w trials and success probability τ_ϵ , its median is bounded below by $\lfloor \tau_\epsilon w \rfloor$ and above by $\lceil \tau_\epsilon w \rceil$. Consequently, evaluating the CDF at or above the mean guarantees a probability of at least 1/2. In a more formal way, for $s = \lceil \tau_\epsilon w \rceil$:

$$F_{\text{bin}}(\lceil \tau_\epsilon w \rceil; w, \tau_\epsilon) \geq \frac{1}{2}. \tag{96}$$

Next, we upper bound the numerator $f_j^*(s; w)$ for $s = \lceil \tau_\epsilon w \rceil$. Under Assumption 2, the success probability is $\theta_j^{\text{orc}} \geq \tau_\epsilon + \alpha$. Thus, observing $s = \lceil \tau_\epsilon w \rceil$ successes represents a substantial lower-tail deviation of at least αw from the expected mean $w(\tau_\epsilon + \alpha)$. By the Chernoff-Hoeffding inequality:

$$f_j^*(s; w) \leq \mathbb{P}(\text{Bin}(w, \theta_j^{\text{orc}}) \leq \tau_\epsilon w) \leq \exp(-w \cdot d(\tau_\epsilon, \theta_j^{\text{orc}})).$$

Applying Pinsker's inequality, $d(p, q) \geq 2(p - q)^2$, we have:

$$d(\tau_\epsilon, \theta_j^{\text{orc}}) \geq 2(\theta_j^{\text{orc}} - \tau_\epsilon)^2 \geq 2\alpha^2.$$

Therefore, $f_j^*(s; w) \leq \exp(-2w\alpha^2)$. Combining the bounds for the numerator and denominator yields the bound for this single term:

$$\Sigma(\lceil \tau_\epsilon w \rceil, \lceil \tau_\epsilon w \rceil) = \frac{f_j^*(\lceil \tau_\epsilon w \rceil; w)}{F_{\text{bin}}(\lceil \tau_\epsilon w \rceil; w, \tau_\epsilon)} \leq \frac{\exp(-2w\alpha^2)}{1/2} = 2\exp(-2w\alpha^2).$$

18.7.2 Bounding the Second Sum in Section 18.7

For this second regime, we want to establish the following bound:

$$\Sigma(\lceil \tau_\epsilon w \rceil + 1, \lfloor (\tau_\epsilon + \alpha/2)w \rfloor) = \sum_{s=\lceil \tau_\epsilon w \rceil + 1}^{\lfloor (\tau_\epsilon + \alpha/2)(w + a_j^0 + b_j^0 - 1) \rfloor} \frac{f_i^*(s; w)}{F_{\text{bin}}(s; w, \tau_\epsilon)} \leq 2\exp\left(-\frac{w\alpha^2}{8}\right).$$

The proof proceeds as follows:

$$\begin{aligned}
\Sigma(\lceil \tau_\epsilon w \rceil + 1, \lfloor (\tau_\epsilon + \alpha/2)(w + a_j^0 + b_j^0 - 1) \rfloor) &= \sum_{s=\lceil \tau_\epsilon w \rceil + 1}^{\lfloor (\tau_\epsilon + \alpha/2)(w + a_j^0 + b_j^0 - 1) \rfloor} \frac{f_j^*(s; w)}{F_{\text{bin}}(s; w, \tau_\epsilon)} \\
&\stackrel{(i)}{\leq} 2 \sum_{s=\lceil \tau_\epsilon w \rceil + 1}^{\lfloor (\tau_\epsilon + \alpha/2)(w + a_j^0 + b_j^0 - 1) \rfloor} f_j^*(s; w) \\
&\stackrel{(ii)}{\leq} 2\mathbb{P}(S_j^w \leq \lfloor (\tau_\epsilon + \alpha/2)(w + a_j^0 + b_j^0 - 1) \rfloor) \\
&\stackrel{(iii)}{\leq} 2\mathbb{P}\left(\frac{S_j^w}{w} \leq \tau_\epsilon + \frac{\alpha}{2} + \frac{(\tau_\epsilon + \alpha/2)(a_j^0 + b_j^0 - 1)}{w}\right) \\
&\stackrel{(iv)}{\leq} 2\exp(-2w\lambda_w^2) \\
&\stackrel{(v)}{\leq} 2\exp\left(-2w\left(\frac{\alpha}{4}\right)^2\right) \\
&= 2\exp\left(-\frac{w\alpha^2}{8}\right).
\end{aligned}$$

where each step uses:

- (i) $\frac{1}{F_{\text{bin}}(s; w, \tau_\epsilon)} \leq 2$, as shown by the logic of (96),
- (ii) The definition of $f_j^*(s; w)$ as the PMF of S_j^w , bounding the partial sum by the CDF,
- (iii) Dividing by the number of trials w to evaluate the empirical mean,
- (iv) Hoeffding's inequality for $S_j^w \sim \text{Bin}(w, \theta_j^{\text{orc}})$. Here, the effective deviation is $\lambda_w := \theta_j^{\text{orc}} - (\tau_\epsilon + \alpha/2) - \frac{(\tau_\epsilon + \alpha/2)(a_j^0 + b_j^0 - 1)}{w}$,
- (v) For sufficiently large w , specifically $w \geq \frac{4(\tau_\epsilon + \alpha/2)(a_j^0 + b_j^0 - 1)}{\alpha}$, the penalty term is at most $\alpha/4$. Since $\theta_j^{\text{orc}} - (\tau_\epsilon + \alpha/2) \geq \alpha/2$ by Assumption 2, we are guaranteed that the effective deviation $\lambda_w \geq \alpha/4$.

18.7.3 Bounding the Third Sum in Section 18.7

Let $X \sim \text{Bin}(a_j^0 + b_j^0 + w - 1, \tau_\epsilon)$, so that the denominator is exactly:

$$F_{\text{bin}}(s; a_j^0 + b_j^0 + w - 1, \tau_\epsilon) = \mathbb{P}(X \leq s) = 1 - \mathbb{P}(X > s).$$

We define the observed deviation from the threshold mean as:

$$\lambda := \frac{s}{w + a_j^0 + b_j^0 - 1} - \tau_\epsilon$$

For every term in this sum, we have

$$s \geq \lfloor (\tau_\epsilon + \alpha/2)(w + a_j^0 + b_j^0 - 1) \rfloor + 1 \geq (\tau_\epsilon + \alpha/2)(w + a_j^0 + b_j^0 - 1)$$

Substituting this into our deviation λ , we bound λ from below as:

$$\lambda \geq \frac{(\tau_\epsilon + \alpha/2)(w + a_j^0 + b_j^0 - 1)}{w + a_j^0 + b_j^0 - 1} - \tau_\epsilon = \frac{\alpha}{2}.$$

We now apply Hoeffding's inequality to the upper tail of the binomial distribution (note that τ_ϵ is the population mean)

$$\begin{aligned} \mathbb{P}(X > s) &= \mathbb{P}\left(\frac{X}{w + a_j^0 + b_j^0 - 1} - \tau_\epsilon > \lambda\right) \\ &\leq \exp\left(-2(w + a_j^0 + b_j^0 - 1)\lambda^2\right) \\ &\leq \exp\left(-2(a_j^0 + b_j^0 - 1)\left(\frac{\alpha}{2}\right)^2\right) \exp\left(-2w\left(\frac{\alpha}{2}\right)^2\right) \\ &= \exp\left(-\frac{(a_j^0 + b_j^0 - 1)\alpha^2}{2}\right) \exp\left(-\frac{w\alpha^2}{2}\right). \end{aligned}$$

Consequently, we can bound the cumulative distribution function from below:

$$\begin{aligned} F_{\text{bin}}(s; a_j^0 + b_j^0 + w - 1, \tau_\epsilon) &= 1 - \mathbb{P}(X > s) \\ &\geq 1 - \exp\left(-\frac{(a_j^0 + b_j^0 - 1)\alpha^2}{2}\right) \exp\left(-\frac{w\alpha^2}{2}\right). \end{aligned}$$

Taking the reciprocal gives us a uniform upper bound for the denominator of every term in our summation:

$$\frac{1}{F_{\text{bin}}(s; a_j^0 + b_j^0 + w - 1, \tau_\epsilon)} \leq \frac{1}{1 - \exp\left(-\frac{(a_j^0 + b_j^0 - 1)\alpha^2}{2}\right) \exp\left(-\frac{w\alpha^2}{2}\right)}.$$

Finally, we substitute this back into our original sum. Using the fact that the probability masses must sum to at most 1 (i.e., $\sum_s f_j^*(s; w) \leq 1$), we arrive at the final bound:

$$\begin{aligned} &\frac{1}{(1 - \tau_\epsilon)^{a_j^0 + b_j^0 - 1}} \sum_{s=\lfloor(\tau_\epsilon + \alpha/2)(w + a_j^0 + b_j^0 - 1)\rfloor + 1}^w \frac{f_j^*(s; w)}{F_{\text{bin}}(s; a_j^0 + b_j^0 + w - 1, \tau_\epsilon)} \\ &\leq \frac{\sum_s f_j^*(s; w)}{1 - \exp\left(-\frac{(a_j^0 + b_j^0 - 1)\alpha^2}{2}\right) \exp\left(-\frac{w\alpha^2}{2}\right)} \\ &\leq 1 + \frac{1}{\exp\left(-\frac{(a_j^0 + b_j^0 - 1)\alpha^2}{2}\right) \exp\left(-\frac{w\alpha^2}{2}\right) - 1}. \end{aligned}$$

19 Details for ISEE Subroutine

This section provides the omitted details for the ISEE subroutine introduced in Section 3.4 of the main text. We proceed in the following steps:

1. Regression relationship on the data sub-vectors (Section 19.1),
2. Key identities on transformed cluster center and noise (Section 19.2).

Without loss of generality, we focus on the case where $i \in G_1^*$. The derivations build on

those in [Fan & Lv \(2016\)](#), with the key distinction that they only consider the special case $\mathbf{c}_1^* = 0$.

19.1 Regression relationship on the data sub-vectors

Using the conditional distribution property of multivariate Gaussians and the Schur complement, we derive the following regression relationship between sub-vectors of $\mathbf{X}_i \sim \mathcal{N}(\mathbf{c}_1^*, \Sigma^*)$ for $i \in G_1^*$, as presented in equation (17) in Section 3.4 of the main text. Given a small index set $A \subset [p]$ of size 2 or 3, we recall the following relationship:

$$\underbrace{\mathbf{X}_{A,i}}_{\text{response} \in \mathbb{R}^{|A|}} = \underbrace{(\mathbf{c}_1^*)_A + (\Omega_{A,A}^*)^{-1} \Omega_{A,A^c}^* (\mathbf{c}_1^*)_{A^c}}_{:= (\alpha_1)_A \in \mathbb{R}^{|A|} \text{ (intercept)}} - \underbrace{(\Omega_{A,A}^*)^{-1} \Omega_{A,A^c}^*}_{\text{slope} \in \mathbb{R}^{|A| \times (p-|A|)}} \underbrace{\mathbf{X}_{A^c,i}}_{\text{predictor} \in \mathbb{R}^{p-|A|}} + \underbrace{\mathbf{E}_{A,i}}_{\text{residual} \in \mathbb{R}^{|A|}},$$

where $\mathbf{E}_{A,i} \sim \mathcal{N}(0, (\Omega_{A,A}^*)^{-1})$. Let us partition $\mathbf{X}_i$, $\mathbf{c}_1^*$, and Ω^* according to :

$$\mathbf{X}_i = \begin{bmatrix} \mathbf{X}_{A,i} \\ \mathbf{X}_{A^c,i} \end{bmatrix}, \quad \mathbf{c}_1^* = \begin{bmatrix} (\mathbf{c}_1^*)_A \\ (\mathbf{c}_1^*)_{A^c} \end{bmatrix}, \quad \Sigma^* = \begin{bmatrix} \Sigma_{A,A}^* & \Sigma_{A,A^c}^* \\ \Sigma_{A^c,A}^* & \Sigma_{A^c,A^c}^* \end{bmatrix}.$$

Then the conditional distribution $\mathbf{X}_{A,i} \mid \mathbf{X}_{A^c,i}$ follows $\mathcal{N}(\bar{\mathbf{c}}, \bar{\Sigma})$, where

$$\bar{\mathbf{c}} := (\mathbf{c}_1^*)_A + \Sigma_{A,A^c}^* (\Sigma_{A^c,A^c}^*)^{-1} (\tilde{\mathbf{X}}_{A^c,i} - (\mathbf{c}_1^*)_{A^c}), \quad (97)$$

and

$$\bar{\Sigma} := \Sigma_{A,A}^* - \Sigma_{A,A^c}^* (\Sigma_{A^c,A^c}^*)^{-1} \Sigma_{A^c,A}^*. \quad (98)$$

Next, to apply the Schur complement, we first introduce the following simplified notation for the matrix partition:

$$\Sigma^* = \begin{bmatrix} \Sigma_{A,A}^* & \Sigma_{A,A^c}^* \\ \Sigma_{A^c,A}^* & \Sigma_{A^c,A^c}^* \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix}, \quad (99)$$

Then by the Schur complement, we have:

$$\Omega^* = (\Sigma^*)^{-1} = \begin{bmatrix} \Omega_{A,A}^* & \Omega_{A,A^c}^* \\ \Omega_{A^c,A}^* & \Omega_{A^c,A^c}^* \end{bmatrix} = \begin{bmatrix} \mathbf{W} & -\mathbf{W}\mathbf{B}\mathbf{D}^{-1} \\ -\mathbf{D}^{-1}\mathbf{C}\mathbf{W} & \mathbf{D}^{-1} + \mathbf{D}^{-1}\mathbf{C}\mathbf{W}\mathbf{B}\mathbf{D}^{-1} \end{bmatrix}, \quad (100)$$

where

$$\mathbf{W} := (\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C})^{-1}. \quad (101)$$

Leveraging this representation of Ω^* , we can derive:

$$-\Sigma_{A,A^c}^* (\Sigma_{A^c,A^c}^*)^{-1} \stackrel{(i)}{=} -\mathbf{B}\mathbf{D}^{-1} \stackrel{(ii)}{=} \mathbf{W}^{-1} (-\mathbf{W}\mathbf{B}\mathbf{D}^{-1}) \stackrel{(iii)}{=} (\Omega_{A,A}^*)^{-1} \Omega_{A,A^c}^*,$$

where step (i) uses (99), step (ii) uses (101), and step (iii) uses (100). This lets us rewrite $\bar{\mathbf{c}}$ in (97) as follows:

$$\begin{aligned} \bar{\mathbf{c}} &= (\mathbf{c}_1^*)_A + \Sigma_{A,A^c}^* (\Sigma_{A^c,A^c}^*)^{-1} (\tilde{\mathbf{X}}_{A^c,i} - (\mathbf{c}_1^*)_{A^c}) \\ &= (\mathbf{c}_1^*)_A - (\Omega_{A,A}^*)^{-1} \Omega_{A,A^c}^* (\tilde{\mathbf{X}}_{A^c,i} - (\mathbf{c}_1^*)_{A^c}) \\ &= \underbrace{(\mathbf{c}_1^*)_A + (\Omega_{A,A}^*)^{-1} \Omega_{A,A^c}^* (\mathbf{c}_1^*)_{A^c}}_{(\alpha_1)_A} - (\Omega_{A,A}^*)^{-1} \Omega_{A,A^c}^* \tilde{\mathbf{X}}_{A^c,i}, \end{aligned}$$

Likewise, we can also simplify $\bar{\Sigma}$ in (98) as follows:

$$\bar{\Sigma} = \Sigma_{A,A}^* - \Sigma_{A,A^c}^* (\Sigma_{A^c,A^c}^*)^{-1} \Sigma_{A^c,A}^* \stackrel{(i)}{=} \mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C} \stackrel{(ii)}{=} \mathbf{W}^{-1} = (\Omega_{A,A}^*)^{-1},$$

where step (i) uses (99), step (ii) uses (101), and step (iii) uses (100). Substituting these expressions into the conditional distribution $\mathbf{X}_{A,i} \mid \mathbf{X}_{A^c,i} \sim \mathcal{N}(\bar{\mathbf{c}}, \bar{\Sigma})$, we obtain the regression relationship (17) in Section 3.4 of the main text.

19.2 Key identities on transformed cluster center and noise

Next, we turn to the two key identities presented in equation (18) in Section 3.4 of the main text. :

$$(\tilde{\mathbf{c}}_1^*)_A = \Omega_{A,A}^*(\alpha_1)_A \quad \text{and} \quad (\tilde{\mathcal{E}}_i)_A = \Omega_{A,A} \mathbf{E}_{A,i},$$

which play a central role in the ISEE subroutine.

We start with the first identity. By pre-multiplying $\Omega_{A,A}^*$ to $(\alpha_1)_A$, we have:

$$\Omega_{A,A}^*(\alpha_1)_A = \Omega_{A,A}^*(\mathbf{c}_1^*)_A + \Omega_{A,A^c}^*(\mathbf{c}_1^*)_{A^c}.$$

Therefore, we can express $(\tilde{\mathbf{c}}_1^*)_A$ as follows:

$$(\tilde{\mathbf{c}}_1^*)_A = \Omega_{A,\cdot}^* \mathbf{c}_1^* = \Omega_{A,A}^*(\mathbf{c}_1^*)_A + \Omega_{A,A^c}^*(\mathbf{c}_1^*)_{A^c} = \Omega_{A,A}^*(\alpha_1)_A.$$

This completes the derivation of the first identity.

Next, we derive the second identity:

$$\begin{aligned} \tilde{\mathcal{E}}_{A,i} &= \Omega_{A,\cdot}^* \mathbf{X}_i - \Omega_{A,\cdot}^* \mathbf{c}_1^* \\ &= \Omega_{A,A}^* \mathbf{X}_{A,i} + \Omega_{A,A^c}^* \mathbf{X}_{A^c,i} - \Omega_{A,A}^*(\mathbf{c}_1^*)_A + \Omega_{A,A^c}^*(\mathbf{c}_1^*)_{A^c} \\ &= \Omega_{A,A}^* \left(\mathbf{X}_{A,i} + (\Omega_{A,A}^*)^{-1} \Omega_{A,A^c}^* \mathbf{X}_{A^c,i} - (\mathbf{c}_1^*)_A + (\Omega_{A,A}^*)^{-1} \Omega_{A,A^c}^*(\mathbf{c}_1^*)_{A^c} \right) \\ &= \Omega_{A,A}^* \mathbf{E}_{A,i}. \end{aligned}$$

where the last equality uses the regression relationship (17) in Section 3.4 of the main text.

This completes the derivation of the second identity.

20 Algorithm Details

This section provides the omitted details of our proposed greedy and TVS algorithms.

- Section 20.1 provides details on the SDP solver,
- Section 20.2 provides the hyperparameter setting for the the high-dimensional regression method used for ISEE,
- Section 20.3 introduces stopping criterion for our greedy algorithm.

20.1 Details on Initial Clustering and SDP K-means Implementation

We use SDP K-means (Chen & Yang 2021) for the initial clustering step, implemented as `sdp_kmeans.R` in our code repository. The SDP is solved using a custom first-order ADMM solver (Boyd et al. 2011) with adaptive penalty updates and an efficient simplex projection algorithm proposed by Duchi et al. (2008). To improve convergence stability and reduce sensitivity to the initial choice of the ADMM penalty parameter ρ , we implement a residual balancing scheme that uses hyperparamters μ and τ . In this scheme, μ serves as the imbalance threshold, ensuring the primal and dual residuals remain within the same order of magnitude. The parameter τ acts as a scaling factor that dictates the magnitude of adjustments to the penalty parameter ρ . This adaptive mechanism enables the solver to dynamically balance the trade-off between objective optimization and constraint satisfaction. For our implementation, we employ the following hyperparameter configuration: $\rho = 1.0$, $\mu = 10.0$, and $\tau = 2.0$, with a convergence tolerance of 10^{-3} and a maximum of 2,000 iterations.

In the final step of the SDP-relaxed K -means, we extract the leading eigenvector (an n -dimensional vector) from the output matrix $\mathbf{Z}$ and apply the `kmeans` function from the R

`stats` package.

20.2 High-dimensional regression for ISEE Subroutine

We use the lasso ([Tibshirani 1996](#)) for high-dimensional regression between nodes in the ISEE subroutine (Algorithm 5). At each iteration, the regularization parameter is tuned by computing the lasso path and selecting the value that minimizes the Akaike Information Criterion (AIC). This procedure is implemented using functions from the R package `glmnet` ([Friedman et al. 2010](#)).

20.3 Stopping Criteria

For the TVS algorithm, we use a fixed $T = 2000$ iterations. This choice is based on empirical observations showing that a larger number of iterations leads to better convergence of the posterior distribution and, consequently, improved clustering accuracy.

In contrast, the greedy algorithm does not involve accumulated learning and we empirically observe that it can enter short cycles, repeatedly visiting a limited set of states. To address this, we introduce an early stopping criterion based on the adjusted Rand index (ARI; [Hubert & Arabie 1985](#)).

The ARI measures the similarity between two clustering assignments by comparing all pairs of observations and evaluating whether they are assigned to the same or different clusters in both partitions. It corrects for agreement expected by chance by normalizing the Rand Index with respect to its null expectation under random labeling. As a result, the ARI takes the value 1 for identical clusterings, 0 for random agreement, and can be negative when agreement is worse than chance, making it a standardized measure of clustering concordance across settings with different numbers of clusters.

For the greedy algorithm, we monitor the ARI between clusterings from consecutive iter-

Table 3: Summary of parameters for (19) for synthetic data with known (Figure 3) and unknown (Figure 4) covariance settings. For all settings, we have $S^* = \{1, \dots, 10\}$.

Figure	Noise	p	Ω^*	n	Δ_{S^*}
3-(a)	Gaussian	50-5000	$\mathbf{I}_p$ (known)	200	4
3-(b), (c)	(a) Gaussian (b) Laplace	1000-30000	$\mathbf{I}_p$ (known)	200	4
4-(a), (b)	(a) Gaussian (b) Laplace	100-1000	Chain graph	200	6
4-(c), (d)	(c) Gaussian (d) Laplace	100-1000	AR(1)	200	4
4-(e), (f)	(a) Gaussian (b) Laplace	100-400	Chain graph	500	3
4-(g), (h)	(c) Gaussian (d) Laplace	100-400	AR(1)	500	3

ations and terminate the procedure when the ARI equals 1 for 10 consecutive iterations.

We compute the ARI using the `adjustedRandIndex` function from the R package `mclust`.

21 Details for Numerical Study

This section provides additional details on the numerical studies, omitted from Section 4.

All simulations are conducted on a Linux machine equipped with an AMD EPYC 7542 processor (32 cores, 2.90 GHz) and 16 GB of RAM. Table 3 presents detailed parameter settings for our synthetic data study in Section 4.

- Section 21.1 lists hyperparameter values used in the simulations and real data analysis.
- Section 21.2 describes sparsity-aware high-dimensional clustering methods used for comparison.
- Section 21.3 provides details on text preprocessing used in real data analysis.

21.1 Hyperparameter Settings

This section specifies the hyperparameter settings used in synthetic and real data study in the main text.

21.1.1.1 Synthetic Data Study

For the greedy algorithm introduced in Section 3.1, the hyperparameters are specified as follows:

- Target false discovery rate (FDR): $q = 0.4$,
- Maximum number of iterations: $T = 100$,
- Stopping criterion: the algorithm terminates if the adjusted Rand index (ARI) between clusterings from consecutive iterations equals 1 for 10 successive iterations,
- FDR control method: the Benjamini–Hochberg (BH) procedure

For the TVS algorithm introduced in Section 3.2.1, the hyperparameters are defined as follows:

- Maximum number of iterations: $T = 300$,
- Significance threshold: $\tau_\epsilon - \alpha = 0.5$,
- Maximum number of misclustered samples: $r = 1$,
- Beta prior parameters: $a_j^0 = b_j^0 = 1$ for all $j = 1, \dots, p$,
- Regularization parameter C : selected via grid search over $\{0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9\}$ based on silhouette index.

21.1.2 Real Data Study

For the greedy algorithm introduced in Section 3.1, the hyperparameters are specified as follows:

- Target false discovery rate (FDR): $q = 0.4$,
- Maximum number of iterations: $T = 100$,

- Stopping criterion: the algorithm terminates if the adjusted Rand index (ARI) between clusterings from consecutive iterations equals 1 for 10 successive iterations,
- FDR control method:
 - Without ISEE: the Benjamini–Hochberg (BH) procedure,
 - With ISEE: the Significance Analysis of Microarrays (SAM) framework ([Tusher et al. 2001](#)).

For the TVS algorithm introduced in Section 3.2.1, the hyperparameters are defined as follows:

- Maximum number of iterations: $T = 2000$,
- Significance threshold: $\tau_\epsilon - \alpha = 0.3$,
- Maximum number of misclustered samples: $r = 5$,
- Beta prior parameters: $a_j^0 = b_j^0 = 1$ for all $j = 1, \dots, p$,
- Regularization parameter C : selected via grid search over $\{0.3, 0.4, 0.5\}$ based on silhouette index.

21.2 Sparsity-aware clustering methods used for benchmarking

This section describes the baseline sparsity-aware methods used for comparison with our algorithms in the numerical studies presented in Section 4.

IFPCA Influential Feature PCA (IFPCA; [Jin & Wang 2016](#)) is a two step method. In the feature selection step, features with the highest Kolmogorov-Smirnov (KS) scores are selected, with the selection threshold determined in a data-driven manner using the Higher Criticism principle ([Donoho & Jin 2004](#)). In the clustering step, spectral clustering is applied: the top $K - 1$ left singular vectors are extracted from the normalized data matrix

after feature selection and used as inputs to the classical K -means algorithm to estimate cluster labels. For simulations, we use the authors' [MATLAB implementation](#).

SAS Sparse Alternate Similarity clustering (SAS; [Arias-Castro & Pu 2017](#)) is an iterative method that uses the fact that the within-cluster dissimilarity of K -means, measured by the squared ℓ_2 distances, can be decomposed coordinate-wise. The algorithm alternates between clustering and support estimation: given a support set, it applies K -means to estimate the clusters; given the cluster assignments, it selects the top $|\hat{S}^*|$ coordinates with the largest average within-cluster dissimilarity. The tuning parameter $|\hat{S}^*|$, which denotes the number of signal features, is selected using the following permutation-based approach: For each $|\hat{S}^*| = 1, \dots, p$, run the SAS algorithm to obtain corresponding cluster and support estimates. For each run, compare the within-cluster dissimilarity of the selected features under the estimated cluster assignment to that under a randomly permuted cluster assignment. The discrepancy between the two is quantified using the gap statistic ([Tibshirani et al. 2001](#)). Then choose the value of $|\hat{S}^*|$ that yields the largest gap statistic. We use the authors' [R implementation](#).

SKM Sparse K -means (SKM; [Witten & Tibshirani 2010](#)) is an iterative method which exploits the fact that the within-cluster dissimilarity in K -means, measured by squared ℓ_2 distances, can be decomposed coordinate-wise. The method assigns a feature weight Y_j to each variable $j = 1, \dots, p$, and constructs a feature-weighted version of the K -means objective. It then optimizes this objective over both feature weights and cluster assignments, subject to a combined lasso and ridge penalty. The proposed algorithm alternates between maximizing the objective with respect to the cluster assignments (given fixed feature weights) and with respect to the feature weights (given fixed cluster assignments). For simulations, we use the authors' [R package sparcl](#).

CHIME CHIME ([Cai et al. 2019](#)) is an Etype iterative method, where the E-step updates

the cluster assignments based on current Gaussian mixture model parameters, and the step updates the model parameters with cluster labels fixed. Notably, CHIME avoids explicit covariance estimation by employing an ℓ_1 -regularized estimation of the sparse discriminative direction $\beta^* = \Omega^*(\mathbf{c}_1^* - \mathbf{c}_2^*)$. After a suitable number of iterations, the estimated β^* is plugged into the Fisher discriminant rule to produce the final cluster assignments. For the simulations, we translated the official [MATLAB implementation](#) into R. This algorithm requires careful initialization of the model parameters. The initialization procedure suggested in the original paper (Hardt-Price) is exponential-time and is not practical in high-dimensional settings. Therefore, in our simulations, we initialize the model parameters using the estimated clusters obtained via SKM.

SCVX Sparse Convex Clustering (SCVX; [Wang et al. 2018b](#)) extends convex clustering to high-dimensional scenarios by addressing the performance degradation caused by uninformative features. The method simultaneously clusters observations and conducts feature selection by formulating convex clustering as a regularization problem, applying an adaptive group-lasso penalty term to the cluster centers. To optimally balance the trade-off between cluster fitting and sparsity, SCVX employs a tuning criterion based on clustering stability. Theoretically, the estimator guarantees a finite sample error bound and establishes variable selection consistency. Code and further details are provided in the authors' online supplementary material.

21.3 Text Preprocessing Details

We use the `tm` package in R.

1. All text were converted to lowercase.
2. Punctuation marks and numeric characters were systematically excised.
3. Structural language tokens are discarded by applying the standard English stop word

dictionary built into the `tm` package. This dictionary, derived from the standard Snowball stemmer list, contains exactly 174 high-frequency grammatical tokens, including pronouns (e.g., “we”, “itself”, “they”), articles (“a”, “the”), conjunctions (“and”, “because”), and auxiliary verbs (“is”, “would”, “having”).

4. Excess whitespace was stripped.

22 Additional Numerical Results

This section supplements Section 4 by presenting additional simulation results.

- Section 22.1 evaluates our greedy algorithm under known non-isotropic covariance setting.
- Section 22.2 demonstrates the benefits of ISEE bypassing the full covariance structure estimation.
- Section 22.3 demonstrates the robustness of our greedy algorithm to homogeneous covariance assumption.
- Section 22.4 shows the robustness of our greedy algorithm to sparsity assumptions.
- Section 22.5 shows the scalability of our algorithms to large sample sizes, via approximate SDP solvers.
- Section 22.6 compares our sparsity-based algorithms to subspace clustering approaches.

Table 4: Summary of Parameter Settings for Model (19) used in Sections 22.1 and 22.2, with $i, j = 1, \dots, p$.

Section	Covariance structure	p	n	Δ_{S^*}
22.1	known, $\Sigma_{i,j}^* = \rho^{ i-j }$, $\rho \in \{0.2, 0.5, 0.8\}$	50-5000	200	3, 4
22.2	unknown, $\Omega_{i,i+1}^* = \Omega_{i+1,i}^* = \rho$, $\Omega_{i,i}^* = 1$, $\Omega_{i,j}^* = 0$ o.w., $\rho = 0.45$	400	500	3, 4

22.1 Evaluation of Greedy Algorithm under Known Non-isotropic Covariance

As discussed in Section 3.1, the subset SDP problem (6) captures the principal sparse clustering signal encoded in $\Omega^*(\mathbf{c}_1^* - \mathbf{c}_2^*)$. To numerically validate this claim, we evaluate the greedy algorithm from Section 3.1 under a non-isotropic covariance setting, where the true covariance matrix is provided to the procedure. Specifically, we implement a block coordinate optimization algorithm that uses Algorithm 2 for the selection step and Algorithm 1 for the clustering step, omitting the ISEE subroutine (Algorithm 5). The synthetic data are generated according to (19), with parameter settings (including the covariance structure) matching those in line 1 of Table 4.

The results in Figure 5 reveal two main findings. First, the method adapts effectively to the sparsity of $\Omega^*(\mathbf{c}_1^* - \mathbf{c}_2^*)$: when the total clustering signal is fixed (in particular, when $\Delta_{S^*} = 4$), performance remains stable as ambient dimension increases with the clustering signal fixed. Second, when the covariance matrix is known, stronger correlations facilitate clustering. As ρ increases, the miclustering rate deteriorates more slowly with increasing sparsity, again with the total clustering signal held fixed. This behavior is expected, since knowledge of a highly correlated structure is effectively equivalent to operating in a lower-dimensional space.

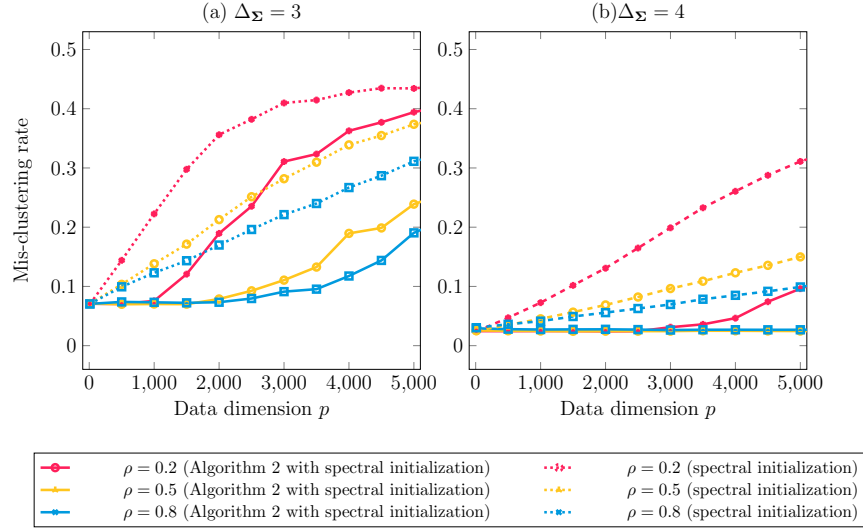


Figure 5: Validation of the subset SDP: under a known non-isotropic (AR(1)) covariance, we solve (6) via a greedy block coordinate method (greedy selection + SDP clustering, without ISEE), with data generated as in (19) and Table 4. The y -axis reports misclustering rates.

22.2 Demonstrating the Benefits of ISEE Bypassing Full Covariance Structure Estimation

We argued in Section 3.4 that a key strength of the greedy algorithm with ISEE is its ability to avoid full estimation of the precision matrix. In this section, we provide numerical evidence supporting this claim by comparing the greedy ISEE-based algorithm with a specially designed variant. At each iteration t , the variant algorithm estimates the full precision matrix using the graphical lasso (Friedman et al. 2008), and then uses this estimate to compute the required quantities, namely $\tilde{\mathbf{X}}$ and $\Omega^*(\mathbf{c}_1^* - \mathbf{c}_2^*)$. The submatrix $\Sigma_{\hat{S}^t, \hat{S}^t}$ is estimated in the same manner as in the greedy algorithm with ISEE. These estimates are subsequently used in the clustering block (Algorithm 1) and selection block (Algorithm 2) of our greedy algorithm. We use synthetic data generated with the setting specified in line 2 of Table 4. The results in Table 5 show that the graphical lasso-based greedy algorithm performs substantially worse than the ISEE-based approach, indicating that avoiding full estimation of the precision matrix leads to improved performance.

Table 5: Miclustering rate of graphical-lasso based and ISEE-based greedy algorithms, with data generated as in (19) and Table 4.

Ambient Dimension (p)	$\Delta_{S^*} = 3$		$\Delta_{S^*} = 4$	
	Graphical Lasso	ISEE	Graphical Lasso	ISEE
50	0.28	0.19	0.06	0.04
100	0.31	0.22	0.07	0.05
150	0.33	0.23	0.10	0.07
200	0.33	0.26	0.14	0.08
250	0.37	0.27	0.18	0.08
300	0.39	0.29	0.22	0.11
350	0.40	0.30	0.24	0.10
400	0.41	0.31	0.25	0.11

22.3 Robustness to the Assumption of Covariance Homogeneity

When cluster covariances differ, our theoretical results no longer apply and the optimal (Bayes) clustering boundary becomes nonlinear (quadratic for Gaussian distributions). Blindly applying our algorithm under covariance heterogeneity presents a common model misspecification challenge, as our algorithm learns a linear boundary. Under mild heterogeneity, we expect our greedy algorithm to retain strong performance. To verify this, we conduct a simulation in which the data distribution remains Gaussian but with two different covariance matrices, $(1 + \delta)\mathbf{I}_p$ and $(1 - \delta)\mathbf{I}_p$. The results, reported in Table 6, show an interesting phenomenon of higher clustering accuracy for larger δ . A plausible explanation is that larger δ corresponds to larger difference in clusters and hence an easier clustering problem. Theoretical understanding of the impact of such heterogeneity is an interesting future research direction.

Table 6: Comparison of clustering accuracy for the greedy algorithm under heterogeneous and homogeneous covariances. All other settings follow line 1 of Table 3 with $\Delta_{S^*} = 4$.

δ	$p = 1000$	$p = 2000$	$p = 3000$	$p = 4000$	$p = 5000$
0 (homogeneous covariance)	0.97	0.93	0.86	0.74	0.68
0.05 (homogeneous covariance)	0.98	0.92	0.84	0.76	0.74
0.1 (heterogeneous covariance)	0.98	0.92	0.91	0.87	0.88

Table 7: Comparison of clustering accuracy for our greedy algorithm under exact and approximate sparsity of the cluster centers. All other settings follow line 1 of Table 3 with $\Delta_{S^*} = 4$.

δ	$p = 1000$	$p = 2000$	$p = 3000$	$p = 4000$	$p = 5000$
0 (exact sparsity)	0.97	0.93	0.86	0.74	0.68
0.02 (approximate sparsity)	0.98	0.95	0.81	0.70	0.63
0.05 (approximate sparsity)	0.97	0.96	0.85	0.75	0.68

22.4 Assumption of Sparsity

Rather than the cluster centers themselves being exactly sparse, we assume that their difference ($\mathbf{c}_1^* - \mathbf{c}_2^*$) is exactly sparse. When this difference is approximately sparse, it implies that all entries have clustering power, just that some have weak signals. Our separation bound in Theorem 1 is derived for the difference $\mathbf{c}_1^* - \mathbf{c}_2^*$, and it does not rule out the approximately sparse scenario where $\|\mathbf{c}_1^* - \mathbf{c}_2^*\|_0 \approx s$. Indeed, our result ensures that for any subset S containing enough signals such that the cumulative signal strength satisfies the inequality in (10), exact recovery of cluster IDs can be achieved.

In practice, having weak signals can cause challenges in feature selection. However, since our goal is clustering and feature selection is conducted only to reduce noise, including a few weak signals does not necessarily make clustering much worse. We conduct additional experiments to support this claim. First, in the known covariance setting of line 1 of Table 3, we add $\delta \in \{0.02, 0.05\}$ to $\lfloor p/10 \rfloor$ noise entries of $\mathbf{c}_1^*$. The result presented in Table 7 shows that our greedy algorithm retains strong performance under approximate sparsity of the cluster centers. Note that the slight increase in clustering accuracy stems from the increased cumulative signal strength from these weak signals.

Under the unknown covariance setting, our greedy algorithm depends on ISEE, which depends on the exact sparsity of the precision matrix. We demonstrate numerically that our greedy algorithm with ISEE remains robust under approximate sparsity of the precision

Table 8: Comparison of clustering accuracy for our greedy algorithm under exact and approximate sparsity of the precision matrix. All other settings follow line 2 of Table 3 with $\Delta_{S^*} = 4$.

δ	$p = 100$	$p = 200$	$p = 300$	$p = 400$
0 (exact sparsity)	0.94	0.95	0.93	0.92
0.02 (added to 10% of zero entries)	0.95	0.96	0.93	0.94
0.05 (added to 10% of zero entries)	0.96	0.91	0.91	0.88
0.001 (added to all zero entries)	0.96	0.95	0.95	0.93

matrix. Starting from an exactly sparse precision matrix Ω^* , we introduce approximate sparsity in two ways. First, we add $\delta \in \{0.02, 0.05\}$ to 10% of the zero entries of Ω^* . Second, we add 0.01 to all zero entries of Ω^* . The result presented in Table 8 shows that our greedy algorithm with ISEE retains strong performance under approximate sparsity of the precision matrix.

22.5 Scalability to Large Samples and Alternative Solvers

Our focus is the high-dimensional, low-sample-size regime, but we also address scalability: for large n , we recommend approximate SDP solvers, and our greedy algorithm shows successful scaling up to $n = 100,000$.

The computational cost of the clustering block of our algorithms (both greedy and TVS) is largely determined by the sample size n , since the decision variable in the SDP is an $n \times n$ matrix. The selection block prunes irrelevant variables, so p primarily affects only the initial clustering stage. However, the ADMbased solver does not scale well beyond $n > 1000$ (see Table 5 of [Sun et al. 2020](#) for details).

For moderately large sample sizes (up to $n = 10,000$), instead of solving the full SDP, we recommend solving its nonnegative low-rank restriction using a nonconvex Burer–Monteiro factorization ([Zhuang et al. 2024](#)). [Zhuang et al. \(2024\)](#) shows that this approach retains

the same statistical clustering guarantees as the full SDP problem. Under the same setting as line 1 of Table 3, we numerically compare the clustering accuracy of our greedy algorithm, where the clustering step is performed either by solving the original SDP formulation or its low-rank restriction. The results in Table 9 show that the low-rank SDP formulation achieves comparable clustering accuracy in the small-sample setting. For $p = 5000$, the average end-to-end runtime was 121.33 s for the original SDP-based iterative algorithm and 26.66 s for the low-rank SDP-based version (Intel Core i7-1270P CPU, 16 GB RAM).

We further verify numerically that the low-rank restriction-based algorithm efficiently handles sample sizes up to $n = 10,000$, achieving reliable clustering accuracy (Table 10). For $n = 10,000$, the average end-to-end runtime of the iterative algorithm was about seven minutes (Intel Core i7-1270P CPU, 16 GB RAM).

To further demonstrate scalability to extremely large sample sizes (e.g., $n = 100,000$), we incorporate the sketch-and-lift technique of Zhuang et al. (2022). This method randomly subsamples observations, solves the smaller clustering problem, and then propagates the clustering results. Zhuang et al. (2022) show that this approach enjoys a similar statistical guarantees as the inner algorithm applied to the sub-problems. In our implementation, we solve the low-rank restriction of the SDP proposed by Zhuang et al. (2024) for the sub-problems of sample size $n^{3/7}$. Table 11 reports strong clustering accuracy and favorable end-to-end timing of this approach, for sample sizes up to $n = 100,000$ (Intel Core i7-1270P CPU, 16 GB RAM). The runtime increases only modestly with n , since feature selection

Table 9: Clustering accuracy of our greedy algorithm with different clustering step formulations. Original SDP solved by ADMbased solver; low-rank restriction solved via Burer–Monteiro factorization. Other settings follow line 1 of Table 3 with $\Delta_{S^*} = 4$.

Clustering step formulation	$p = 1000$	$p = 2000$	$p = 3000$	$p = 4000$	$p = 5000$
Original SDP	0.97	0.93	0.86	0.74	0.68
Low-rank restriction of SDP	0.97	0.93	0.83	0.74	0.66

Table 10: Clustering accuracy of our greedy algorithm with clustering step formulated as low-rank restriction of SDP, under varying n with fixed ambient dimension $p = 10000$. Low-rank restriction solved via Burer–Monteiro factorization. Other settings follow line 1 of Table 3 with $\Delta_{S^*} = 3$.

Clustering step formulation	$n = 7000$	$n = 8000$	$n = 9000$	$n = 10000$
Low-rank restriction of SDP	0.92	0.92	0.92	0.93

Table 11: Scalability of our greedy algorithm with known covariance, with clustering step formulated as sketch-and-lift with low-rank restriction of SDP, for fixed ambient dimension $p = 10,000$ and varying sample sizes. Other settings follow line 1 of Table 3 with $\Delta_{S^*} = 3$.

	$n = 60,000$	$n = 80,000$	$n = 100,000$
Clustering accuracy	0.957	0.962	0.969
End-to-end timing (s)	196.80	277.58	345.03
Iterations until stop	11.26	10.54	10.34

and the sketch-and-lift technique keep the per-iteration cost manageable. Moreover, the algorithm tends to stop earlier for larger sample sizes, further reducing the overall runtime.

22.6 Comparison with Subspace Clustering Approach

While both subspace clustering and sparse clustering involve dimensionality reduction, they differ fundamentally. Subspace clustering achieves dimensionality reduction via linear combinations of features, whereas feature selection explicitly identifies and prunes individual variables. Beyond this conceptual distinction, we empirically compared our method against a subspace clustering approach with theoretical guarantees: the Johnson-Lindenstrauss (JL) transform-based K -means (JL K -means; Boutsidis et al. 2010). After applying their JL transform, we ran the SDP-relaxed K -means (Chen & Yang 2021) on the dimension-reduced data. This JL transform is theoretically guaranteed to preserve the K -means objective up to a constant. Under the setting of line 1 of Table 3), which has 10 signal

Table 12: Comparison of clustering accuracy of our greedy algorithm with JL K -means with different intrinsic dimension parameter t . Other settings follow line 1 of Table 3 with $\Delta_{S^*} = 4$.

	$p = 2000$	$p = 3000$	$p = 4000$	$p = 5000$
JL K -means ($t = 10$)	0.53	0.54	0.53	0.52
JL K -means ($t = 20$)	0.53	0.53	0.52	0.54
Our Greedy Algorithm	0.93	0.86	0.74	0.68

features, we provided the JL algorithm with intrinsic dimension parameters $t = 10$ and $t = 20$. The results, displayed in Table 12, show that our greedy algorithm outperforms JL K -means in terms of clustering accuracy.